

Supporting Information for:

Antiaromatic Covalent Organic Frameworks Based on Dibenzopentalenes

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S1. Materials and Methods

Reactions were performed using reagent grade chemicals, purchased from various suppliers (ABCR, BLDpharm, fluorochem, Sigma-Aldrich, and TCI) and used without further purification, unless otherwise specified. Solvents utilized for synthesis were dried using a Pure Solv Micro Solvent Purification System (Innovative Technology) and stored over molecular sieves 3–4 Å. All non-aqueous reactions were performed in oven-dried glassware and under an Ar atmosphere. Automated Medium Pressure Column Chromatography (MPLC) was performed on a Teledyne ISCO CombiFlash Rf+ system, equipped with a 200–800 nm UV/Vis variable wavelength detector. Thin layer chromatography (TLC) was conducted on aluminum sheets coated with SiO₂-60 F254 (Merck); visualized with a UV lamp (254 or 366 nm). Evaporation in vacuo was performed at 40–60 °C and 700–10 mbar. All products were dried under vacuum (ca. 1 mbar) before analytical characterization. Reported yields refer to spectroscopically and chromatographically pure compounds, unless otherwise specified.

Nuclear magnetic resonance (NMR) spectra were recorded using a Bruker Avance II 300 (300 MHz for ¹H and 75 MHz for ¹³C) and a Bruker Avance II 500 (500 MHz for ¹H and 126 MHz for ¹³C) at 298 K and are reported as follows: chemical shift (δ) in ppm (multiplicity, coupling constant *J* in Hz, number of protons; assignment). The residual deuterated solvent was used as the internal reference (CDCl₃: δ_H = 7.26 ppm, CD₂Cl₂: δ_H = 5.32 ppm, CD₃OD: δ_H = 3.31 ppm, (CD₃)₂SO: δ_H = 2.50 ppm; CDCl₃: δ_C = 77.16 ppm, CD₂Cl₂: δ_C = 54.00 ppm, CD₃OD: δ_C = 49.00 ppm, (CD₃)₂SO: δ_C = 39.52 ppm; The resonance multiplicity is described as s (singlet), d (doublet), t (triplet), q (quartet), quint (quintet), sept. (septet), m (multiplet), and br. (broad).

Solid-state ¹³C cross-polarization magic angle spinning (CP-MAS) NMR spectra were recorded at room temperature using an AVANCE 400 spectrometer and a 4 mm probe head under magic angle spinning conditions. All experiments were performed with a rotation frequency of 10 kHz, a contact time of 1 ms and a recycle delay of 5 s. ¹³C chemical shift values are given with respect to TMS and were determined against adamantane as a secondary standard (δ = 29.5 ppm).

Infrared (IR) spectra were recorded on a Vertex 70v spectrometer by Bruker equipped with a diamond ATR attachment and are baseline-corrected. The spectra were measured between 4000 and 400 cm⁻¹. Absorption bands are reported in wavenumbers (cm⁻¹) and their relative intensities described as s (strong), m (medium), or w (weak).

Ultraviolet–visible (UV–vis) absorbance spectroscopy was performed on Agilent Cary 50 and Cary 60 instruments connected to a cryostat from Unisoku Scientific Instruments (temperature accuracy ±0.1 K) in 10 × 10 mm quartz cuvettes with 3 mL volume. Weighing of small quantities was performed on a Sartorius ME5 analytical microbalance.

Electrochemical Measurements were performed using an Autolab PGSTAT 128N potentiostat (Deutsche METROHM GmbH & Co. KG, Germany) with software NOVA 2.1 for data evaluation. A three-electrode configuration contained in a non-divided cell, consisting of a platinum disc (*d* = 1 mm) as working electrode, a platinum plate as counter-electrode and a saturated calomel electrode (SCE) with

an agar-agar-plug in a Luggin capillary with a diaphragm as reference electrode was used (HMDE: 663 VA Stand, Metrohm, Swiss-made). Measurements were carried out in 1×10^{-3} M solutions in CH_2Cl_2 (HPLC-grade, dried over calcium hydride and distilled) containing 0.1 M Bu_4NPF_6 using a scan rate of $dE/dt = 100 \text{ mV s}^{-1}$. The data is given in reference to the ferrocene redox couple (Fc/Fc^+), which was used as an external standard.

Diffuse reflectance UV-vis spectroscopy was carried out on a Perkin Elmer Lambda 950 spectrometer. COF samples were mixed with BaSO_4 prior to measurements.

High-resolution mass spectrometry was measured on following set-ups: a Thermo Fisher Orbitrap XL via MALDI in DHB matrix with a resolution of 70 000 in positive ion mode; a Bruker Solarix via MALDI using trans-2-[3-(4-tert-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB) as matrix; a Bruker Solarix (Hybrid 7T FT-ICR) via ESI.

Supercritical CO_2 drying of COFs was performed using a samdri-PVT-3B supercritical point dryer. COFs were placed into teabags (drawstring teabag, Amazon) while wet, the teabags were sealed with a staple and stored in MeOH for at least 18 h before drying. The tea bags containing the material were placed in the drying chamber and the chamber was topped up with absolute EtOH. The apparatus was cooled by opening the “cool” valve and the chamber was filled with liquid CO_2 by opening the “fill” valve. By opening the “purge” valve the chamber was flushed to replace the EtOH with CO_2 . The “fill” valve was opened and closed at least five times consecutively. The “purge” valve was then closed and the pressure was allowed to rise to 58–62 bar. Next, the “fill” valve was closed and the heating was switched on. The temperature and pressure were allowed to rise above the critical point but also below 100 bar by venting using the “purge” valve. The chamber was held above the critical point of CO_2 ($T_c = 31^\circ\text{C}$, $p_c = 73.8 \text{ bar}$) for 30 min. By opening the “purge” valve, the pressure was released as slowly as possible. The dried COF material was collected from the tea bags. 1–4 COF-containing teabags were dried in the chamber simultaneously.

Small- / wide-angle X-ray scattering (SAXS/WAXS) measurements (further referred to as PXRD) were conducted using the MOUSE (Methodology Optimization for Ultrafine Structure Exploration) at the Bundesanstalt für Materialforschung und -prüfung (BAM).¹ X-rays were generated from a microfocus X-ray tube, followed by multilayer optics to parallelize and monochromatize the X-ray beams to wavelengths of $\text{Cu K}\alpha$ ($\lambda = 0.154 \text{ \AA}$). Scattered radiation was detected on an *in-vacuum* Eiger 1M detector (Dectris, Switzerland), which was placed at multiple distances between 55–2307 mm from the sample. The resulting data has been processed using the DAWN software package in a standardized complete 2D correction pipeline with uncertainty propagation.^{2,3} The experimental patterns were fitted using Pawley refinement to computational COF models constructed using Materials Studio program package.⁴

Nitrogen sorption isotherms were collected using a Micromeritics ASAP 2020 Accelerated Surface Area and Porosity Analyzer. 10–40 mg samples were transferred to dried and tared analysis tubes equipped with filler rods and capped with a seal frit. The samples were heated to 40°C at a rate of 1°C min^{-1} and evacuated at 40°C for 60 min, then heated to 120°C at a rate of 1°C min^{-1} and

evacuated at 100 °C for 10 h. The tube was weighed again to determine the mass of the activated sample and transferred to the analysis port of the instrument. N₂ gas (99.999% purity) was used for all adsorption measurements. N₂ isotherms were measured from a relative pressure p/p_0 of 0.01 to 1 in a liquid nitrogen (77 K) bath. Brunauer–Emmett–Teller (BET) surface areas were calculated from the N₂ isotherm at 77 K within the pressure range p/p_0 of 0.05–0.07.

Pore size distributions were evaluated using NL-DFT employing the SAIEUS program by Micromeritics. N₂-adsorption isotherms with data points down to 10⁻⁶ p/p_0 were used.

Structural Models of COFs were constructed using the Accelrys Materials Studio software package. *P6/m* symmetry was applied for all COFs. The structure models were optimized using the Forcite module of Materials Studio performing an “ultra-fine” optimization using Universal force field and “Smart” algorithm including optimizing the cell. Structural refinements were carried out using the Reflex module with a Pawley-refinement.

Scanning electron microscopy (SEM) was carried out using a Thermo Scientific Phenom Pharos benchtop microscope using a 10 kV voltage. COF flakes were placed on conductive tape and coated with a 20–30 nm gold layer. The morphology of all COFs was found to be consistent throughout the sample and pictures were recorded from representative areas of the sample.

Low-dose, high-resolution transmission electron microscopy (HR-TEM): COF samples for HRTEM were sonicated in ethanol for 15 min and an aliquot dispersion (4 µL) was applied to Lacey carbon-coated copper TEM grids (200 mesh, Science Services) and allowed to air-dry. At room temperature, the grids were loaded into a cryogenic transfer holder (Gatan 914, Gatan, Munich, Germany) that was cooled with liquid nitrogen after the transfer to the TEM. Imaging was performed with a low dose acquisition scheme using SerialEM⁵ on JEM-2100 (JEOL GmbH, Echting, Germany) operated at 200 kV and equipped with a 4 k × 4 k CMOS digital camera (TVIPS TemCam-F416). HRTEM images were acquired at a magnification of 100,000×, or 500,000× corresponding to a pixel size of 0.1198 nm and 0.0229 nm respectively at the specimen level, while keeping the total electron dose below 10 e Å⁻². All imaging was carried out at temperatures around 90 K.

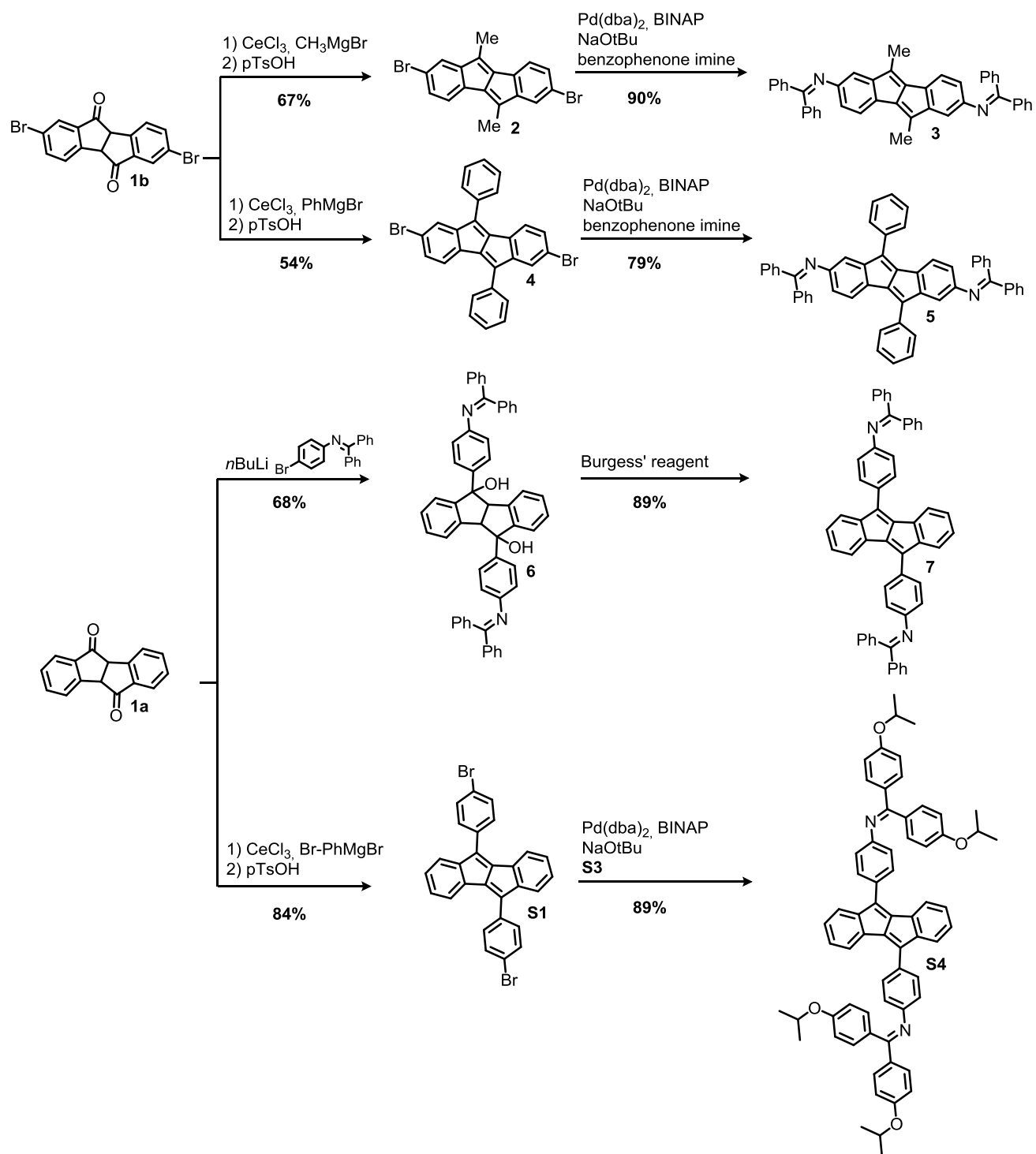
COF films were obtained by mechanical or acid-mediated exfoliation. For mechanical exfoliation, COFs were dispersed in MeOH (~ 2 mg mL⁻¹) and sonicated for 1 h using a BANDELIN HD 2200/-U (200 W, HF 20 kHz) sonotrode setup with a MS72 sonotrode with the amplitude set to 10% in continuous mode. For acid-mediated exfoliation, COFs were dispersed in MeCN/THF/TFA 7:3:2 (1 mg mL⁻¹) and stirred for 18 h. The dispersions were each drop-cast onto an Ossila interdigitated ITO substrate and dried in high-vacuum for 18 h. The film thickness was determined by scratching into the film with a needle and measuring the height using an Olympus LEXT laser scanning microscope.

Conductivity and photoconductivity measurements were conducted on COF films on Ossila interdigitated ITO substrates. Thorlabs LEDs were focused with a Thorlabs collimator adapter (SM1U25) on the substrate and were controlled by a Keithley 2461 in pulse mode at the specified voltage. The interdigitated substrate was contacted by an Ossila board “Push-Fit Test Board for Photovoltaic Substrates (8 pixel)” and a source measure unit (4200A SCS parameter analyser; Keithley) was used

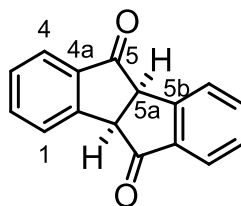
for current measurement. The five interdigitated devices were measured in parallel at a bias of 20 V. The LEDs have the following specifications: Thorlabs' M375L4 mounted LED: 375 nm, bandwidth = 9 nm, $P_{\text{out}} = 1540$ mW, $V_F = 3.6$ V; Thorlabs' M405L4 mounted LED: 405 nm, bandwidth = 12.5 nm, $P_{\text{out}} = 1300$ mW, $V_F = 3.4$ V; Thorlabs' M505L4 mounted LED: 505 nm, bandwidth = 30 nm, $P_{\text{out}} = 440$ mW, $V_F = 3.3$ V; Thorlabs' M660L4 mounted LED: 660 nm, bandwidth = 20 nm, $P_{\text{out}} = 1050$ mW, $V_F = 2.6$ V.

S2. Experimental Procedures

Scheme S1 Overview of synthetic routes

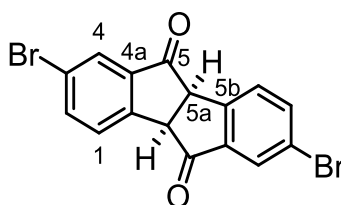


4b,9b-Dihydroindeno[2,1-a]indene-5,10-dione (**1a**)⁶



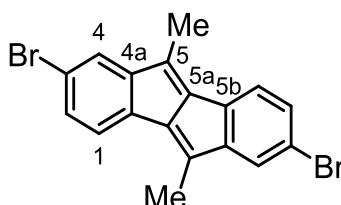
1a was synthesized in three steps from ethyl phenylacetate according to literature.⁶ The product of the last step was purified by column chromatography (silica gel, 0%→20% EtOAc in cyclohexane). The characterization of **1a** and its precursors matches literature.⁶

2,7-Dibromo-4b,9b-dihydroindeno[2,1-a]indene-5,10-dione (**1b**)⁷



1b was synthesized in three steps from ethyl 2-(4-bromophenyl)acetate according to literature.⁷ The characterization of **1b** and its precursors matches literature.⁷

2,7-Dibromo-5,10-dimethylindeno[2,1-a]indene (**2**)⁸



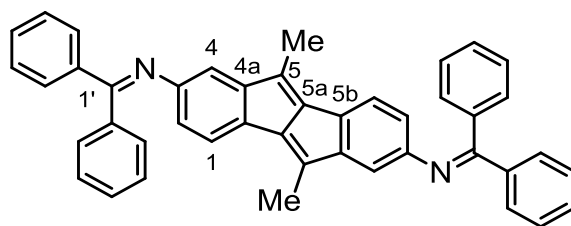
Based on a reported procedure,⁸ anh. CeCl_3 (1.531 g, 6.21 mmol, 3 eq.) was dried at 130 °C under vacuum in a Schlenk flask for 2 h. THF (13 mL) was added and the resulting suspension was stirred for 20 min. After cooling to 0 °C, MeMgBr solution (3 M in diethyl ether, 6.21 mmol, 2.07 mL, 3 eq.) was added and the mixture was stirred at 0 °C for 10 min. 2,7-Dibromo-4b,9b-dihydroindeno[2,1-a]indene-5,10-dione (**1b**, 812 mg, 2.07 mmol, 1 eq.) was added in portions and the mixture was allowed to warm to room temperature. After 4 h, the mixture was diluted with sat. aq. NH_4Cl (20 mL) and extracted with CH_2Cl_2 (3 × 50 mL). The combined organic phases were washed with brine (100 mL), dried over MgSO_4 and the solvent was removed under reduced pressure. The crude product was dissolved in toluene (40 mL), treated with *para*-toluenesulfonic acid (1.05 g, 5.56 mmol, 3 eq.) and heated to reflux in a Dean–Stark apparatus for 4 h. The mixture was allowed to cool to room temperature and the resulting red needles were filtered. The filtrate was concentrated under reduced pressure and acetone (100 mL) was added. The mixture was sonicated and the resulting red powder was filtered. The two product

fractions were combined to give **2** (534 mg, 67% over both reaction steps) as a red solid. Characterization matches literature.⁸

¹H NMR (500 MHz, CDCl₃) δ = 7.07 (dd, J = 7.8, 1.8 Hz, 2H, H–C(2)), 7.01 (d, J = 1.7 Hz, 2H, H–C(4)), 6.98 (d, J = 7.8 Hz, 2H, H–C(1)), 2.14 ppm (s, 6H, CH₃);

¹³C NMR (126 MHz, CDCl₃) δ = 153.0 (C_q), 143.3 (C_q), 136.1 (C_q), 133.9 (C_q), 129.7 (C(2)), 124.3 (C(4)), 122.9 (C(1)), 121.5 (C_q), 11.9 ppm (CH₃).

***N,N'*-(5,10-Dimethylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) (**3**)**



In a pressure flask, a solution of [Pd(dba)₂] (317 mg, 0.552 mmol, 30 mol%) and *rac.* BINAP (516 mg, 0.828 mmol, 45 mol%) in toluene (40 mL) was degassed by three freeze–pump–thaw cycles, backfilled with Ar, and heated to 120 °C for 30 min. The mixture was allowed to cool to room temperature and treated with NaOtBu (530 mg, 5.52 mmol, 3 eq.), benzophenone imine (1.167 g, 1.08 mL, 6.44 mmol, 3.5 eq.), and 2,7-dibromo-5,10-dimethylindeno[2,1-*a*]indene (**2**, 714 mg, 1.84 mmol, 1 eq.). The mixture was stirred at 100 °C for 2 h, filtered through a cotton plug while hot, and the filtrate was concentrated in vacuo. EtOH (200 mL) was added to the residue, the mixture was first sonicated and then heated to reflux, and filtered while hot to obtain **3** (970 mg, 90 %) as a brown powdery solid.

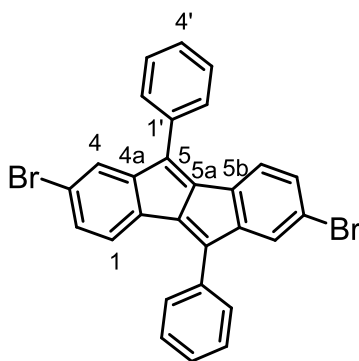
¹H NMR (500 MHz, CD₂Cl₂): δ = 7.73–7.71 (m, 4H, benzophenone), 7.49–7.46 (m, 2.6H, benzophenone), 7.42–7.39 (m, 4.7H benzophenone), 7.34–7.32 (m, 6H, benzophenone), 7.19–7.17 (m, 4H, benzophenone), 6.86 (d, J = 7.7 Hz, 2H, H–C(1)), 6.31 (d, J = 1.8 Hz, 2H, H–C(4)), 6.16 (dd, J = 7.7, 1.9 Hz, 2H, H–C(2)), 1.97 ppm (s, 6H, CH₃), rotamers at the NMR time scale cause a more complex spectrum due to the rotation of the benzophenone imine residue at the C–N bond;

¹³C NMR (126 MHz, CD₂Cl₂): δ = 168.3 (C(1')), 152.5 (C_q), 151.5 (C_q), 143.9 (C_q), 140.2 (C_q), 137.1 (C_q), 135.1 (C_q), 131.2 (benzophenone), 130.7 (C(3)), 129.9 (benzophenone), 129.8 (benzophenone), 129.2 (benzophenone), 128.7 (benzophenone), 128.6 (benzophenone), 121.9 (C(1)), 118.9 (C(2)), 115.1 (C(4)), 11.8 ppm (CH₃);

FT-IR (ATR): $\tilde{\nu}_{\text{max}}$ = 3052 (m), 2360 (s), 2342 (m), 1590 (m), 1492 (w), 1435 (s), 1315 (w), 959 (w), 827 (w), 462 (w), 692 cm^{–1} (s);

HR-MS (ESI) m/z : [M + H]⁺ Calcd for C₄₄H₃₃N₂⁺ 589.2638; Found: 589.2642.

2,7-Dibromo-5,10-diphenylindeno[2,1-a]indene (**4**)



Adapted from a literature procedure,⁹ anhydrous CeCl_3 (5.658 g, 22.96 mmol, 3 eq.) was dried at 130 °C under vacuum in a Schlenk flask for 4 h and was then allowed to cool to room temperature. THF (50 mL) was added and the resulting suspension was stirred for 18 h. In a separate Schlenk flask, Mg turnings (558 mg, 22.96 mmol, 3 eq.) and bromobenzene (3.604 g, 2.417 mL, 22.96 mmol, 3 eq.) were slowly stirred in THF (8 mL) until the Mg turnings had fully dissolved. The CeCl_3 suspension was cooled to 0 °C and treated with the Grignard solution and the mixture was stirred for 1.5 h at 0 °C. 2,7-Dibromo-4b,9b-dihydroindeno[2,1-a]indene-5,10-dione (**1b**, 3.00 g, 7.652 mmol, 1 eq.) was added in portions to the reaction mixture at 0 °C, the mixture was allowed to warm to room temperature, and was stirred for 20 min at that temperature. The mixture was diluted with sat. aq. NH_4Cl (50 mL) and extracted with CH_2Cl_2 (3 $\times$ 50 mL). The combined organic phases were washed with brine (150 mL), dried over MgSO_4 , and concentrated in vacuo. The crude product was dissolved in toluene (120 mL), treated with *para*-toluenesulfonic acid (3.316 g, 19.26 mmol, 3 eq.), and heated to reflux in a Dean–Stark apparatus for 4 h. The solution was concentrated in vacuo and MeOH (150 mL) was added. After sonication, the suspension was refluxed and subsequently filtered while hot, and the solid was collected. This procedure was repeated with EtOH (100 mL) to afford **4** (2.091 g, 54% over 2 steps) as a brown crystalline solid.

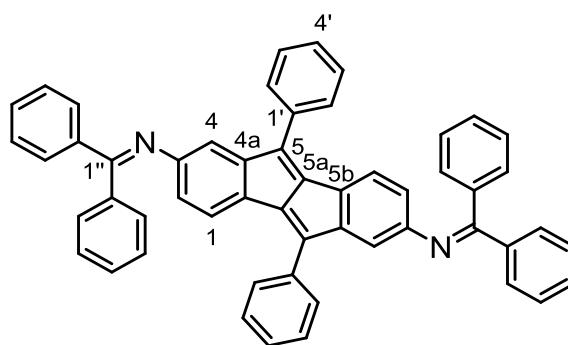
^1H NMR (500 MHz, CDCl_3): δ = 7.61–7.60 (m, 4H, H–C(2')), 7.56–7.53 (m, 4H, H–C(3')), 7.49–7.46 (m, 2H, H–C(4')), 7.14 (d, J = 1.7 Hz, 2H, H–C(4)), 7.04 (d, J = 7.9 Hz, 2H, H–C(1)), 6.99 ppm (dd, J = 7.9, 1.7 Hz, 2H, H–C(2));

^{13}C NMR (101 MHz, CDCl_3): δ = 151.8 (C(4a)), 143.2 (C(5a)), 140.6 (C(5)), 133.6 (C(5b)), 133.1 (C(1')), 130.3 (C(2)), 129.5 (C(4')), 129.1 (C(3')), 128.4 (C(2')), 126.0 (C(4)), 123.2 (C(1)), 122.2 ppm (C(3));

FT-IR (ATR): $\tilde{\nu}_{\text{max}}$ = 3052 (m), 2359 (m), 1614 (w), 1571 (m), 1488 (m), 1426 (s), 1289 (w), 1062 (m), 818 (w), 710 cm^{-1} (m);

HRMS (MALDI) m/z : $[\text{M}]^+$ Calcd for $\text{C}_{28}\text{H}_{16}\text{Br}_2^+$ 511.9593; Found: 511.9591.

***N,N'*-(5,10-Diphenylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) (**5**)**



A degassed solution of [Pd(dba)₂] (404 mg, 0.703 mmol, 30 mol%) and *rac.* BINAP (656 mg, 1.054 mmol, 45 mol%) in toluene (50 mL) was heated to 120 °C for 30 min in a sealed pressure flask. The mixture was allowed to cool to room temperature, was treated with NaOtBu (675 mg, 7.028 mmol, 3 eq.), benzophenone imine (1.486 g, 1.38 mL, 8.199 mmol, 3.5 eq.), and 2,7-dibromo-5,10-diphenylindeno[2,1-*a*]indene (**4**, 1.200 g, 2.342 mmol, 1 eq.) and the mixture was stirred at 100 °C for 1 h. The mixture was filtered through a cotton plug while hot and the filtrate was concentrated in vacuo. The residue was sonicated in CHCl₃/EtOH 1:1 (200 mL) and the resulting suspension was heated to reflux. The solid was filtered while hot to afford **5** (1.311 g, 79%) as a brown powdery solid.

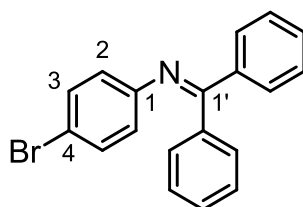
¹H NMR (500 MHz, CDCl₃) δ = 7.68–7.66 (m, 4H), 7.45–7.42 (m, 2H), 7.39–7.32 (m, 16H), 7.30–7.28 (m, 4H), 7.17–7.15 (m, 4H), 6.99 (d, *J* = 7.8 Hz, 2H, H–C(1)), 6.32 (d, *J* = 1.8 Hz, 2H, H–C(4)), 6.28 ppm (dd, *J* = 7.8, 1.9 Hz, 2H, H–C(2));

¹³C NMR (126 MHz, CDCl₃) δ = 168.2, 151.2, 150.5, 143.7, 139.9, 139.1, 136.7, 133.9, 130.8, 130.2, 129.7, 129.5, 128.9, 128.6, 128.5, 128.3, 128.3, 122.1 (C(1)), 120.2 (C(2)), 116.8 ppm (C(4));

FT-IR (ATR): $\tilde{\nu}_{\text{max}}$ = 3055 (m), 3029 (m), 1615 (s), 1587 (s), 1573 (s), 1488 (m), 1432 (s), 1315 (m), 1271 (m), 951 (w), 831 (w), 734 (w), 692 (s), 664 (w), 643 cm^{–1} (w);

HR-MS (MALDI) *m/z*: [M + H]⁺ Calcd for C₅₄H₃₇N₂⁺ 713.2952; Found: 713.2917.

***N*-(4-Bromophenyl)-1,1-diphenylmethanimine (**A**)¹⁰**



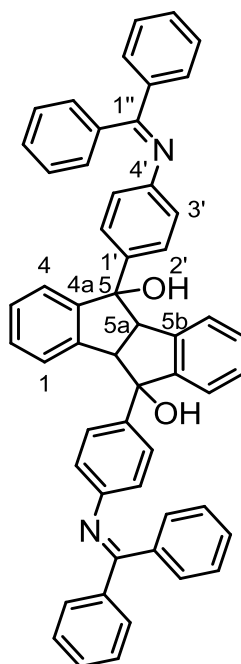
Adapted from a literature procedure,¹¹ 4-bromoaniline (8.00 g, 46.5 mmol, 1 eq), benzophenone (8.05 g, 44.2 mmol, 0.95 eq), tetraethyl orthosilicate (11.63 g, 55.8 mmol, 1.2 eq), and H₂SO₄ (4 mg, 46.5 μ mol) were added to a flask equipped with a Liebig condenser and a distillation collecting flask. After heating for 20 h at 160 °C, the reaction mixture was cooled to room temperature and diluted with aq. KOH (1.0 M,

50 mL). The resulting mixture was vigorously stirred for 10 min and extracted with CH₂Cl₂ (3 × 50 mL). The combined organic layers were washed with H₂O (50 mL) and brine (50 mL), dried over MgSO₄ and the solvent was removed under reduced pressure. The resulting sticky black solid was washed with ice-cold ethanol (5 × 20 mL) to give **A** (10.80 g, 69%) as yellow solid. Characterization matches literature.¹⁰

¹H NMR (400 MHz, CDCl₃): δ = 7.72–7.69 (m, 2H, benzophenone), 7.47–7.43 (m, 1H, benzophenone), 7.40–7.38 (m, 2H, benzophenone), 7.31–7.27 (m, 3H, benzophenone), 7.25–7.21 (m, 2H, H–C(3)), 7.08–7.06 (m, 2H, benzophenone), 6.58–6.56 ppm (m, 2H, H–C(2));

¹³C NMR (101 MHz, CDCl₃): δ = 169.0 (C(1')), 150.4 (C(1)), 139.5 (benzophenone), 136.0 (benzophenone), 131.7 (benzophenone), 131.1 (C(3)), 129.6 (C), 129.5 (C), 129.0 (benzophenone), 128.4 (benzophenone), 128.2 (benzophenone), 122.9 (C(2)), 116.4 ppm (C(4)).

5,10-Bis(4-((diphenylmethylene)amino)phenyl)-4b,5,9b,10-tetrahydroindeno[2,1-a]-indene-5,10-diol (6**)**



A solution of *N*-(4-bromophenyl)-1,1-diphenylmethanimine (**A**, 2.153 g, 6.403 mmol, 3 eq.) in THF (20 mL) was treated dropwise with *n*BuLi (2.5 M in hexanes, 2.6 mL, 6.403 mmol, 3 eq.) at –78 °C. 4b,9b-Dihydroindeno[2,1-a]indene-5,10-dione (**1a**, 500 mg, 2.134 mmol, 1 eq) was added and the reaction mixture was allowed to warm to room temperature and stirred for 3 h. The reaction mixture was diluted with sat. aq. NH₄Cl (30 mL) and extracted with CH₂Cl₂ (3 × 60 mL). The combined organic layers were washed with sat. aq. NaHCO₃ (100 mL) and brine (100 mL), dried over MgSO₄, and the solvent was removed under reduced pressure. Column chromatography (SiO₂, 4%→30% EtOAc in cyclohexane, with 1% NEt₃) afforded diol **6** (1.083 g, 68%) as a yellow solid.

¹H NMR (500 MHz, (CD₃)₂SO): δ = 7.70–7.69 (m, 4H, benzophenone), 7.56–7.53 (m, 2H, benzophenone), 7.50–7.47 (m, 4H, benzophenone), 7.37–7.36 (m, 6H, benzophenone), 7.24–7.22 (m,

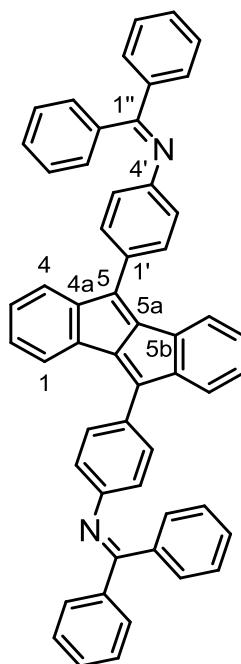
4H, H-C(1,2,3,4)), 7.19–7.17 (m, 4H, benzophenone), 7.16–7.14 (m, 4H, H-C(2')), 6.83–6.77 (m, 4H, H-C(1,2,3,4), 6.72 (d, $J = 8.5$ Hz, 4H, H-C(3')), 5.69 (s, 2H, OH), 3.94 ppm (s, 2H, H-C(5a)).

^{13}C NMR (126 MHz, $(\text{CD}_3)_2\text{SO}$): $\delta = 167.7$ (C(1'')), 150.4 (C(4')), 149.9 (C_q), 140.6 (C_q), 140.4 (C_q), 138.8 (C_q), 136.0 (C_q), 131.0 (C–H), 129.0 (C–H), 128.8 (C–H), 128.5 (C_q), 128.4 (C–H), 128.0 (C–H), 127.6 (C–H), 126.1 (C(2')), 125.3 (C–H), 124.6 (C–H), 120.0 (C(3')), 82.7 (C(5)), 63.5 ppm (C(5a));

FT-IR (ATR): $\tilde{\nu}_{\text{max}} = 3711$ (w), 3671 (w), 3657 (w), 3619 (w), 3589 (w), 3527 (w), 3436 (m), 3422 (m), 3386 (m), 3305 (w), 3263 (w), 3231 (w), 3218 (w), 3189 (w), 3064 (m), 3026 (m), 2928 (w), 2370 (w), 2352 (m), 2339 (m), 2328 (m), 1700 (w), 1615 (s), 1596 (s), 1572 (m), 1500 (m), 1476 (w), 1446 (m), 1316 (m), 1292 (m), 1224 (w), 1176 (w), 1142 (w), 1055 (w), 1001 (w), 959 (w), 840 (w), 783 (w), 762 (m), 739 (w), 717 (w), 697 (s), 635 (w), 618 cm^{-1} (w);

HR-MS (MALDI) m/z : $[\text{M} + \text{H}]^+$ Calcd for $\text{C}_{54}\text{H}_{41}\text{N}_2\text{O}_2^+$ 749.3163; Found: 749.3161.

***N,N'*-(Indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-diphenylmethanimine) (7)**



A solution of 5,10-bis(4-((diphenylmethylene)amino)phenyl)-4b,5,9b,10-tetrahydroindeno[2,1-*a*]indene-5,10-diol (**6**, 717 mg, 957 μmol 1 eq) in toluene (10 mL) was treated with 1-methoxy-*N*-triethylammoniosulfonyl-methanimidate (Burgess reagent, 912.6 mg, 3.83 mmol, 4 eq.) and the mixture was stirred at 110 °C for 1 h. Another portion of 1-methoxy-*N*-triethylammoniosulfonyl-methanimidate (228 mg, 957 μmol , 1 eq.) was added and after another 30 min, the mixture was filtered while hot. The filter cake was rinsed with CH_2Cl_2 (5 $\times$ 20 mL) to give **7** (609 mg, 89%) as a red solid. Signals in the ^1H NMR spectrum that do not correspond to the compound stem from trace impurities in the NMR solvent. Very low solubility and poor signal intensity over-emphasize the trace impurities in comparison. A blank spectrum of the deuterated solvent is shown

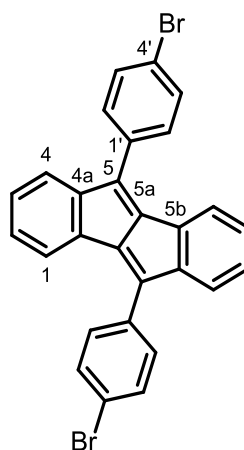
as reference. Due to low solubility in common deuterated organic solvents, ^{13}C NMR, ^1H , ^{13}C HSCQ, and ^1H , ^{13}C HMBC were not successful.

^1H NMR (500 MHz, $\text{C}_2\text{D}_2\text{Cl}_4$): δ = 7.82 (d, J = 7.3 Hz, 4H), 7.54 (t, J = 7.4 Hz, 2H), 7.49–7.44 (m, 8H), 7.36–7.26 (m, 7H), 7.20–7.16 (m, 5H), 7.01 (t, J = 7.3 Hz, 3H), 6.92 (t, J = 7.5 Hz, 2H), 6.87–6.83 ppm (m, 5H);

FT-IR (ATR): $\tilde{\nu}_{\text{max}}$ = 3854 (w), 3839 (w), 3746 (m), 3711 (w), 3617 (w), 3052 (m), 2349 (m), 2328 (m), 1607 (s), 1590 (s), 1570 (s), 1495 (m), 1444 (m), 1426 (m), 1317 (m), 1294 (m), 1225 (w), 1156 (w), 1143 (w), 957 (m), 853 (m), 772 (w), 753 (m), 714 (w), 698 (s), 674 (w), 647 cm^{-1} (w);

HR-MS (MALDI) m/z : $[\text{M} + \text{H}]^+$ Calcd for $\text{C}_{54}\text{H}_{37}\text{N}_2^+$ 713.2952; Found: 713.2936.

5,10-Bis(4-bromophenyl)indeno[2,1-a]indene (**S1**)¹²



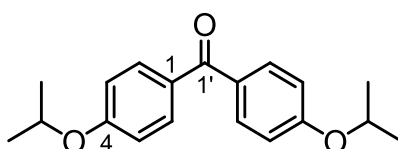
Adapted from a literature procedure,⁸ anhydrous CeCl_3 (3.602 g, 14.6 mmol, 3 eq) was dried at 135 °C under vacuum in a Schlenk flask for 2 h and was then allowed to cool to room temperature. THF (37 mL) was added and the resulting suspension was stirred for 14 h. In a separate Schlenk flask Mg turnings (355.2 mg, 14.6 mmol, 3 eq) and 1,4-dibromobenzene (3.447 g, 14.6 mmol, 3 eq) were slowly stirred in THF (12 mL) until the Mg turnings had fully dissolved. The CeCl_3 suspension was cooled to 0 °C, treated with the Grignard solution and the mixture was stirred at 0 °C for 1.5 h. 4b,9b-Dihydroindeno[2,1-a]indene-5,10-dione (**1a**, 1.141 g, 4.9 mmol, 1 eq) was added in portions to the reaction mixture at 0 °C, the mixture was allowed to warm to room temperature and was stirred for 6 h at that temperature. The mixture was diluted with sat. aq. NH_4Cl (40 mL) and extracted with CH_2Cl_2 (3 $\times$ 100 mL). The combined organic phases were washed with sat. aq. NaHCO_3 (150 mL) and brine (150 mL), dried over MgSO_4 , and concentrated in vacuo. The crude product was dissolved in toluene (60 mL), treated with *para*-toluenesulfonic acid (2.779 g, 14.6 mmol, 3 eq) and heated to reflux in a Dean–Stark apparatus for 3 h. The solution was concentrated in vacuo and MeOH (50 mL) was added. After sonication, the suspension was refluxed and filtered while hot affording **S5** (2.094 g, 84%) as a brown solid. Characterization matches literature.¹²

^1H NMR (500 MHz, CDCl_3): δ = 7.65 (d, J = 8.4 Hz, 4H, H–C(3')), 7.53 (m, J = 8.4 Hz, 4H, H–C(2')), 7.15 (d, J = 7.2 Hz, 2H, H–C(1')), 6.96 (d, J = 6.9 Hz, 2H, H–C(4')), 6.90 (td, J = 7.5, 1.1 Hz, 2H, H–C(3)), 6.85 ppm (td, J = 7.4, 1.1 Hz, 2H, H–C(2));

¹³C NMR (125 MHz, CDCl₃): δ = 149.4 (C(4a)), 143.5 (C(5a)), 139.8 (C(5)), 134.9 (C(5b)), 132.8 (C(1')), 132.1 (C(3')), 130.2 (C(2')), 128.3 (C(3)), 127.9 (C(2)), 123.1 (C(4')), 122.5 (C(4)), 122.1 ppm (C(1));

FT-IR (ATR): $\tilde{\nu}_{\text{max}}$ = 2350 (s), 2251 (w), 1769 (m), 1759 (m), 1650 (w), 1485 (w), 1432 (w), 1396 (w), 1245 (m), 743 (s) cm⁻¹.

Bis(4-isopropoxyphenyl)methanone (**S2**)¹³

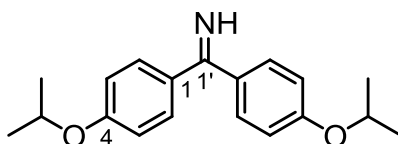


Based on a literature procedure,¹³ a solution of bis(4-hydroxyphenyl)methanone (6.88 g, 32.1 mmol, 1 eq.) and K₂CO₃ (17.8 g, 128 mmol, 4 eq.) in DMF (20 mL) was stirred for 5 min. 2-Iodopropane (16.4 g, 9.6 mL, 96.3 mmol, 3 eq.) was added and the mixture was stirred for 64 h. The mixture was diluted with H₂O (40 mL) and extracted using EtOAc (3 x 50 mL). The combined organic phases were washed with brine (100 mL), dried over MgSO₄, and the solvent was removed under reduced pressure. The crude product was purified via automated flash column chromatography (120 g spherical SiO₂ Teledyne ISCO; 0%→40% EtOAc in cyclohexane) affording **S2** (5.713 g, 60%) as a white solid. Characterization matches literature.¹³

¹H NMR (300 MHz, CDCl₃): δ = 7.77 (d, *J* = 8.8 Hz, 4H, H-C(2)), 6.93 (d, *J* = 8.8 Hz, 4H, H-C(3)), 4.66 (sept, *J* = 6.1 Hz, 1H, CH), 1.38 ppm (d, *J* = 6.1 Hz, 12H, CH₃).

¹³C NMR (101 MHz, CDCl₃): δ = 194.6 (C(1')), 161.4 (C(4)), 132.4 (C(2)), 130.6 (C(1)), 115.0 (C(3)), 70.2 (CH), 22.1 ppm (CH₃).

Bis(4-isopropoxyphenyl)methanimine (**S3**)



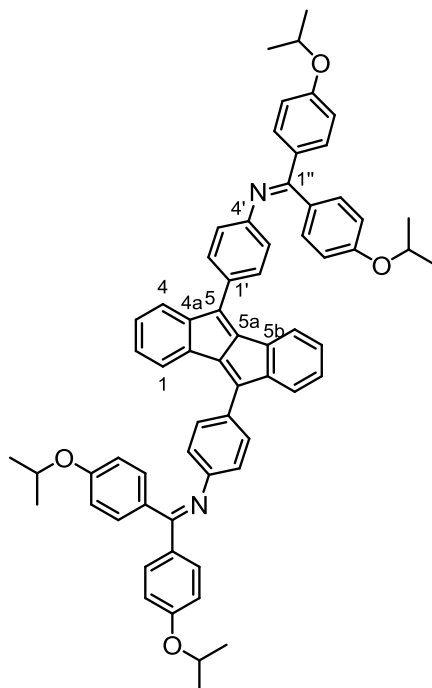
Adapted from literature,¹⁴ bis(4-isopropoxyphenyl)methanone (3.500 g, 11.73 mmol, 1 eq.), Sc(OTf)₃ (288 mg, 0.58 mmol, 5 mol%) and 1,1,1,3,3,3-hexamethylidisilazan (2.08 g, 2.69 mL, 12.90 mmol, 1.1 eq) were mixed in chlorobenzene (15 mL) and heated to 90 °C. After 40 h, the mixture was allowed to cool to room temperature. The mixture was diluted with CH₂Cl₂ (30 mL) and extracted using aq. sat. NaHCO₃ (30 mL) and brine (30 mL), dried over MgSO₄, and the solvent was removed in vacuo, affording **S3** (3.5 g, 100%) as a pale-yellow viscous oil (solid at 4°C). The splitting of signals is due to rotameric species on the NMR-timescale (not observed at ¹H NMR spectra at 300 MHz, 25 °C).

¹H NMR (500 MHz, CD₂Cl₂): δ = 7.78–7.76 (m, 1.38 H, H-C(2)), 7.57–7.55 (m, 2.58 H, H-C(2)), 6.93–6.91 (m, 4H, H-C(3)), 4.68–4.61 (m, 2H, CH), 1.39–1.37 ppm (m, 12H, CH₃);

¹³C NMR (126 MHz, CD₂Cl₂): δ = 177.9 (C(1')), 161.4 (C(4)), 160.9 (C(4)), 132.4 (C(2)), 131.3 (C(2)), 130.6 (C(1)), 129.9 (C(1)), 115.5 (C(3)), 115.0 (C(3)), 70.3 (CH), 70.2 (CH), 22.1 ppm (CH₃);

FT-IR (ATR): $\tilde{\nu}_{\text{max}}$ = 3251 (m), 3203 (m), 2975 (s), 2934 (m), 2360 (m), 1589 (s), 1502 (m), 13773 (m), 1299 (w), 1243 (s), 1104 (m), 948 (m), 893 (m), 836 (s), 752 (m), 603 cm⁻¹ (m);

***N,N'*-(Indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-bis(4-isopropoxyphenyl)methanimine) (**S4**)**



A solution of Pd(dba)₂ (199 mg, 0.347 mmol, 30 mol%) and rac. BINAP (324 mg, 0.520 mmol, 45 mol%) in toluene (23 mL) was degassed by three cycles of freeze–pump–thaw and then heated to 120 °C for 30 min. The mixture was allowed to cool to room temperature and NaOtBu (333 mg, 3.47 mmol, 3 eq.), bis(4-isopropoxyphenyl)methanimine (**S3**, 1.20 g, 4.04 mmol, 3.5 eq.), and 5,10-bis(4-bromophenyl)indeno[2,1-*a*]indene (**S1**, 592 mg, 1.16 mmol, 1 eq.) were added and the reaction was stirred at 100 °C for 3 h. The mixture was filtered through a cotton plug while hot and the filtrate was concentrated in vacuo. The residue was recrystallized from EtOH (150 mL) affording **S4** (970 mg, 89%) as a red-brown solid.

¹H NMR (500 MHz, CD₂Cl₂): δ = 7.72 (d, *J* = 8.5 Hz, 4H), 7.46 (d, *J* = 8.5 Hz, 4H), 7.09 (d, *J* = 8.6 Hz, 6H), 7.00 (d, *J* = 7.2 Hz, 2H), 6.91 (d, *J* = 8.8 Hz, 4H), 6.88 (dd, *J* = 7.4, 1.1 Hz, 2H), 6.85–6.81 (m, 6H), 6.79 (d, *J* = 8.7 Hz, 4H), 4.70–4.61 (sept, *J* = 6 Hz, 2H), 4.59–4.50 (sept, *J* = 6 Hz, 2H), 1.36 (d, *J* = 6.0 Hz, 12H), 1.30 ppm (d, *J* = 6.0 Hz, 12H);

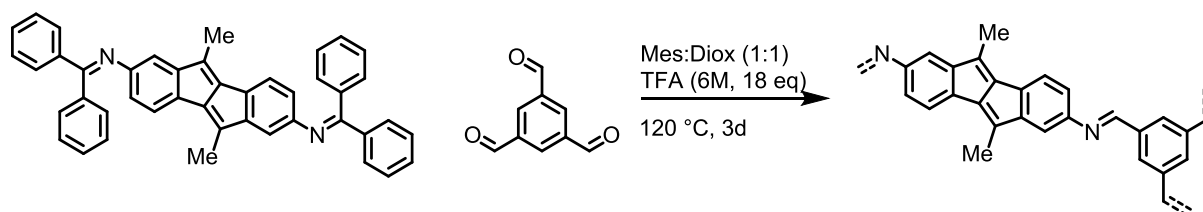
¹³C NMR (101 MHz, CDCl₃): δ = 168.2 (C_q), 160.4 (C_q), 158.4 (C_q), 152.7 (C_q), 149.7 (C_q), 142.6 (C_q), 140.5 (C_q), 135.6 (C_q), 132.4 (C_q), 131.8, 131.5, 129.1, 128.5, 128.2, 127.5, 127.4, 122.5, 121.8, 115.3, 115.0, 70.1 (CH), 70.0 (CH), 22.2 ppm (CH₃);

FT-IR (ATR) $\tilde{\nu}_{\max}$ = 3412 (w), 3059 (m), 2976 (s), 2933 (m), 2359 (m), 1584 (s), 1505 (s), 1430 (m), 1384 (m), 1303 (m), 1244 (s), 1180 (m), 1138 (m), 1105 (s), 949 (m), 835 (m), 747 (m), 695 (w), 624 cm^{-1} (w);

HR-MS (MALDI) m/z : $[M + H]^+$ Calcd for $\text{C}_{66}\text{H}_{61}\text{N}_2\text{O}_4^+$ 945.4626; Found: 945.4606.

S3. Synthesis of COFs

MeDBP-TFB COF



In an oven-dried pressure flask (10 mL), a suspension of *N,N'*-(5,10-dimethylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) (**3**, 84.2 mg, 0.143 mmol, 1 eq.) and benzene-1,3,5-tricarbaldehyde (15.5 mg, 0.0958 mmol, 0.67 eq.) in mesitylene (500 μL) and dioxane (500 μL) was heated to 70 °C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and aq. trifluoroacetic acid (6 M, 429 μL , 18 eq.) was added. The reaction mixture was degassed by three cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 °C for 3 d without stirring. The mixture was allowed to cool to room temperature and the solid was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then with MeOH, while making sure that the filter cake would never run dry. The solid was filled into a tea bag, sealed with a staple, and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO_2 yielding **MeDBP-TFB COF** (42 mg, 88%) as a dark-brown solid. Reproducibility was confirmed in 4 repetitions, yielding the same BET surface area within < 10% deviation from the average.

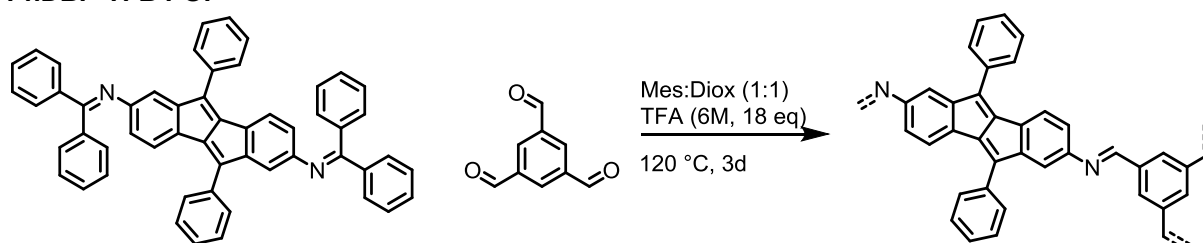
Screening of Conditions

For improving the COF quality, we screened solvent mixture, concentration, type of acid, acid concentration and reaction time. The BET surface area was chosen as a prime indicator for the quality of the COF. Crystallinity was confirmed by PXRD afterwards.

Table S1 Condition screening for **MeDBP-TFB** COF

Exp. No	Solvent, ratio	Acid	Conc. [mmol L ⁻¹]	SA _{BET} [m ² g ⁻¹]	Crystallinity
1	Mes:Diox 1:1	Acetic acid 6 M, 6 eq	153	256	yes
2	Mes:Diox 1:4	Acetic acid 6 M, 6 eq	153	61	yes
3	Mes:Diox 4:1	Acetic acid 6 M, 6 eq	153	3	yes
4	Mes:BnOH 9:1	Acetic acid 6 M, 6 eq	153	56	yes
5	Mes:EtOH 9:1	Acetic acid 6 M, 6 eq	153	215	yes
6	Mes:Diox 1:1	Acetic acid 19 M, 37 eq	143	469	no
7*	Mes:Diox 1:1	Acetic acid 19 M, 37 eq	143	482	no
8	Mes:Diox 1:1	Acetic acid 19 M, 37 eq	71.5	197	low
9	Mes:Diox 1:1	TFA, 6 M, 18 eq	143	734	yes
10	Mes:Diox 1:1	Trifluoromethanesulfonic acid, 6 eq	143	No material	-

* 6 d reaction time.

PhDBP-TFB POP

In an oven-dried pressure flask, a suspension of *N,N'*-(5,10-diphenylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) (**5**, 102 mg, 0.143 mmol, 1 eq.) and benzene-1,3,5-tricarbaldehyde (15.5 mg, 0.0958 mmol, 0.67 eq.) in mesitylene (500 μ L) and dioxane (500 μ L) was heated to 70 °C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and aq. trifluoroacetic acid (6 M, 429 μ L, 18 eq.) was added. The reaction mixture was degassed with 3 cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 °C for 3 d without stirring. The mixture was allowed to cool to room temperature and the solid was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then with MeOH, while making sure that the filter cake would never run dry. The solid was filled into a tea bag, sealed with a staple and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO₂ yielding **PhDBP-TFB** POP (60.1 mg, 92%) as a dark-brown solid. Reproducibility was confirmed in 4 repetitions, yielding the same BET surface area within < 5% deviation from the average.

Screening of Conditions

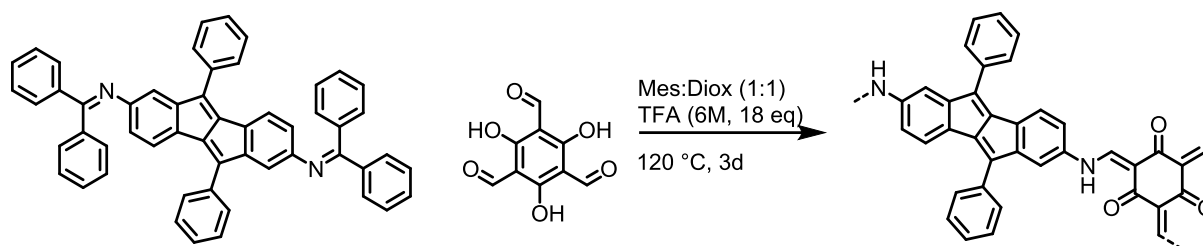
For improving the COF quality, we screened solvent mixture, type of acid, and acid concentration. Initially, the BET surface area was chosen as a prime indicator for the quality of the materials. In a further screening crystallinity was examined. None of the obtained materials showed crystallinity hence we concluded that a POP but not a COF can be obtained from these monomers.

Table S2 Condition screening for **PhDBP-TFB POP**

Exp. No	Solvent, ratio	Acid	Conc. [mmol L⁻¹]	SA_{BET} [m² g⁻¹]	Crystallinity
1	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.143	172	no
2	Mes:Diox 1:1	Acetic acid, 10 M, 36 eq	0.143	436	no
3	Mes:Diox 1:1	TFA, 6 M, 18 eq	0.143	669	no
4	Mes:Diox 4:1	TFA, 6 M, 18 eq	0.143	540	no
5	Mes:Diox 1:4	TFA, 6 M, 18 eq	0.143	588	no
6	Mes:Diox 1:1	TFA, 28 eq	0.143	538	no
7	Mes:Diox 1:1	TFA, 6 M, 18 eq	0.143	n.d.	no
8	Mes:BnOH 1:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
9	Mes:BnOH 4:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
10	Mes:BnOH 1:4	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
11	DCB:BnOH 1:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
12	DCB:BnOH 4:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
13	DCB:BnOH 1:4	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
14	DCB:Diox 1:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
15	DCB:Diox 4:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
16	DCB:Diox 1:4	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
17	nBuOH:DCB 1:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
18	Mes:MeOH 1:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
19	DCB:DMF 1:1	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
20	THF	Acetic acid, 10 M, 36 eq	0.143	n.d.	no
21	DMF	Acetic acid, 10 M, 36 eq	0.143	n.d.	no

* 6 d reaction time

PhDBP-TFP COF



In an oven-dried pressure flask, a suspension of *N,N'*-(5,10-diphenylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) (**5**, 102 mg, 0.143 mmol, 1 eq.) and 2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde (20.1 mg, 0.0958 mmol, 0.67 eq.) in mesitylene (500 μ L) and dioxane (500 μ L) was heated to 70 $^{\circ}$ C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and aq. trifluoroacetic acid (8 M, 500 μ L, 28 eq.) was added. The reaction mixture was degassed with 3 cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 $^{\circ}$ C for 3 d without stirring. The mixture was allowed to cool to room temperature and the solid was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then MeOH, while making sure that the filter cake would never run dry. The solid was filled into a tea bag, sealed with a staple, and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO₂ yielding **PhDBP-TFP COF** (54.3 mg, 77%) as a dark-brown solid. Reproducibility was confirmed in 4 repetitions, yielding the same BET surface area within < 10% deviation from the average.

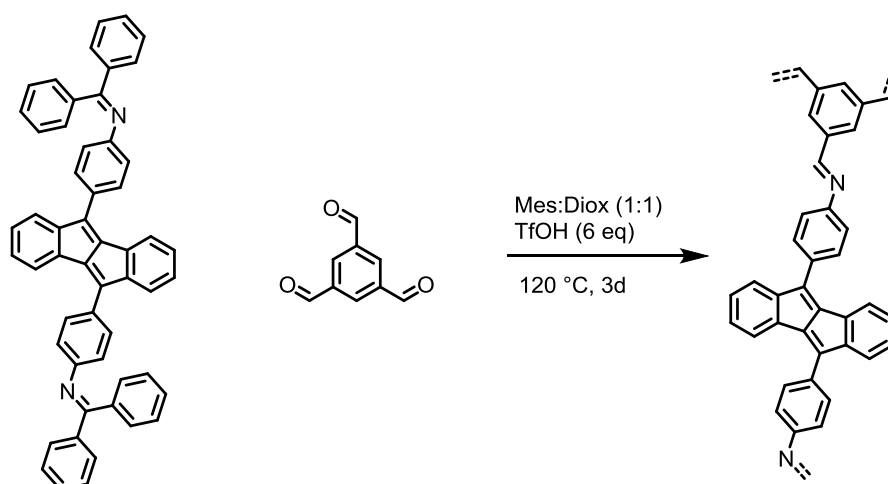
Screening of Conditions

For improving the COF quality, we screened solvent mixture, type of acid, and acid concentration. The BET surface area was chosen as a prime indicator for the quality of the COF. Crystallinity was confirmed by PXRD afterwards.

Table S3 Condition Screening for **PhDBP-TFP COF**

Exp. No	Solvent, ratio	Acid	Conc. [mmol L ⁻¹]	SA _{BET} [m ² g ⁻¹]	Crystallinity
1	Mes:Diox 1:1	TFA 6 M, 18 eq	0.143	926	no
2	Mes:Diox 1:1	Acetic acid, 10 M, 36 eq	0.143	82	no
3	Mes:Diox 1:1	TFA 8 M, 28 eq	0.143	1022	yes
4	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.143	135	no

DBP-Ph-TFB COF



In an oven-dried pressure flask, a suspension of *N,N'*-(indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-diphenylmethanimine) (**7**, 79 mg, 0.111 mmol, 1 eq.) and benzene-1,3,5-tricarbaldehyde (15.5 mg, 0.0958 mmol, 0.67 eq.) in mesitylene (500 μ L) and dioxane (500 μ L) was heated to 70 $^{\circ}$ C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and conc. trifluoromethanesulfonic acid (76 μ L, 6 eq.) was added. The reaction mixture was degassed with 3 cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 $^{\circ}$ C for 3 d without stirring. The mixture was allowed to cool to room temperature and the material was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then MeOH, while making sure that the filter cake would never run dry. The solid was filled into tea bag, sealed with a staple, and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO₂ yielding **DBP-Ph-TFB COF** (46.2 mg, 71%) as a dark brown solid.

Screening of Conditions

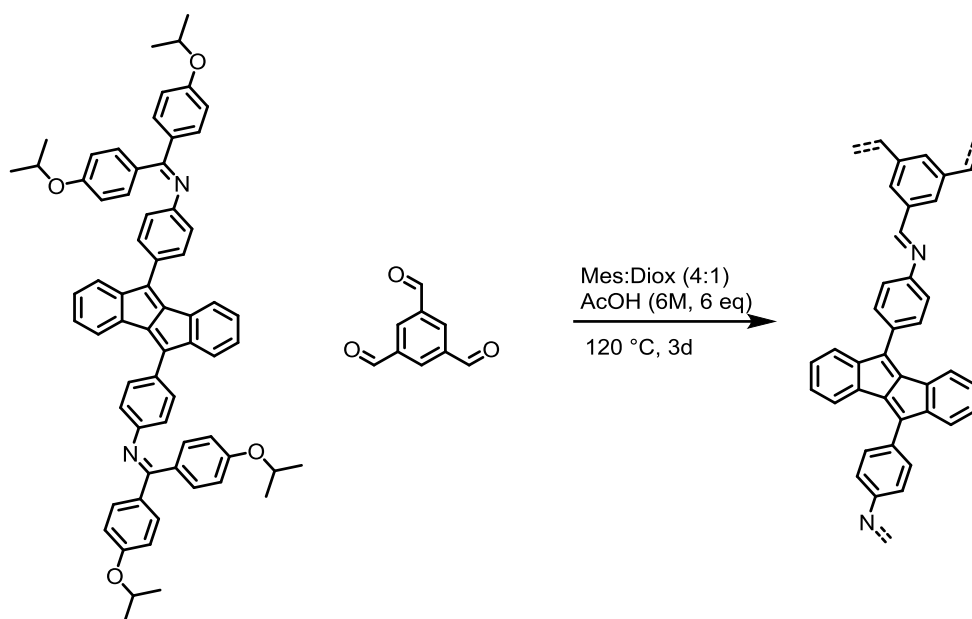
For improving the COF quality, we screened solvent mixture, type of acid, and acid concentration. The BET surface area was chosen as a prime indicator for the quality of the COF. Crystallinity was confirmed by PXRD afterwards.

Table S4 Condition Screening for **DBP-Ph-TFB COF**

Exp. No	Solvent, ratio	Acid	Conc. [mmol L ⁻¹]	S _{BET} [m ² g ⁻¹]	Crystallinity
1	Mes:Diox 1:1	Acetic acid, 6 M, 36 eq	0.143	225	yes
2	Mes:Diox 1:1	TFA 6 M, 130 eq	0.143	92	no
3	Mes:Diox 1:1	Acetic acid, 6 M, 130 eq	0.143	no product	-
4	Mes:Diox 1:1	TFA, 6 M, 18 eq	0.143	30	no
5	Mes:Diox 1:1	Acetic acid, 6 M, 37 eq	0.111	no product	-
6	Mes:Diox 1:1	Trifluoromethanesulfonic acid, conc., 6 eq	0.143	1691	yes

In an alternative approach, we synthesized *para*-(*i*PrO)Ph-substituted benzophenoneimines. Imines **S8** could be obtained straightforward from the corresponding bromide **S1** in a Buchwald–Hartwig coupling and was easier to synthesize than **7** which is not accessible via Buchwald–Hartwig coupling (see main text). As far as we know, a COF synthesis using a *para*-substituted benzophenone imine is not known to literature. The resulting COFs show excellent crystallinity, however a diminished porosity of 146 m² g⁻¹.

Synthetic Procedure



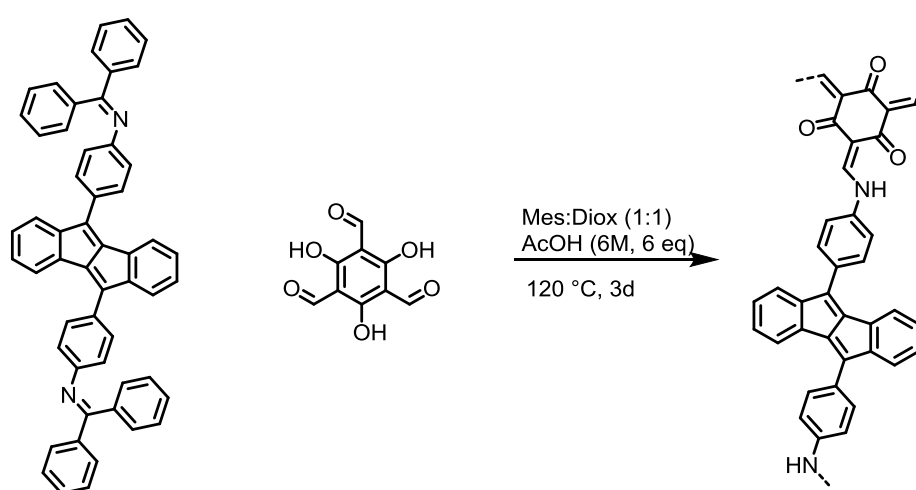
In an oven-dried pressure flask, a suspension of *N,N'*-(indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-bis(4-isopropoxyphenyl)methanimine) **S4** (135 mg, 0.143 mmol, 1 eq.) and benzene-1,3,5-tricarbaldehyde (15.5 mg, 0.0958 mmol, 0.67 eq.) in mesitylene (800 μ L) and dioxane (200 μ L) were added and the mixture was heated to 70 $^{\circ}$ C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and conc. acetic acid (300 μ L, 36 eq.) and water (200 μ L, 78 eq.) were added. The reaction mixture was degassed with 3 cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 $^{\circ}$ C for 3 d without stirring. The mixture was allowed to cool to room temperature and the solid was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then MeOH, while making sure that the filter cake would never run dry. The solid was filled into a tea bag, sealed with a staple, and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO₂ yielding **DBP-Ph-TFB** COF (45.2 mg, 69%) as a dark brown solid.

Screening of Conditions

For improving the COF quality, we screened solvent mixture, concentration, type of acid, and acid concentration. The BET surface area was chosen as a prime indicator for the quality of the COF. Crystallinity was confirmed by PXRD afterwards.

Table S5 Condition Screening for **DBP-Ph-TFB COF** using solubilized monomer **S4**

Exp. No	Solvent, ratio	Acid	Conc. [mmol L ⁻¹]	SA _{BET} [m ² g ⁻¹]	Crystallinity
1	Mes:Diox 1:1	TFA 6 M, 18 eq	0.143	31	no
2	Mes:Diox 1:1	Acetic acid, 10 M, 36 eq	0.0715	82	no
3	Mes:Diox 4:1	Acetic acid, 10 M, 36 eq	0.0715	146	yes
4	Mes:Diox 1:4	Acetic acid, 10 M, 36 eq	0.0715	15	no
5	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.127	120	yes
6	Mes:Diox 4:1	Acetic acid, 6 M, 6 eq	0.127	24	yes
7	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.153	134	yes
8	Mes:Diox 4:1	Acetic acid, 6 M, 6 eq	0.153	99	yes

DBP-Ph-TFP POP

In an oven-dried pressure flask, a suspension of *N,N'*-(indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-diphenylmethanimine) (**7**, 79 mg, 0.111 mmol, 1 eq.) and 2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde (15.6 mg, 0.0742 mmol, 0.67 eq.) in mesitylene (500 μ L) and dioxane (500 μ L) was heated to 70 °C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and aq. trifluoroacetic acid (6 M, 333 μ L, 18 eq.) was added. The reaction mixture was degassed with 3 cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 °C for 3 d without stirring. The mixture was allowed to cool to room temperature and the solid was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then MeOH, while making sure that the filter cake would never run dry. The solid was filled into a tea bag, sealed with a staple, and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO₂ yielding **DBP-Ph-TFP POP** (53.1 mg, 76%) as a bright red solid. Reproducibility was confirmed in 3 repetitions, yielding the same BET surface area within < 10% deviation from the average.

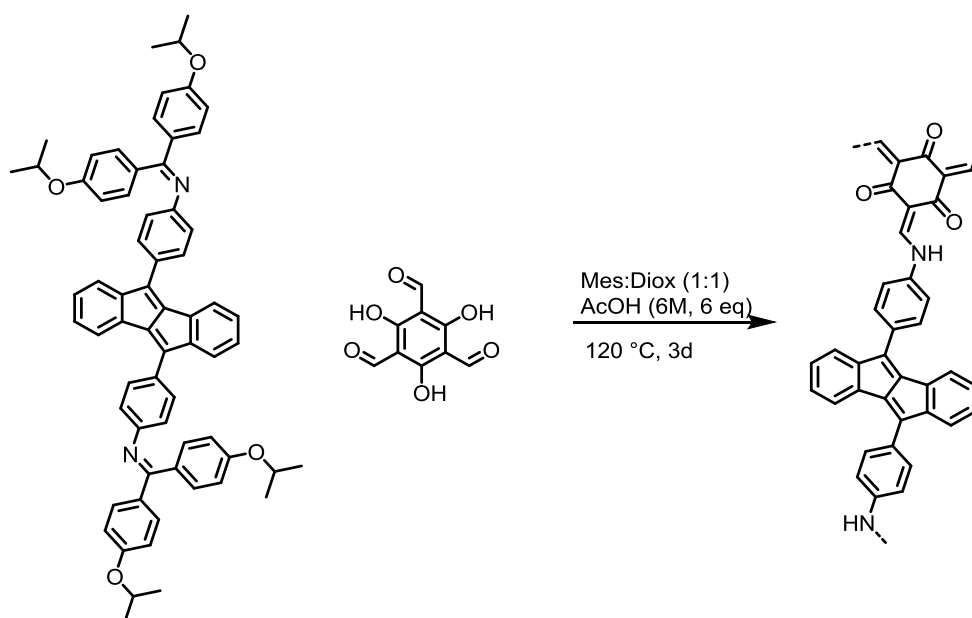
Screening of Conditions

We screened solvent mixture, concentration, type of acid, and acid concentration. The BET surface area was chosen as a prime indicator for the quality of the material.

Table S6 Condition screening for **DBP-Ph-TFP** POP

Exp. No	Solvent, ratio	Acid	Conc. [mmol L ⁻¹]	SA _{BET} [m ² g ⁻¹]	Crystallinity
1	Mes:Diox 1:1	Acetic acid, 10 M, 36 eq	0.111	512	no
2	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.111	497	no
3	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.111	89	no
4	Mes:Diox 1:1	TFA, 6M, 18 eq	0.143	871	no
5	Mes:Diox 1:1	Trifluoromethanesulfonic acid, 6 eq	0.143	no product	-

Synthesis via solubilized monomer



In an oven-dried pressure flask, a suspension of *N,N'*-(indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-bis(4-isopropoxyphenyl)methanimine) (**S4**, 135 mg, 0.143 mmol, 1 eq.) and 2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde (20.1 mg, 0.0958 mmol, 0.67 eq.) in mesitylene (500 μ L) and dioxane (500 μ L) was heated to 70 $^{\circ}$ C for 20 min. After cooling to room temperature, the mixture was sonicated for 2 min and aq. acetic acid (6 M, 143 μ L, 6 eq.) was added. The reaction mixture was degassed with 3 cycles of freeze–pump–thaw and backfilled with Ar. The tube was covered in aluminum foil and heated at 120 $^{\circ}$ C for 3 d without stirring. The mixture was allowed to cool to room temperature and the material was transferred to a PTFE membrane filter on a glass-sintered vacuum filter funnel. The filter cake was washed with copious amounts of acetone, then MeOH, while making sure that the filter cake would never run dry. The solid was filled into a tea bag, sealed with a staple, and submerged in MeOH for 24 h. The material in the tea bag was dried using supercritical CO₂ yielding **DBP-Ph-TFP** POP (59.9 mg, 85%) as a bright red solid.

Screening of Conditions

We screened the type of acid and acid concentrations. The BET surface area was chosen as a prime indicator for the quality of the material.

Table S7 Condition screening for **DBP-Ph-TFP** POP via solubilized monomer **S4**

Exp. No	Solvent, ratio	Acid	Conc. [mmol L ⁻¹]	SA _{BET} [m ² g ⁻¹]	Crystallinity
1	Mes:Diox 1:1	Acetic acid, 10 M, 36 eq	0.143	52	no
2	Mes:Diox 1:1	TFA 6 M, 18 eq	0.143	337	no
3	Mes:Diox 1:1	Acetic acid, 6 M, 6 eq	0.143	630	no

Comment to Crystallinity of materials

Of the five materials we produced, three were optimized to highly porous, highly crystalline COFs and for two materials crystallinity could not be achieved despite high porosity. From this systematic series of materials, no direct trend is deductible. It cannot be concluded that TFP or TFB is an inherently superior node molecule for COF synthesis. The system seems to be multifaceted and not every combination of linker and node easily leads to a crystalline material, at least not in a reasonably large screening of conditions. This is especially difficult if the monomers are not readily available but have to be synthesized in multistep procedures. In this work, advanced six step synthesis is necessary to synthesize the COF building blocks.

S4. Gas Adsorption and BET Surface Area Analysis

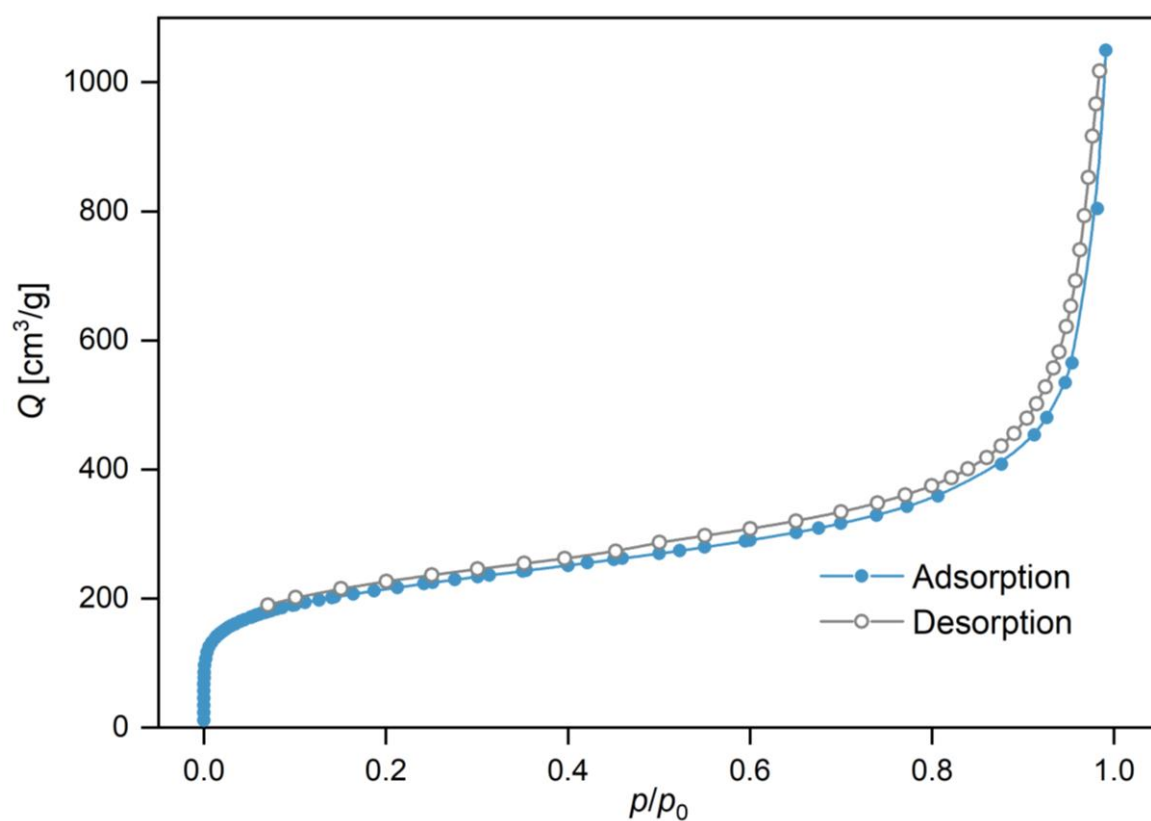


Figure S1 Adsorption-Desorption isotherm of **MeDBP-TFB** COF

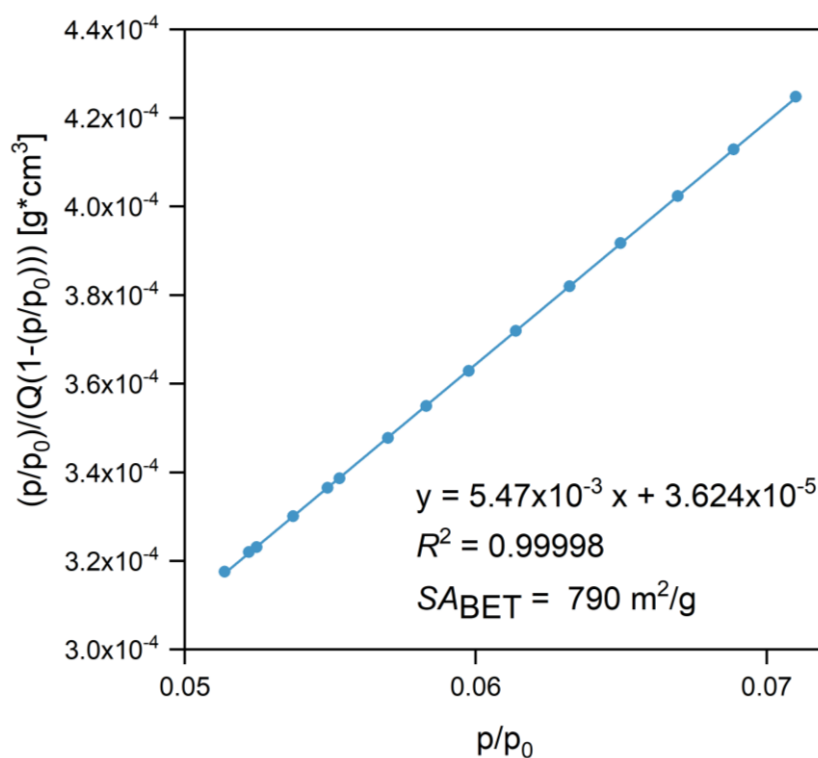


Figure S2 BET plot of **MeDBP-TFB** COF

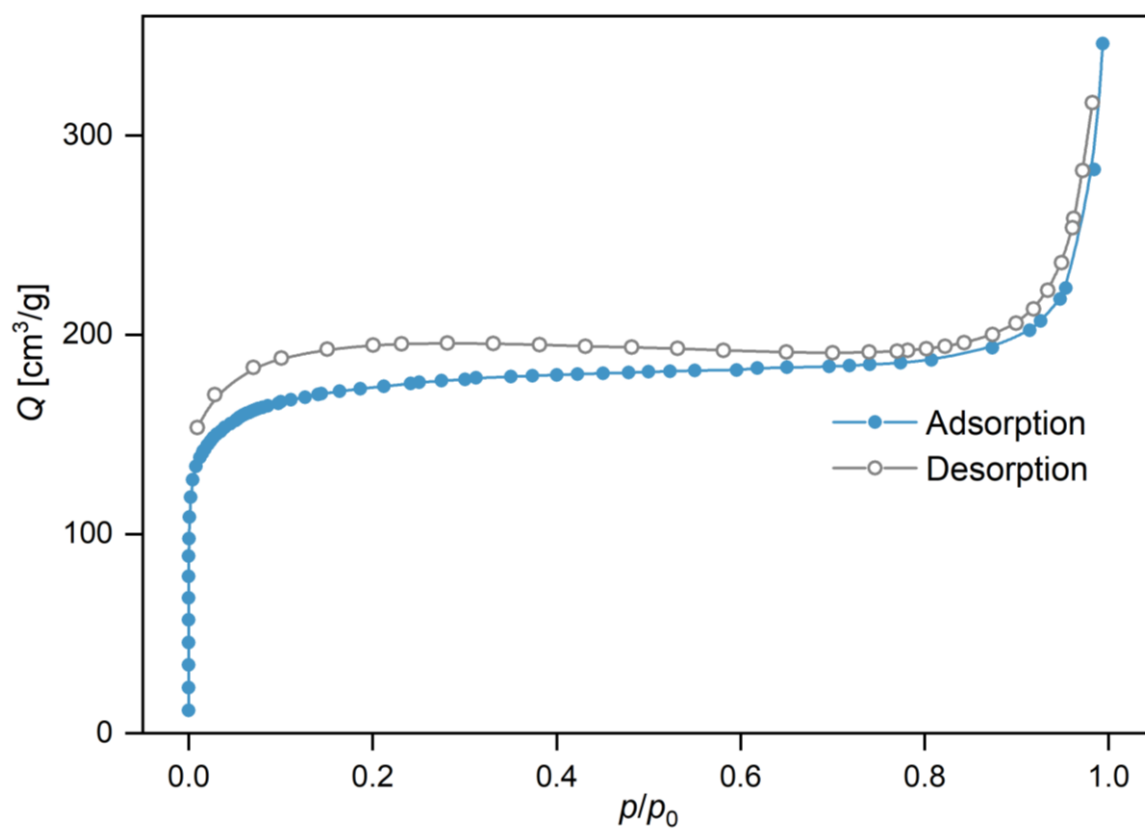


Figure S3 Adsorption-Desorption isotherm of **PhDBP-TFB** POP

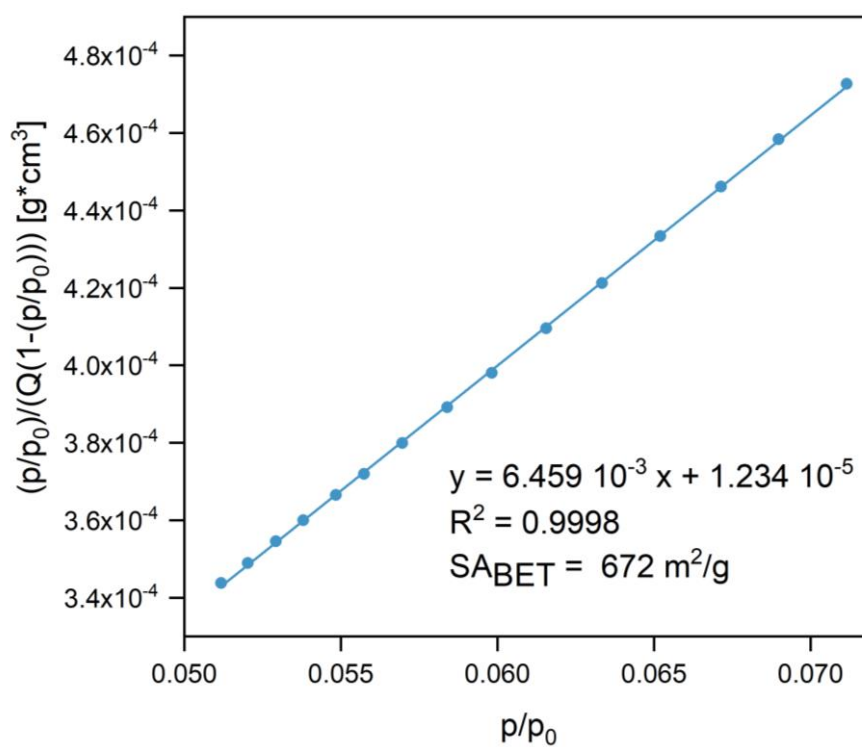


Figure S4 BET plot of **PhDBP-TFB** POP

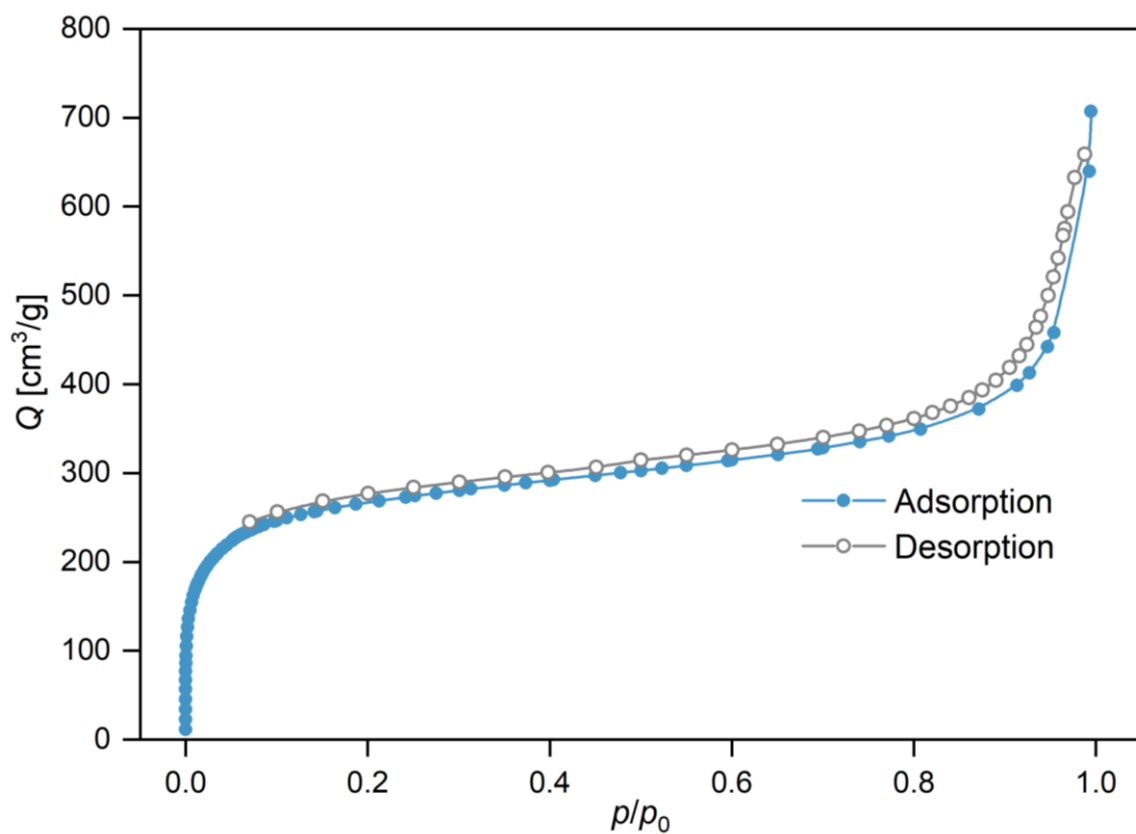


Figure S5 Adsorption-Desorption isotherm of **PhDBP-TFP** COF

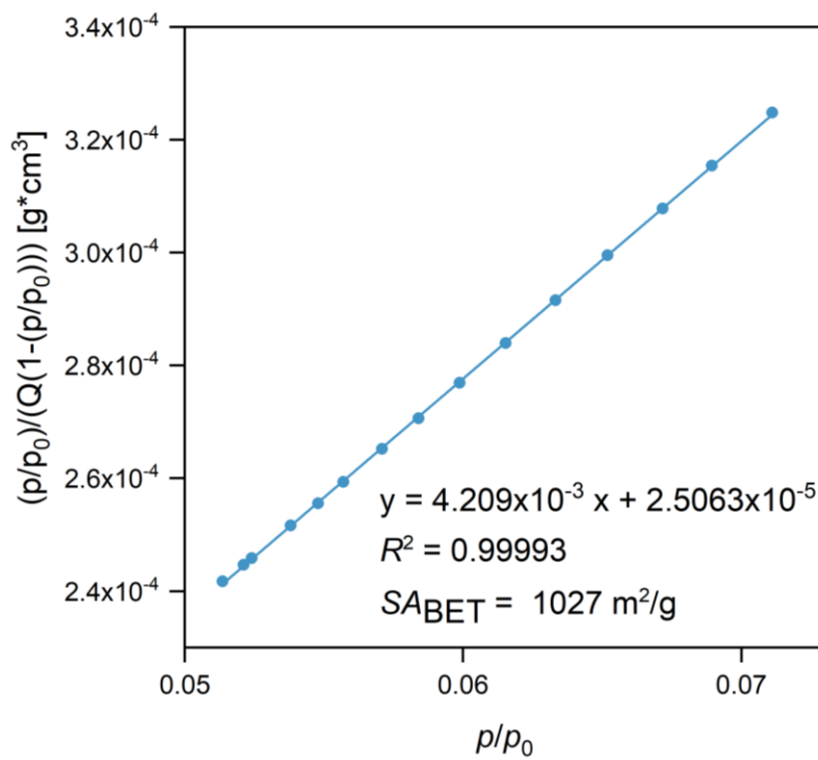


Figure S6 BET plot of **PhDBP-TFP** COF

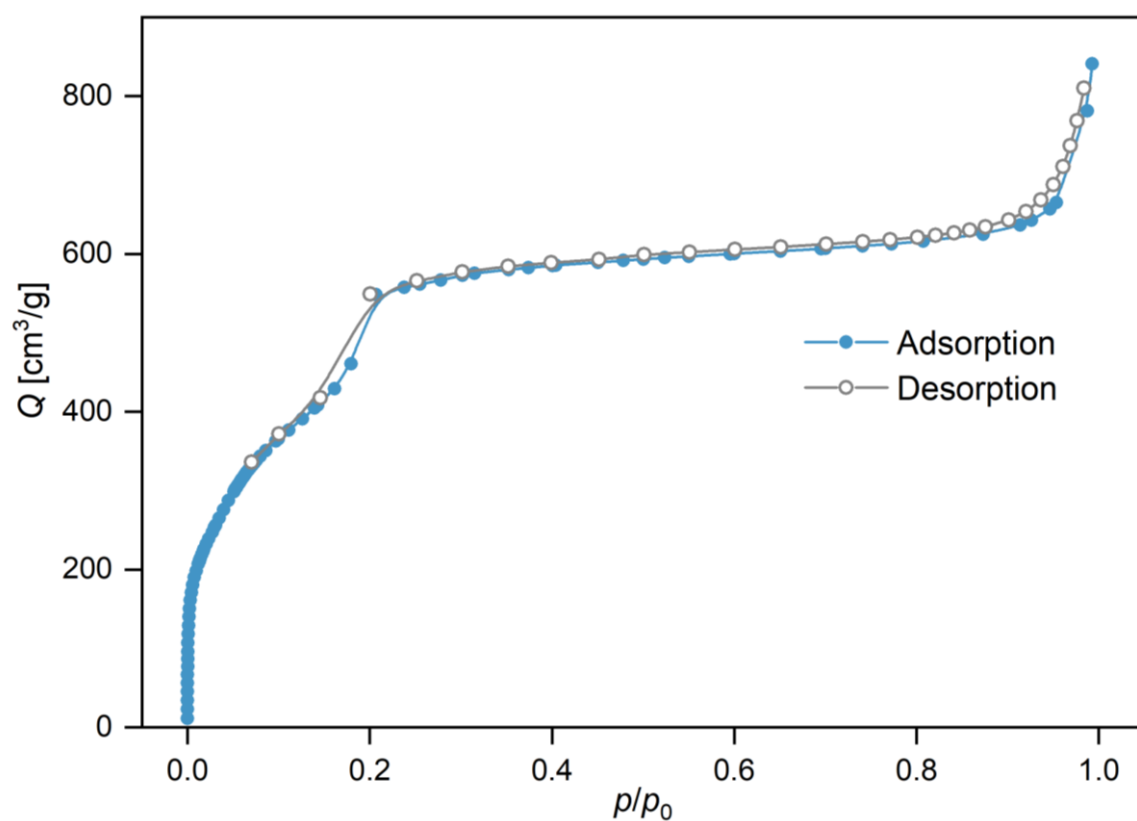


Figure S7 Adsorption-Desorption isotherm of **DBP-Ph-TFB** COF

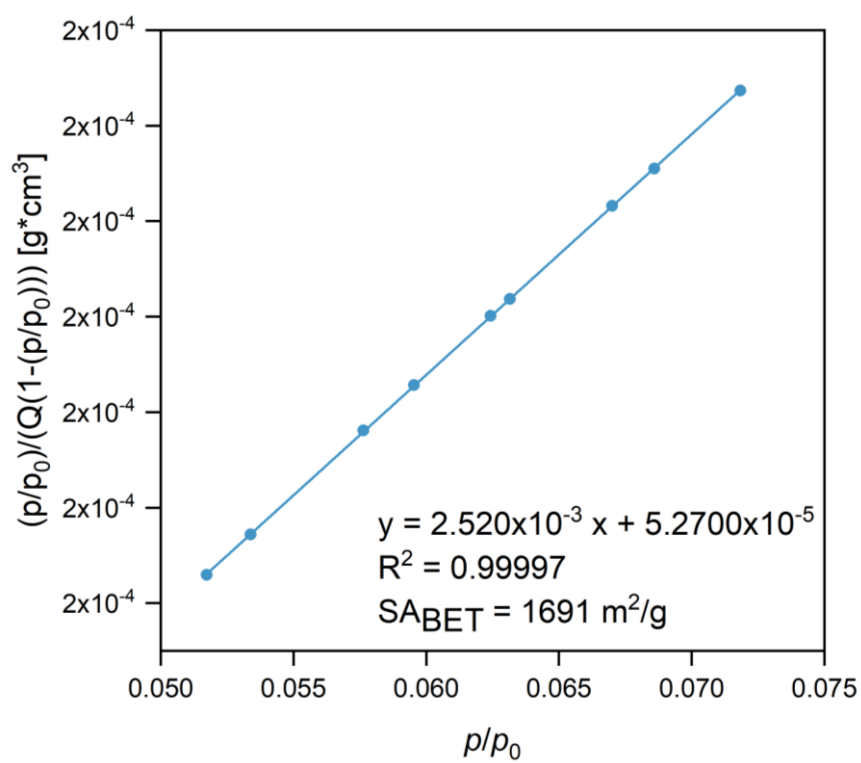


Figure S8 BET plot of **DBP-Ph-TFB** COF

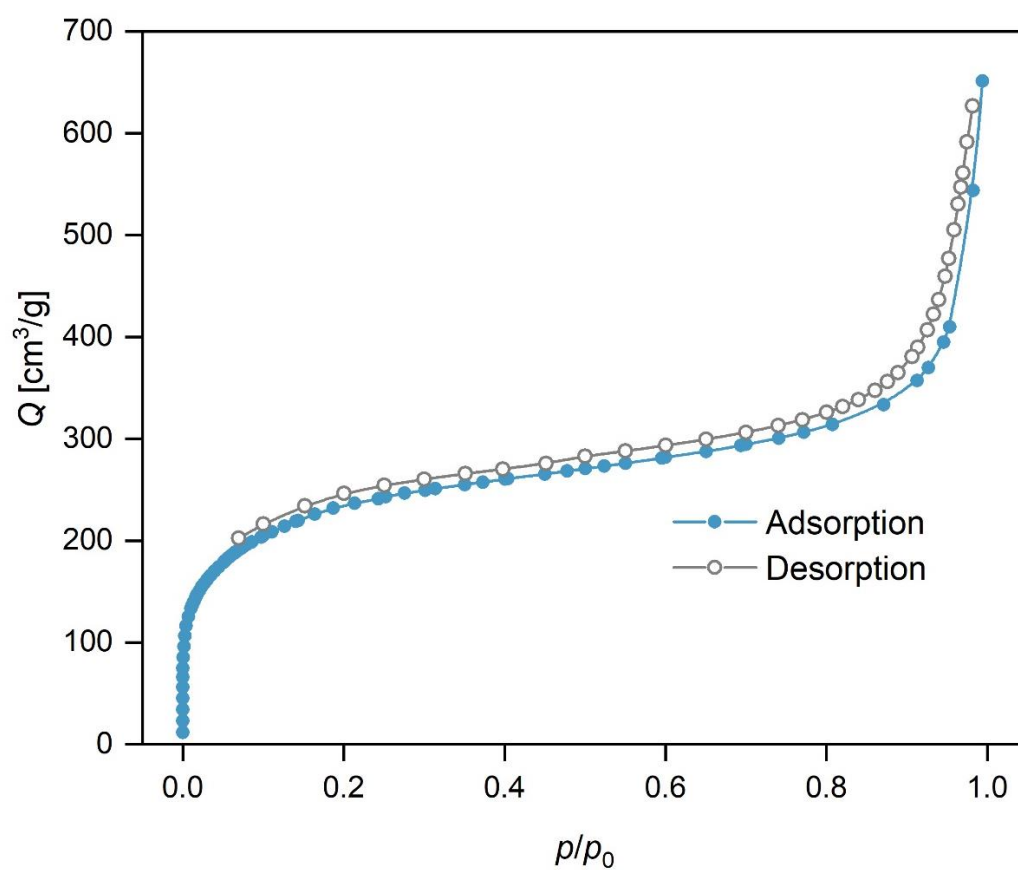


Figure S9 Adsorption-Desorption isotherm of **DBP-Ph-TFP** POP

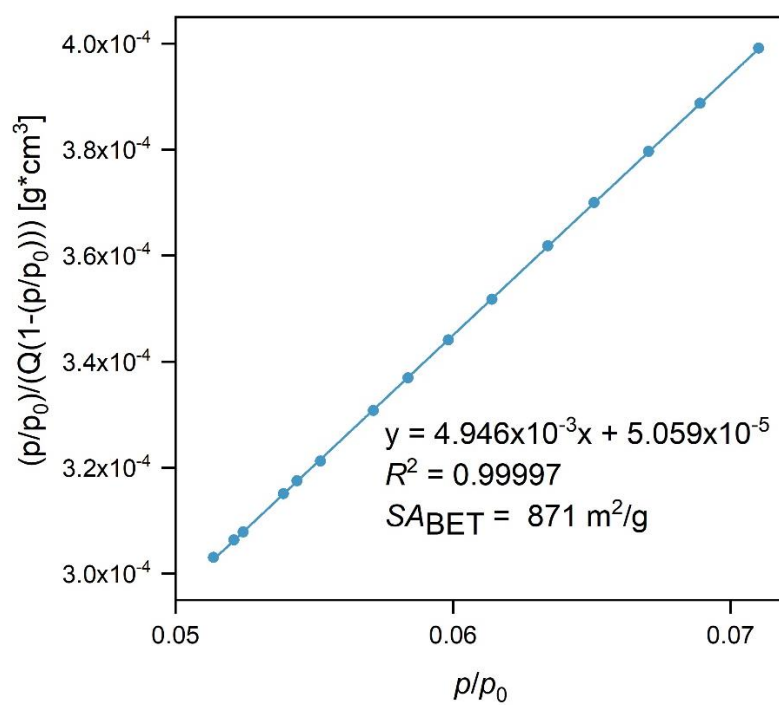


Figure S10 BET plot of **DBP-Ph-TFP** POP

S5. Pore Size Distribution

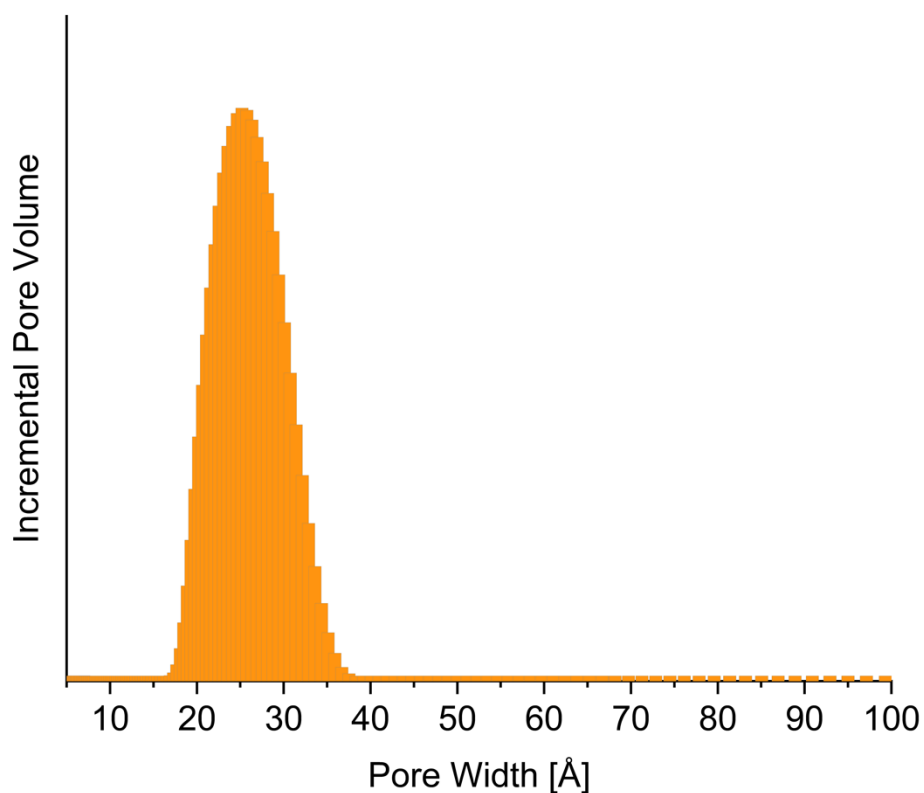


Figure S11 Pore-size distribution of **MeDBP-TFB** COF; kernel: Carbon N₂, 77 K, multiwall sphere

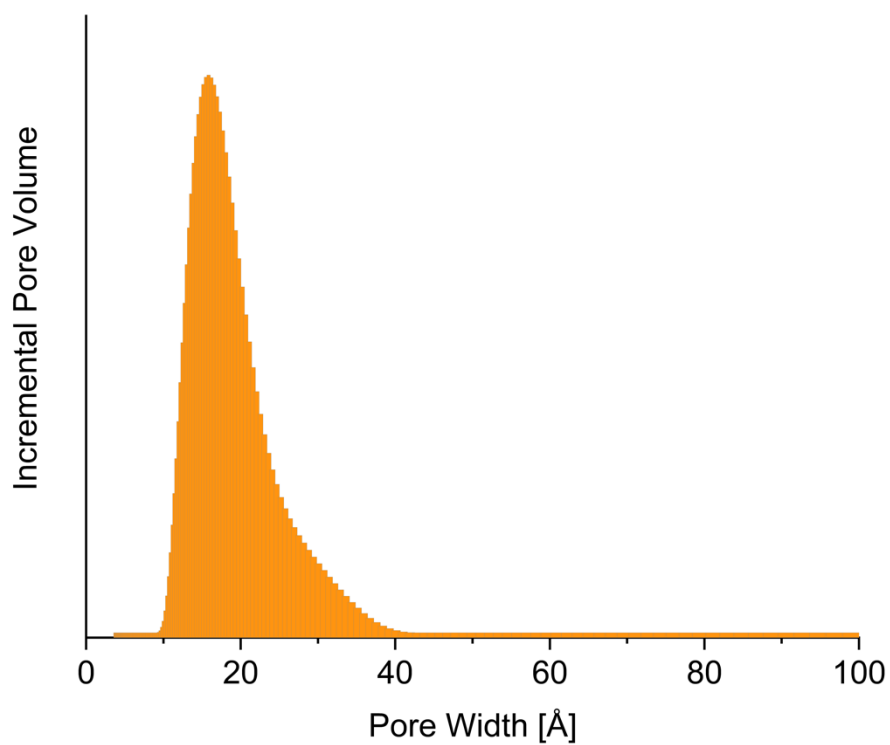


Figure S12 Pore-size distribution of **PhDBP-TFB** POP; kernel: Carbon N₂, 77 K, cylindrical model

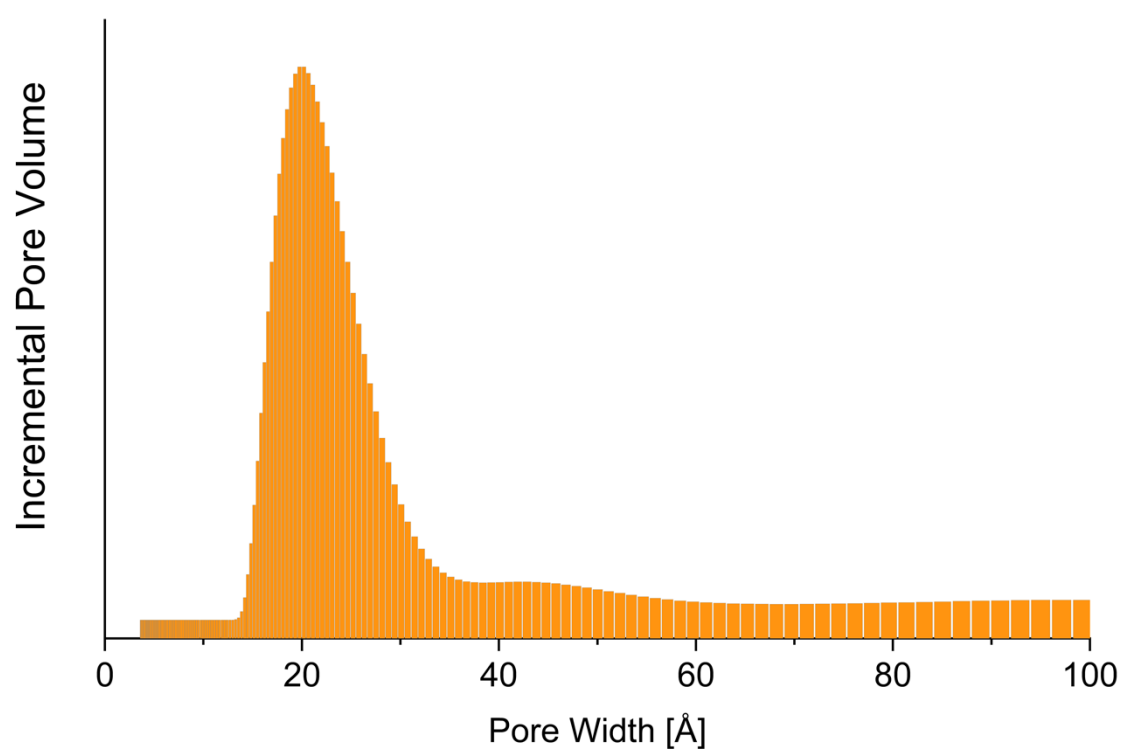


Figure S13 Pore-size distribution of **PhDBP-TFP** COF; kernel: carbon, N₂, 77 K, cylindrical model

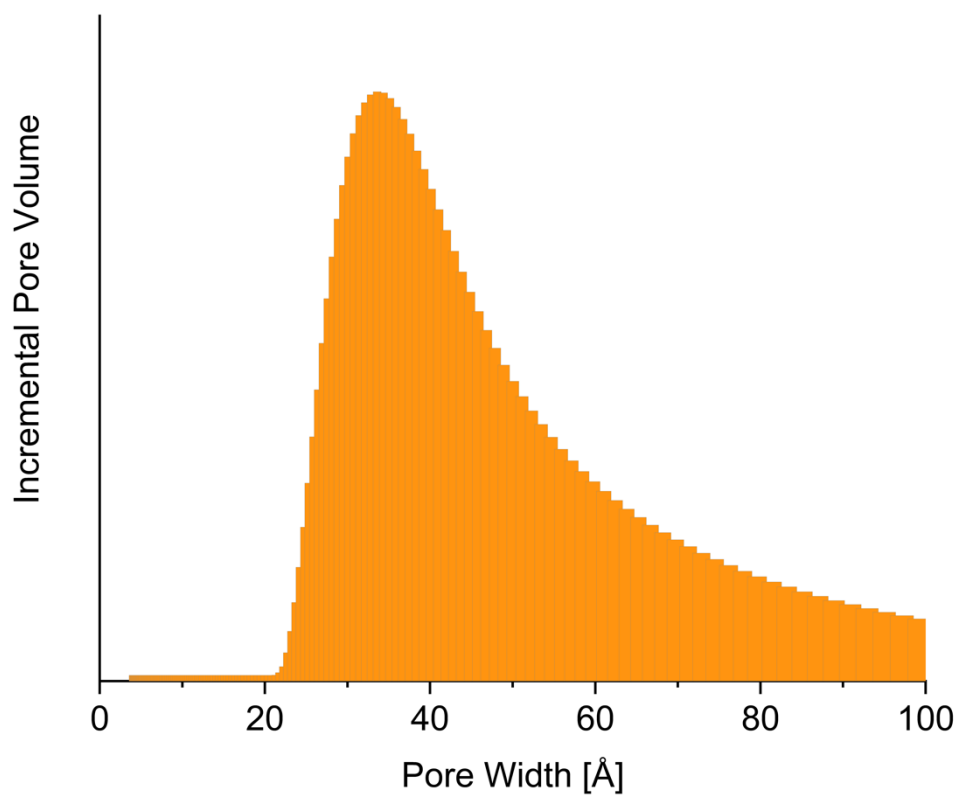


Figure S14 Pore-size distribution of **DBP-Ph-TFB** COF; kernel: carbon, N₂, 77 K, heterogeneous surface

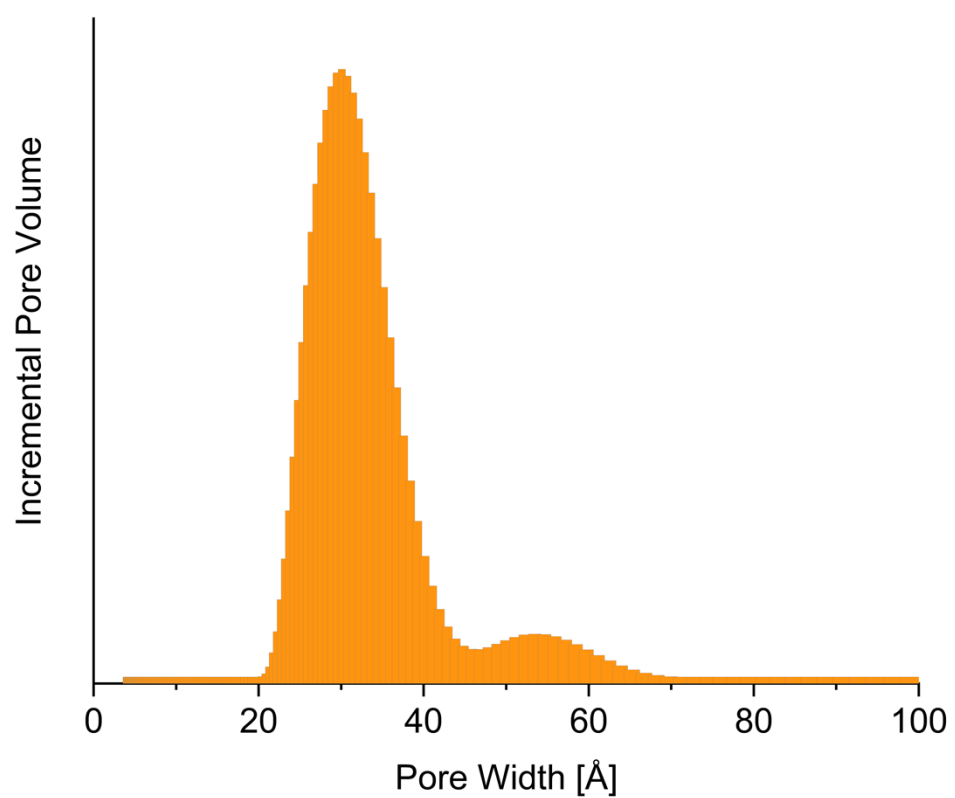


Figure S15 Pore-size distribution of **DBP-Ph-TFP** POP; kernel: carbon, N₂, 77 K, multi-wall sphere

S6. PXRD and Structure Modelling of COFs

MeDBP-TFB COF

Unit cell parameters:

Lattice type: 3D hexagonal

Space group: $P6/m$

Lengths: $a = b = 32.47 \text{ \AA}$, $c = 3.84 \text{ \AA}$

Angles: $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

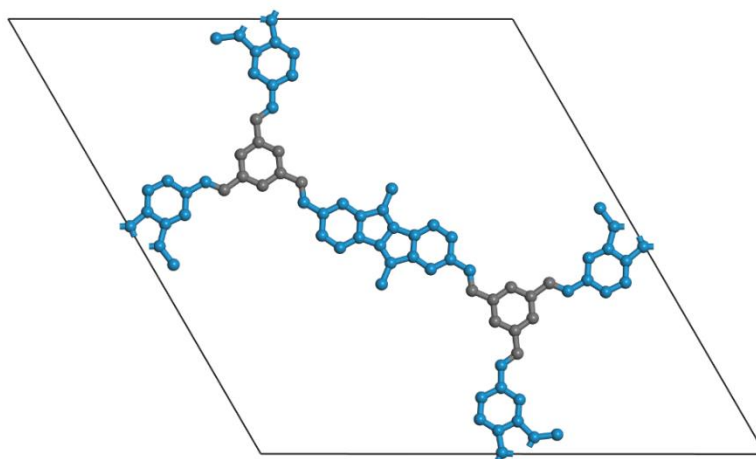


Figure S16 Unit cell of **MeDBP-TFB** COF optimized in Materials Studio; Hydrogens are omitted for clarity

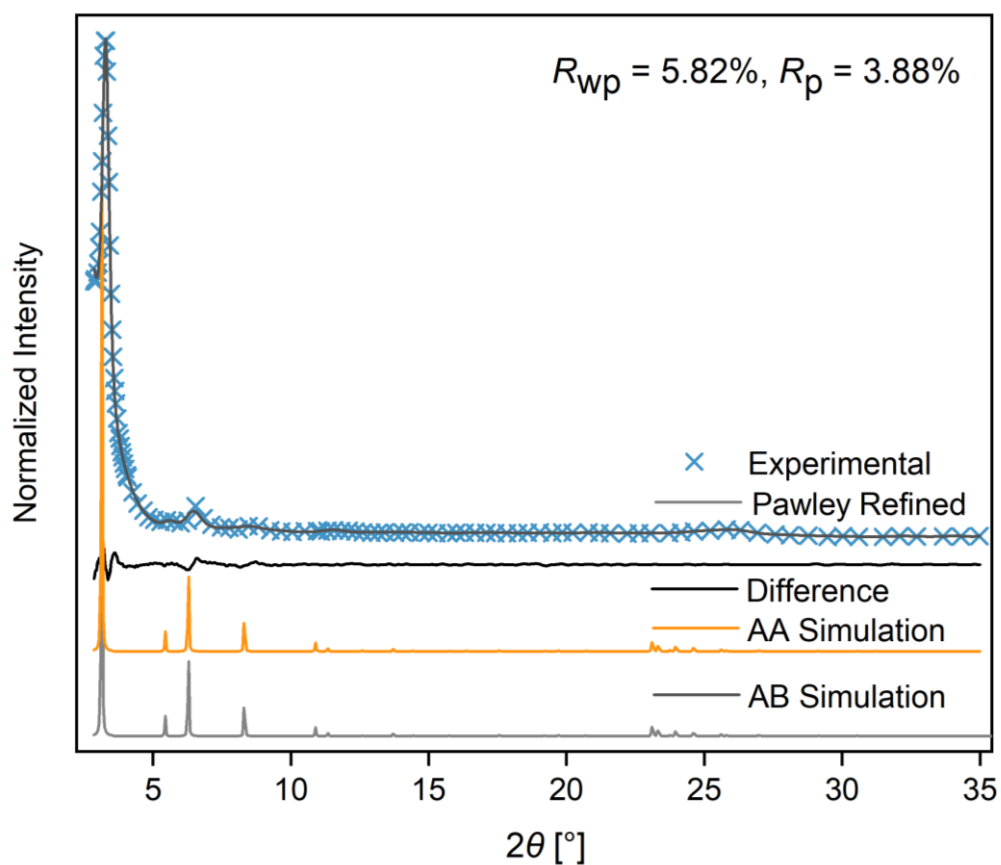


Figure S17 Experimental and calculated PXRD data of **MeDBP-TFB** COF

PhDBP-TFB POP

Unit cell parameters:

Lattice type: 3D hexagonal

Space group: $P6/m$

Lengths: $a = b = 32.90 \text{ \AA}$, $c = 3.44 \text{ \AA}$

Angles: $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

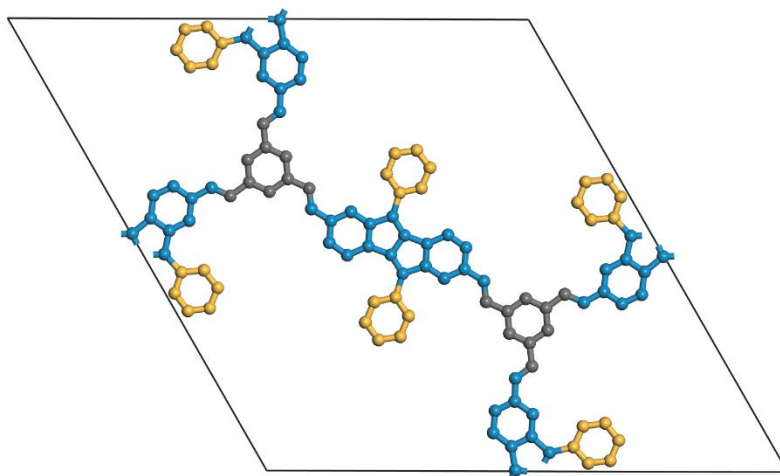


Figure S18 Unit cell of hypothetical crystalline **PhDBP-TFB** POP optimized in Materials Studio; Hydrogens are omitted for clarity

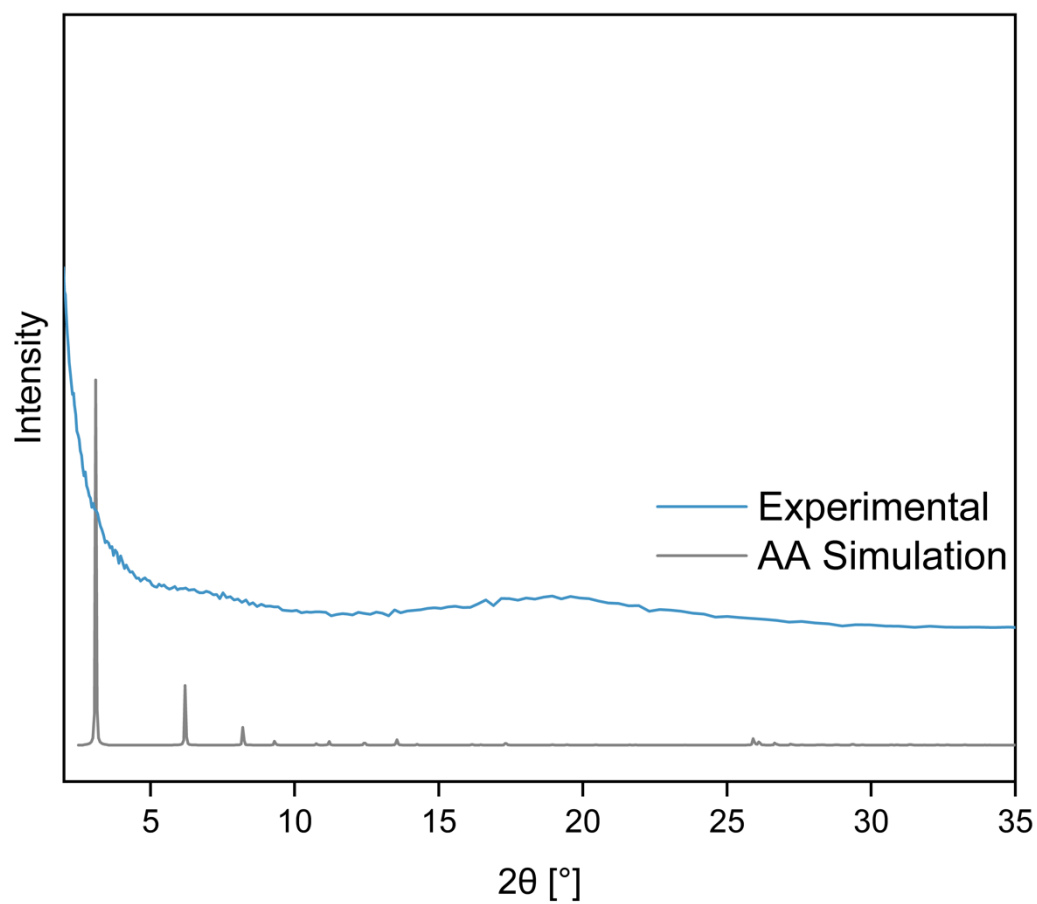


Figure S19 Experimental and calculated PXRD data of **PhDBP-TFB** POP

PhDBP-TFP COF

Unit cell parameters:

Lattice type: 3D hexagonal

Space group: $P6/m$

Lengths: $a = b = 32.90 \text{ \AA}$, $c = 3.44 \text{ \AA}$

Angles: $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

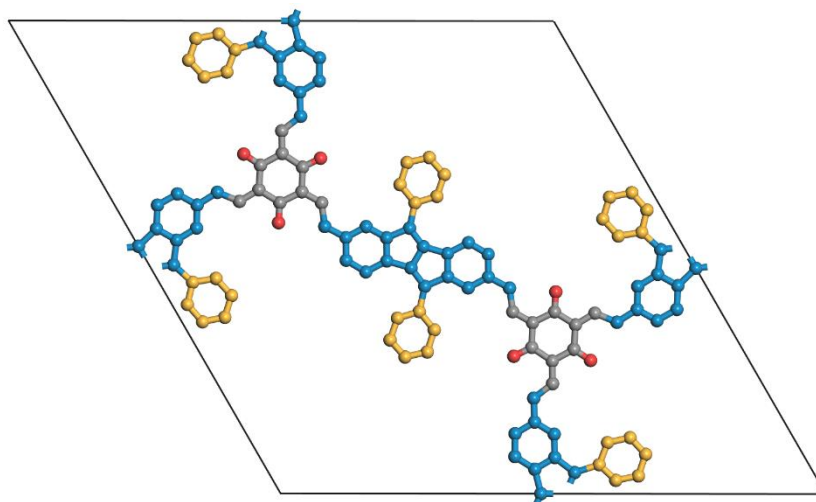


Figure S20 Unit cell of **PhDBP-TFP** COF optimized in Materials Studio; Hydrogens are omitted for clarity

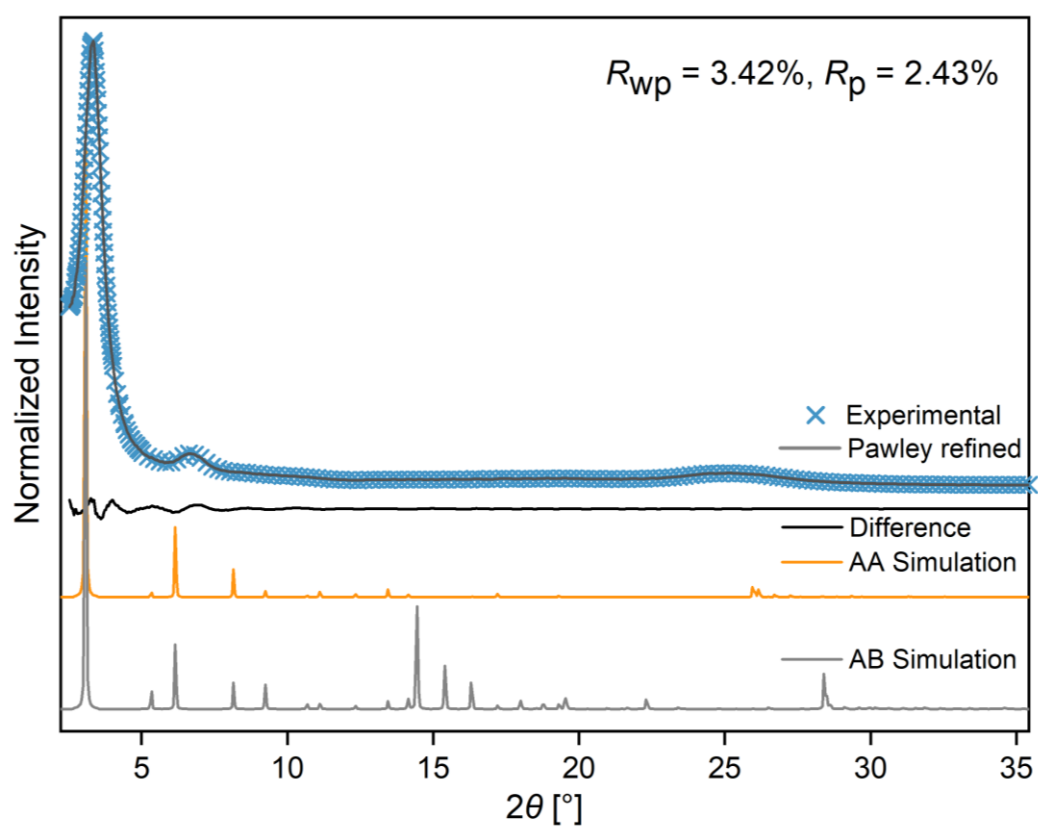


Figure S21 Experimental and calculated PXRD data of **PhDBP-TFP** COF

DBP-Ph-TFB COF

Unit cell parameters:

Lattice type: 3D hexagonal

Space group: $P6/m$

Lengths: $a = b = 38.60 \text{ \AA}$, $c = 3.48 \text{ \AA}$

Angles: $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

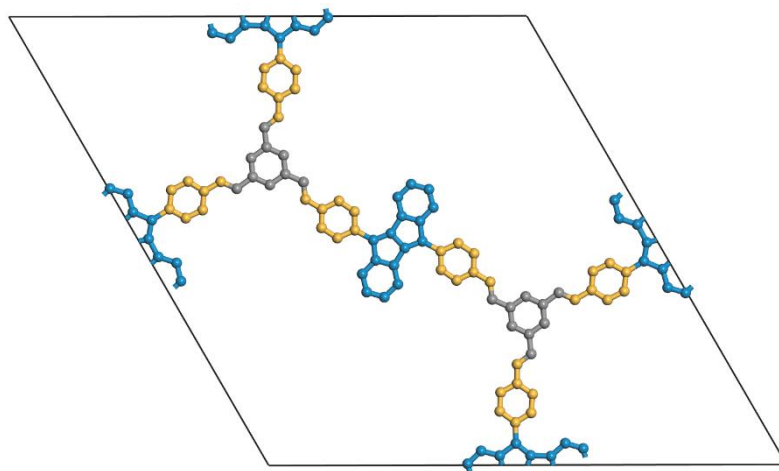


Figure S22 Unit cell of **DBP-Ph-TFB** COF optimized in Materials Studio; Hydrogens are omitted for clarity

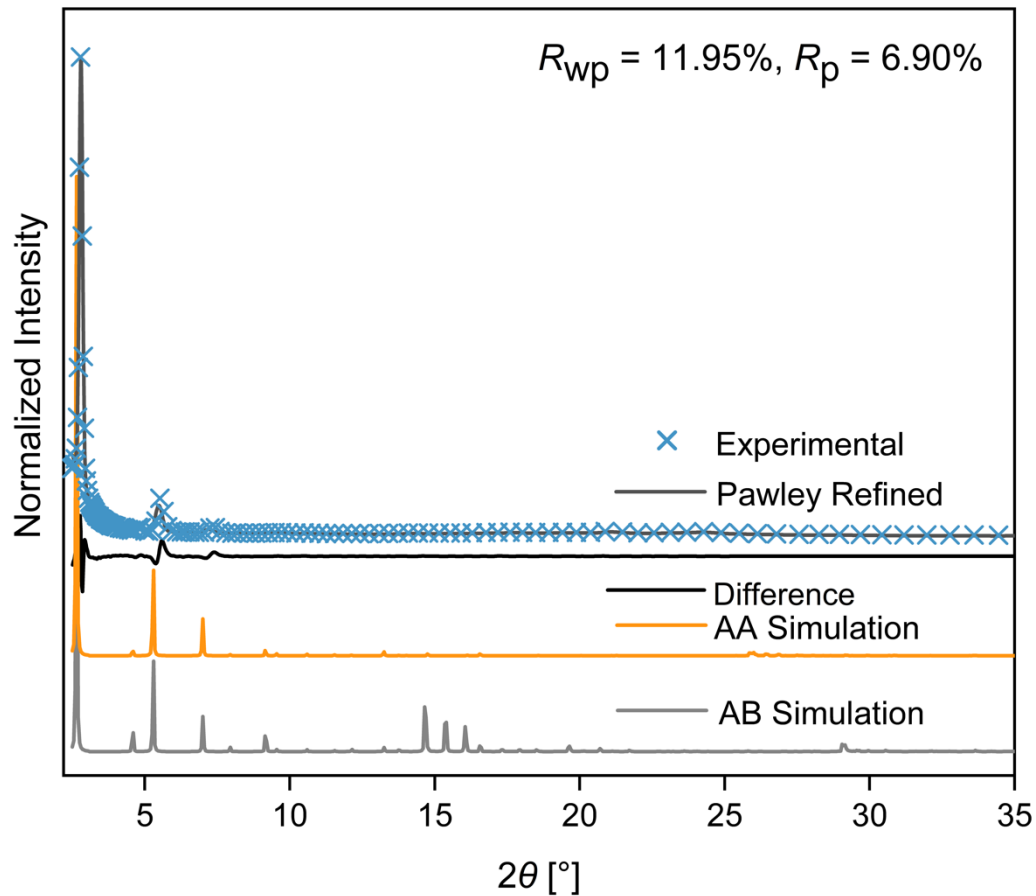


Figure S23 Experimental and calculated PXRD data of **DBP-Ph-TFB** COF

DBP-Ph-TFP POP

Unit cell parameters:

Lattice type: 3D hexagonal

Space group: $P6/m$

Lengths: $a = b = 39.52 \text{ \AA}$, $c = 3.43 \text{ \AA}$

Angles: $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

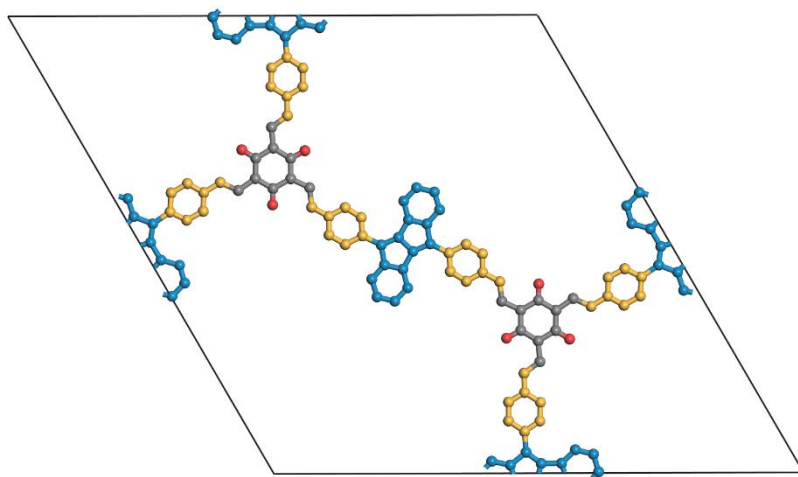


Figure S24 Unit cell of hypothetical crystalline **DBP-Ph-TFP** POP optimized in Materials Studio; Hydrogens are omitted for clarity

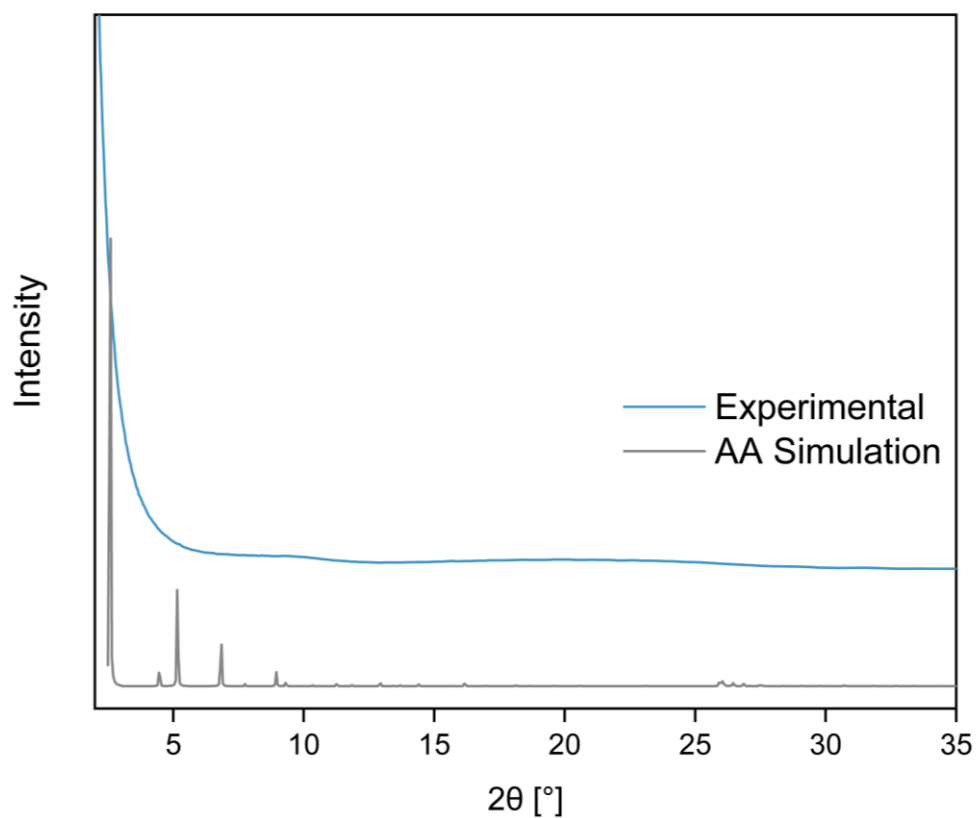


Figure S25 Experimental and calculated PXRD data of **DBP-Ph-TFP** POP

S7. CP-MAS NMR

CP-MAS NMR spectra of all materials show sharp signals indicative of a degree of periodicity within the material. Each material contains multiple carbons in a similar chemical environment leading to overlap of some signals. For **MeDBP-TFB** COF the CH₃-carbon is well resolved at 9.48 ppm. For **PhDBP-TFP** COF and **DBP-Ph-TFP** POP, the signals at 183.98 ppm and 184.63 ppm, respectively, can be assigned to the carbonyl carbon. The spinning side-bands are marked with asterisks.

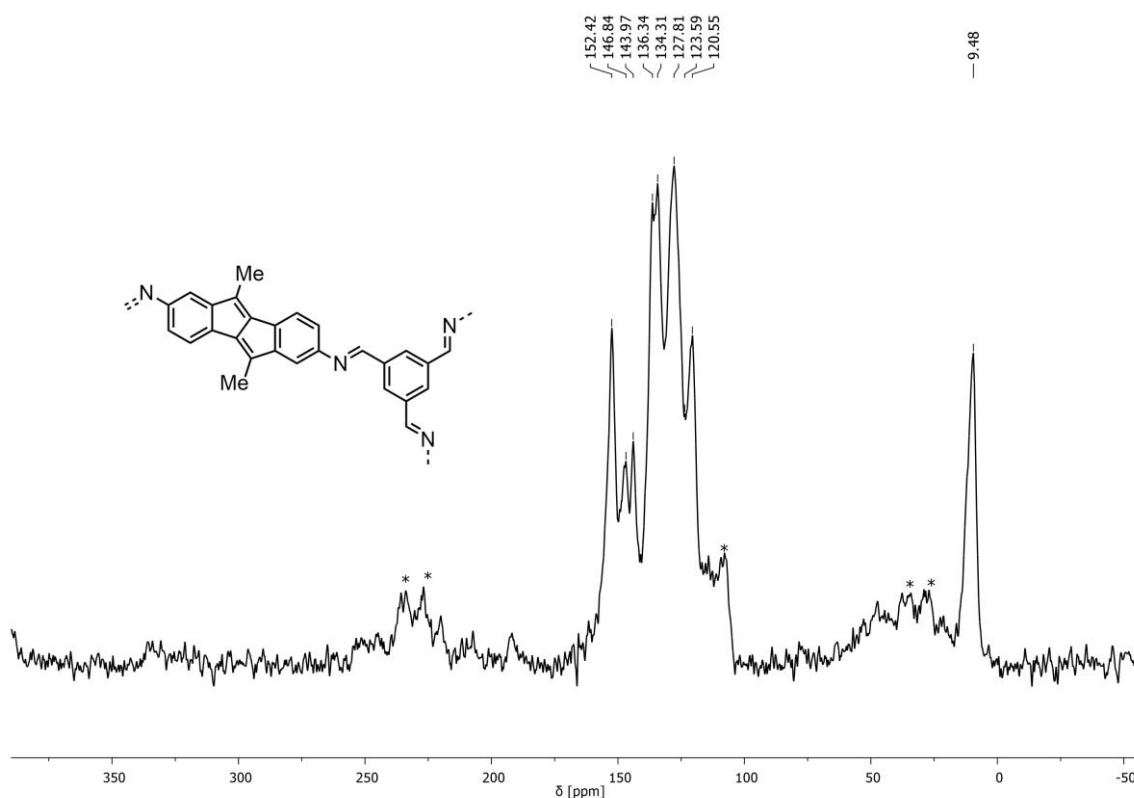


Figure S26 CP-MAS NMR of **MeDBP-TFB** COF; spinning side-bands are marked with asterisks

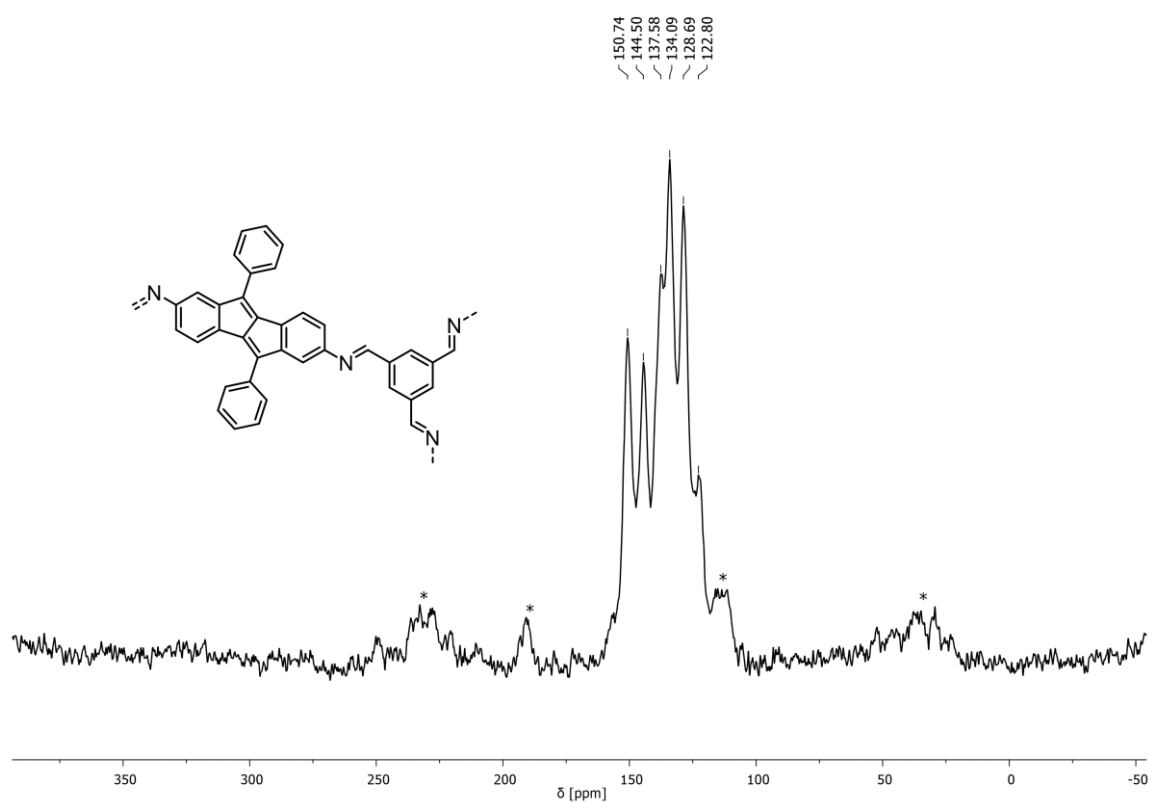


Figure S27 CP-MAS NMR of **PhDBP-TFB** POP; spinning side-bands are marked with asterisks

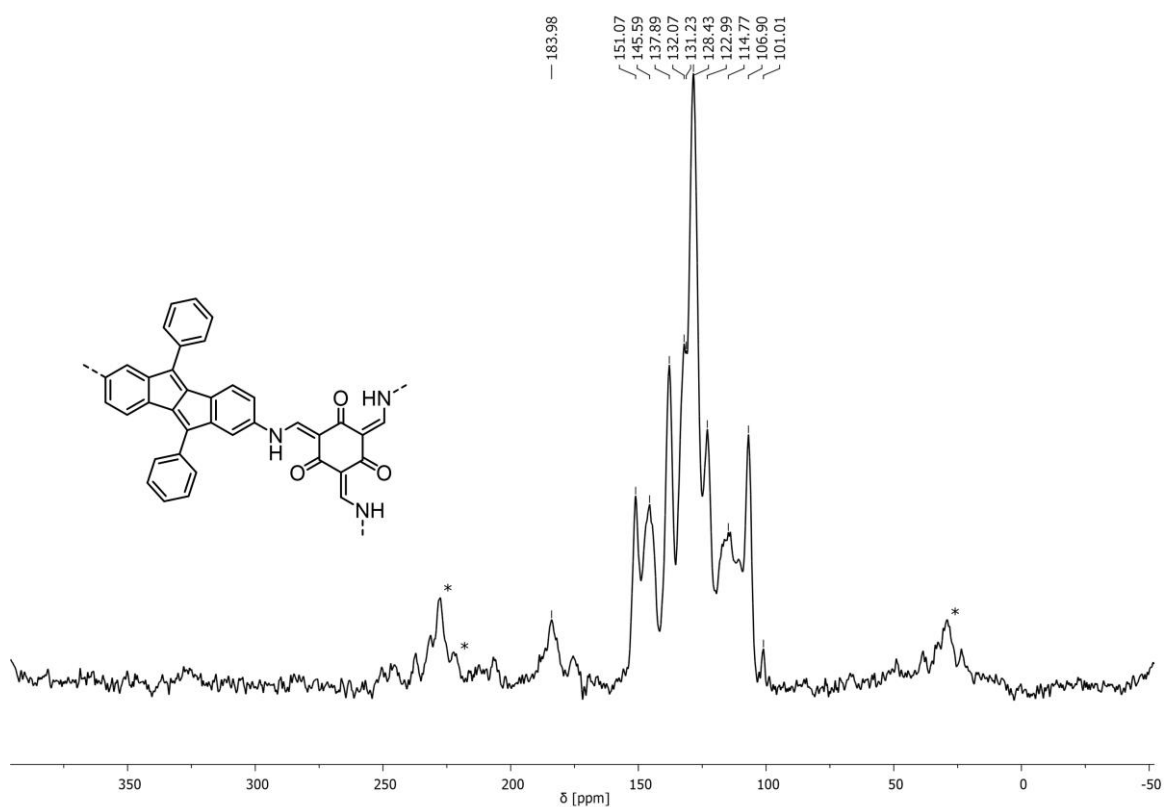


Figure S28 CP-MAS NMR of **PhDBP-TFP** COF; spinning side-bands are marked with asterisks

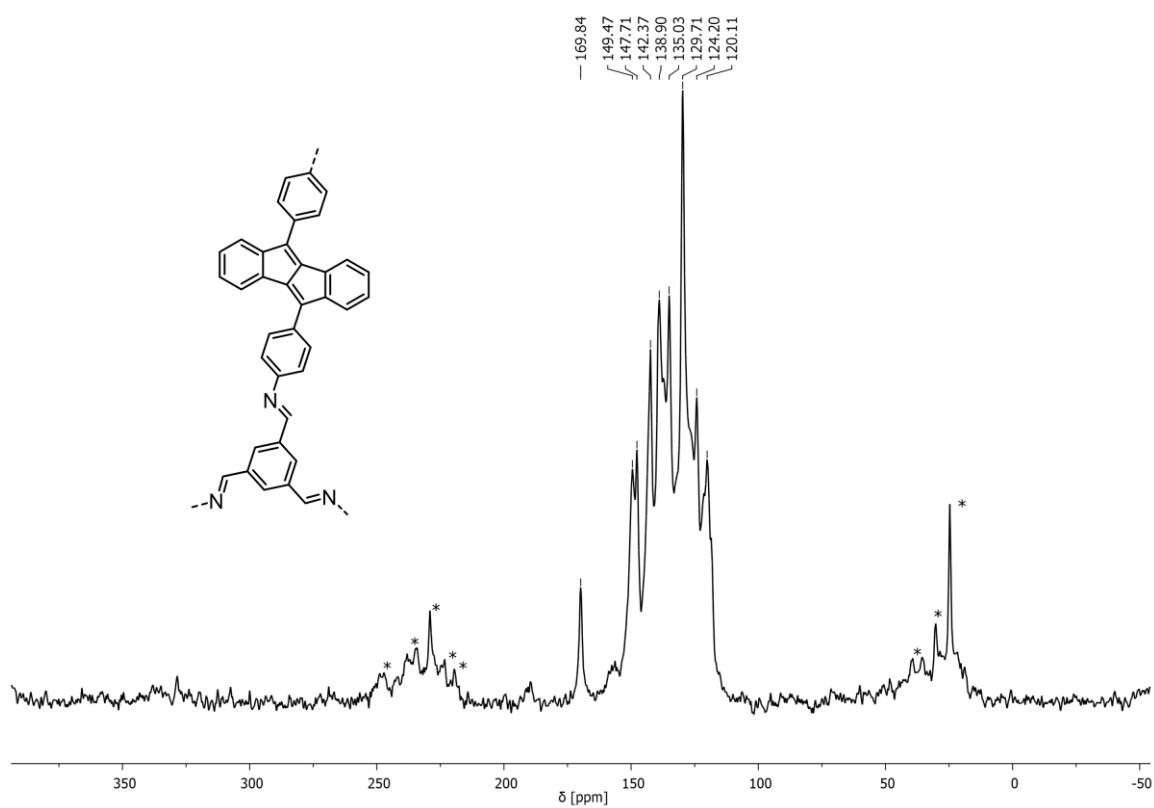


Figure S29 CP-MAS NMR of **DBP-Ph-TFB** COF; spinning side-bands are marked with asterisk

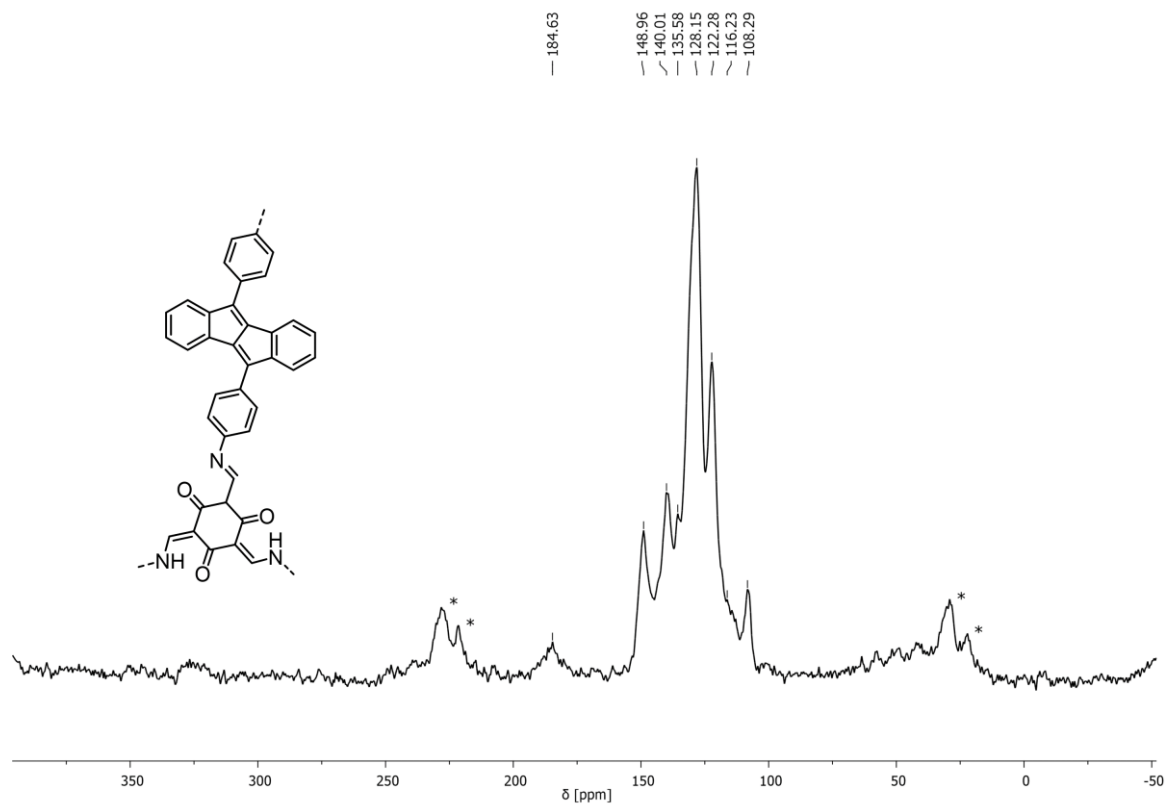


Figure S30 CP-MAS NMR of **DBP-Ph-TFP** POP; spinning side-bands are marked with asterisks

S8. FT-IR

The ATR-FT-IR spectra of all COF and POP materials and monomers show that no significant monomer functionalities are left in the materials. The DBP monomers exhibit spectra very similar to those of the materials, as both contain the same functionalities. For all TFB-linked materials, **MeDBP-TFB** COF, **PhDBP-TFB** POP, and **DBP-Ph-TFB** COF, the signals of the TFB monomer are not visible: The C–H (aldehyde) stretch band at 2870 cm^{-1} is not visible, as well as the C=O stretch band at 1695 cm^{-1} . For all three materials, the characteristic C=N stretch is visible around 1600 cm^{-1} (1595 cm^{-1} for **MeDBP-TFB** COF, 1595 cm^{-1} for **PhDBP-TFB** POP, and 1588 cm^{-1} for **DBP-Ph-TFB** COF). For the ketoenamine-linked materials **PhDBP-TFP** COF and **DBP-Ph-TFP** POP, no signals of TFP monomer are visible. The C=O-stretch of the TFP-aldehyde at 1638 cm^{-1} cannot be detected, while the C=O-stretch of the keto-enamine is visible around 1570 cm^{-1} (1573 cm^{-1} for **PhDBP-TFP** COF, 1578 cm^{-1} for **DBP-Ph-TFP** POP) and the C–N-stretch of the ketoenamine around 1270 cm^{-1} (1269 cm^{-1} for **PhDBP-TFP** COF, 1276 cm^{-1} for **DBP-Ph-TFB TFP** POP).

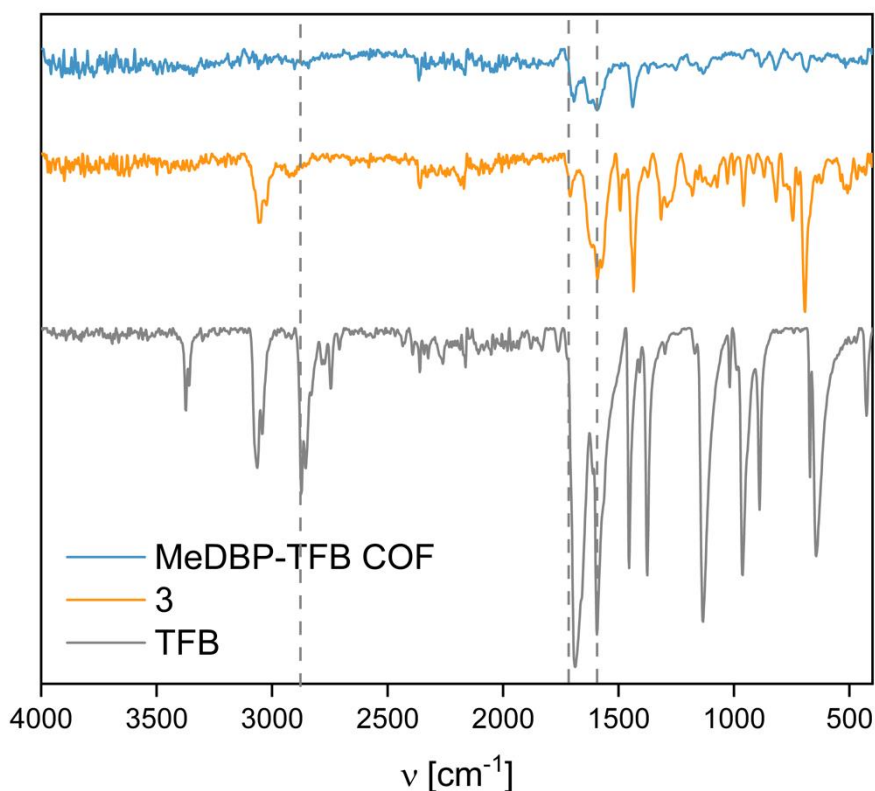


Figure S31 FT-IR of **MeDBP-TFB** COF (blue) in comparison with **3** (orange) and **TFB** (grey)

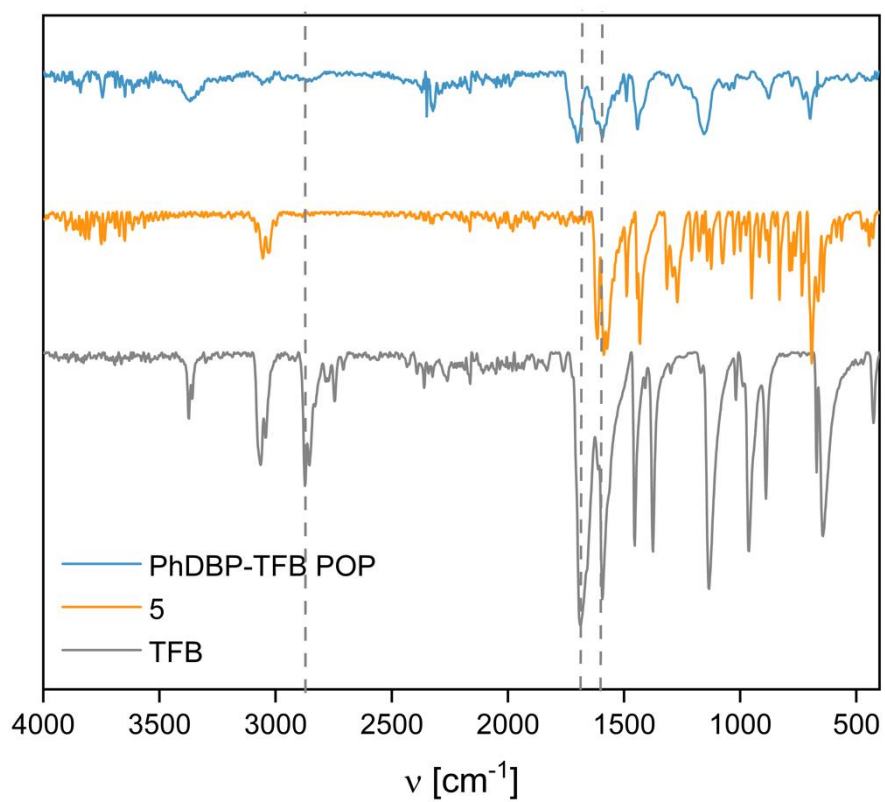


Figure S32 FT-IR of **PhDBP-TFB POP** (blue) in comparison with **5** (orange) and **TFB** (grey)

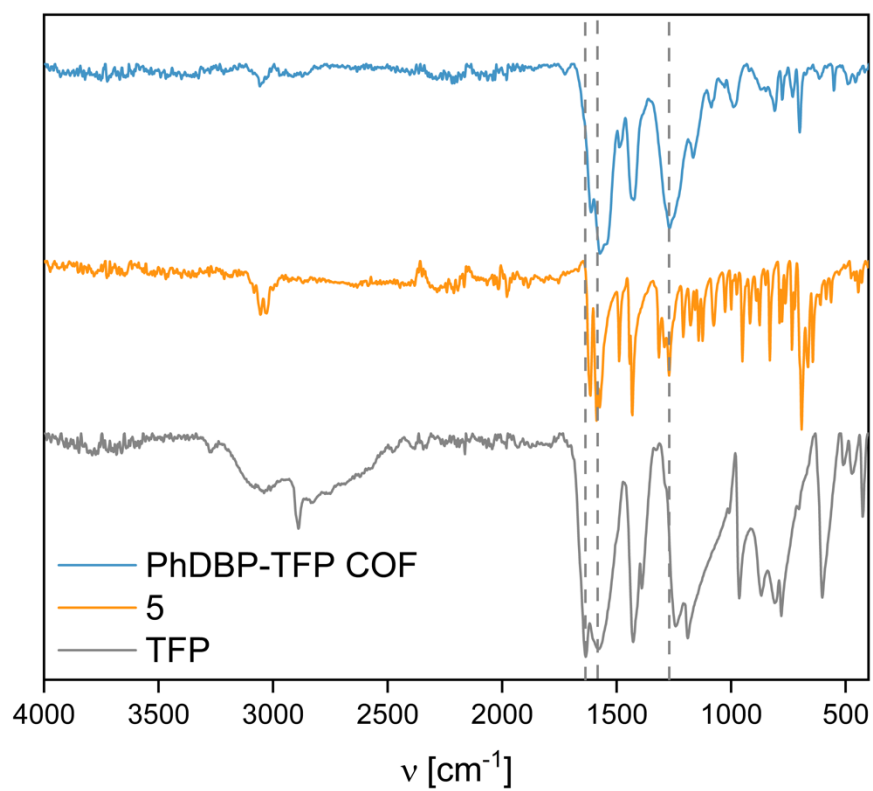


Figure S33 FT-IR of **PhDBP-TFP COF** (blue) in comparison with **5** (orange) and **TFP** (grey)

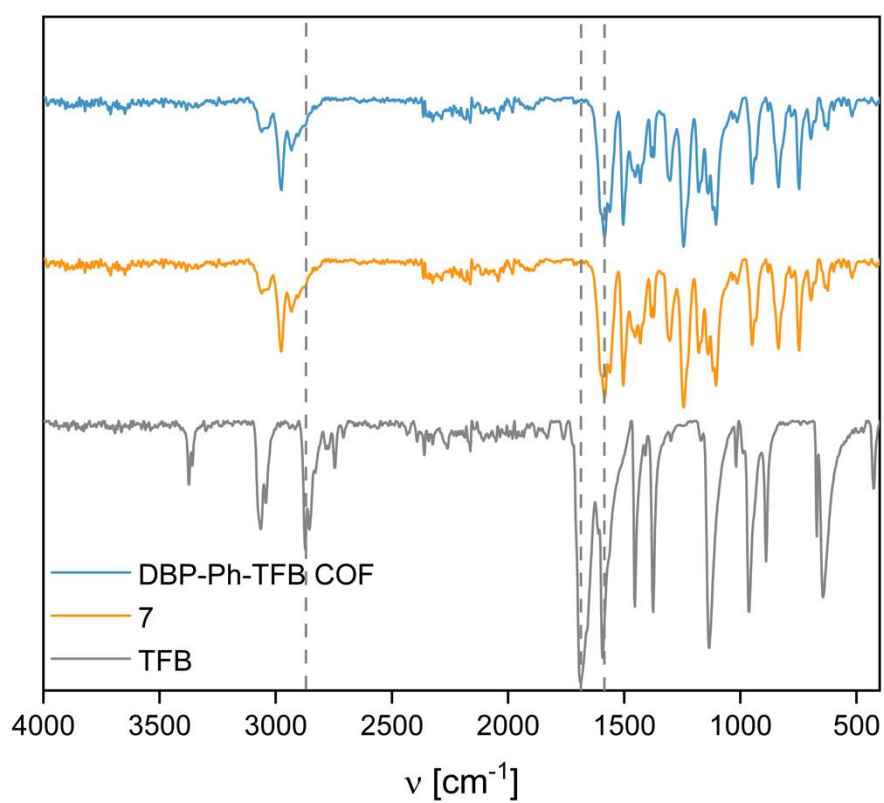


Figure S34 FT-IR of **DBP-Ph-TFB** COF (blue) in comparison with **7** (orange) and **TFB** (grey)

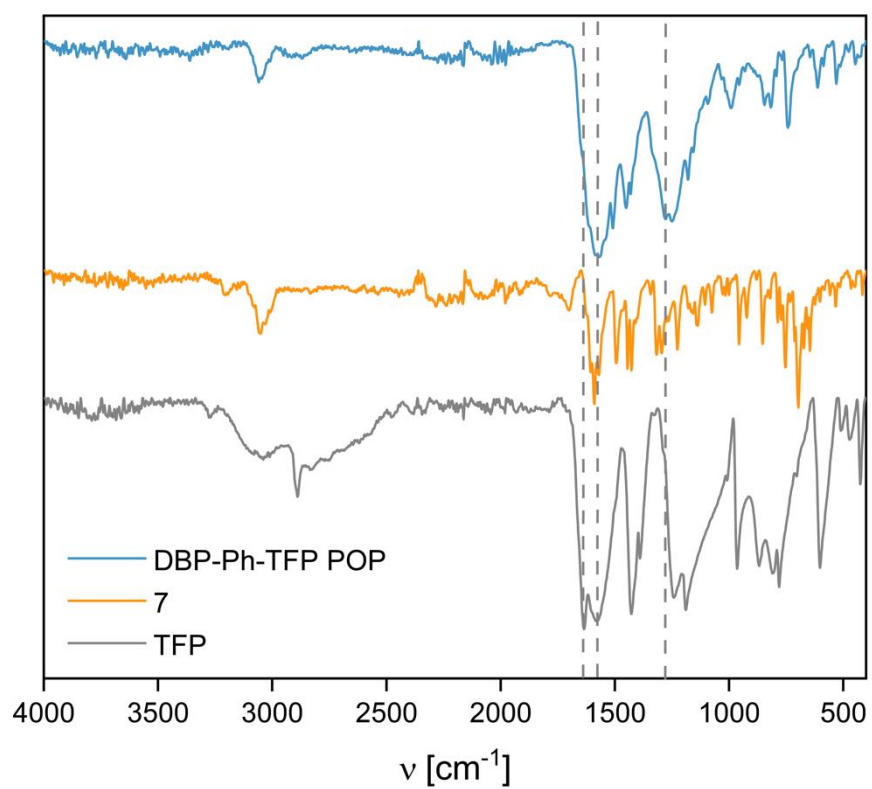


Figure S35 FT-IR of **DBP-Ph-TFP** POP (blue) in comparison with **7** (orange) and **TFP** (grey)

S9. High-Resolution Transmission Electron Microscopy

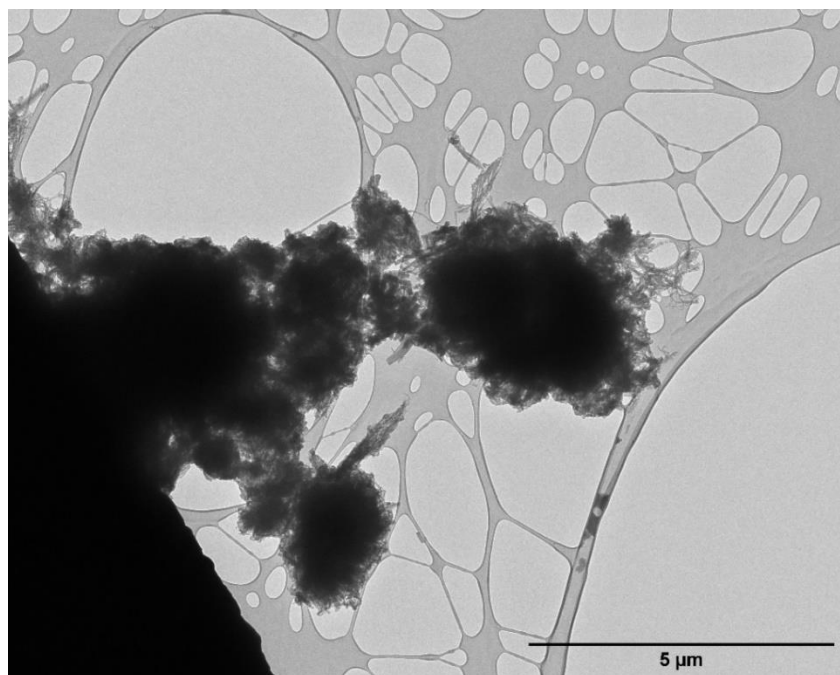


Figure S36 TEM image of **PhDBP-TFP** COF at 3k magnification. Accelerating voltage: 200 kV, temperature 90 K.

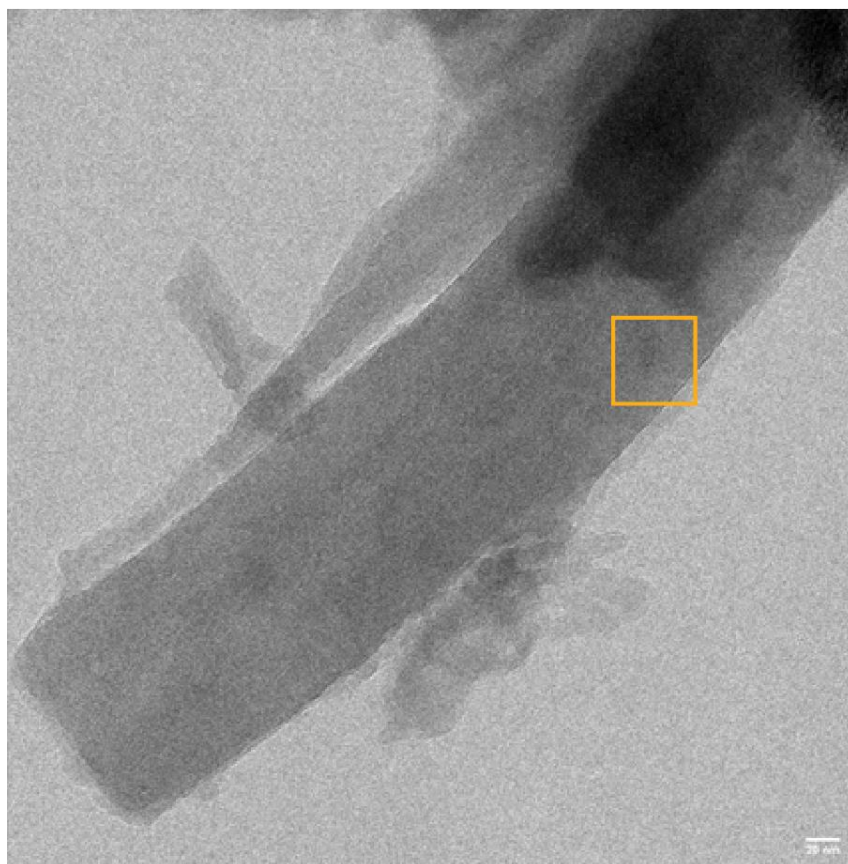


Figure S37 Low-dose TEM image of **PhDBP-TFP** COF at 100k magnification. Accelerating voltage: 200 kV, temperature 90 K.

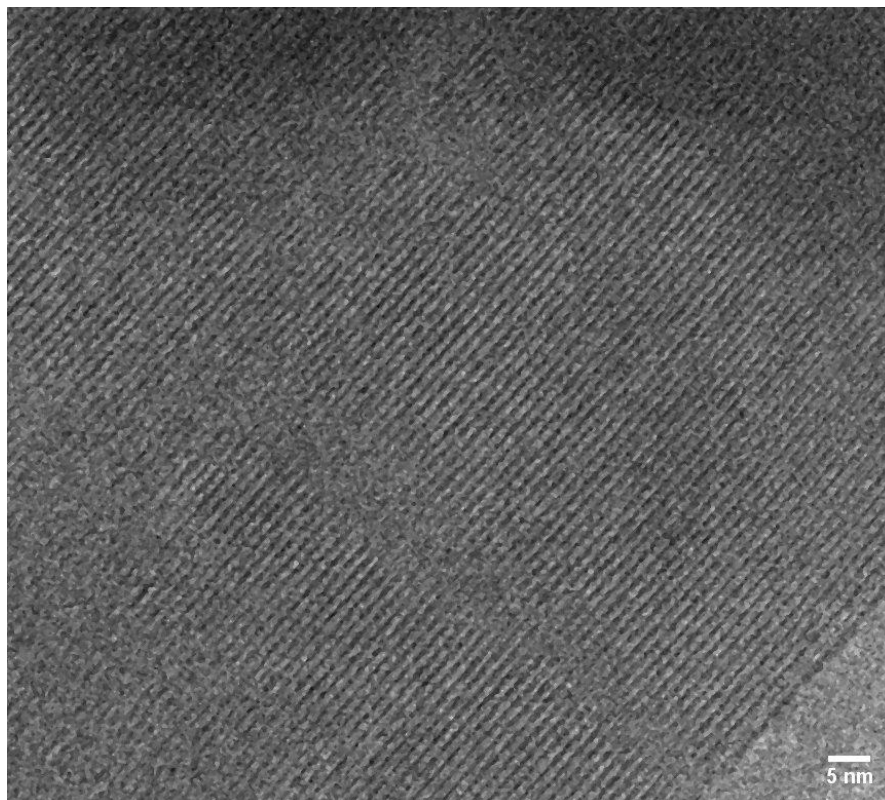


Figure S38 Zoom-in on orange marked part of Figure 37. Low-dose TEM image of **PhDBP-TFP** COF at 100k magnification. Accelerating voltage: 200 kV, temperature 90 K.

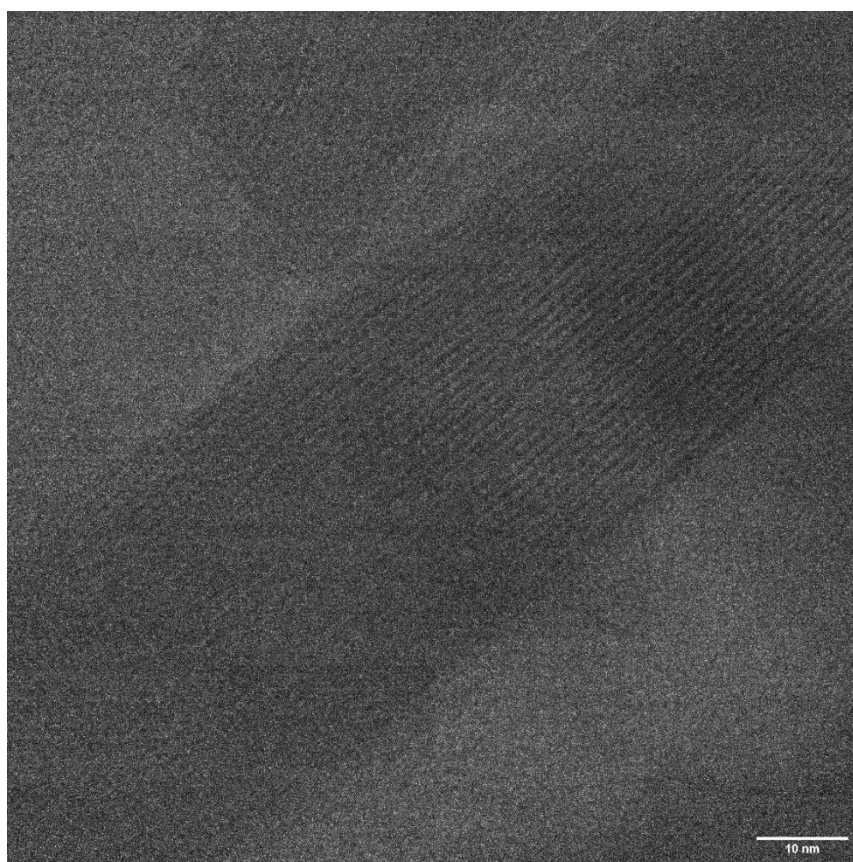


Figure S39 Low-dose HR-TEM image of **PhDBP-TFP** COF at 500k magnification. Accelerating voltage: 200 kV, temperature 90 K.

S10. Scanning Electron Microscopy

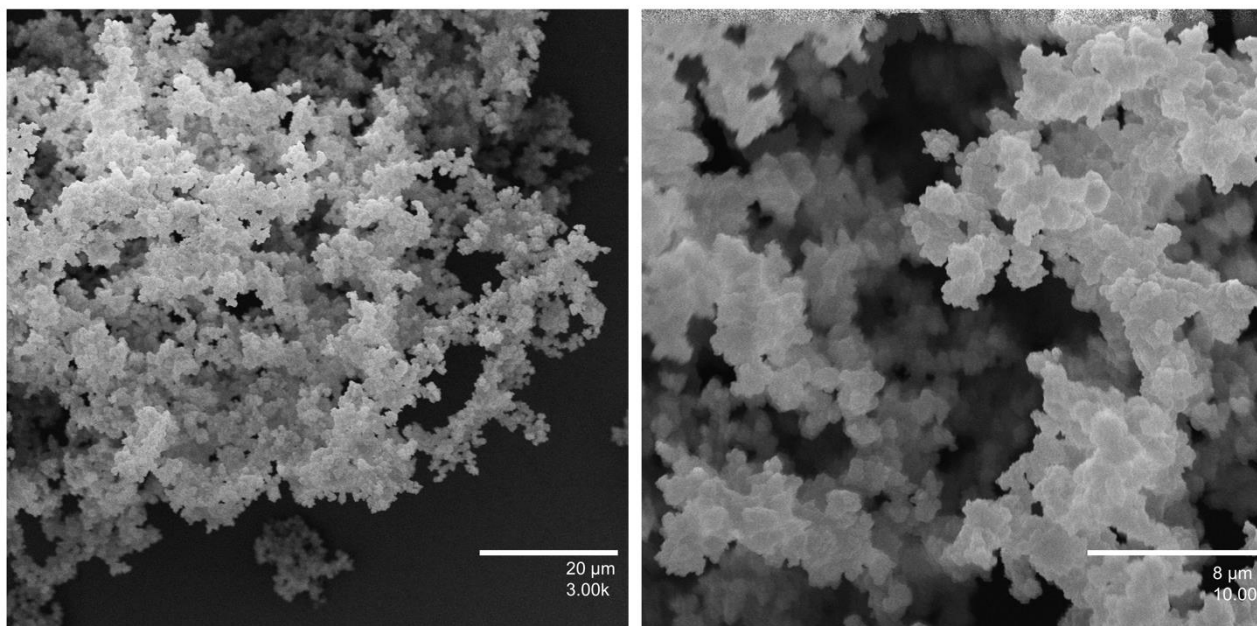


Figure S40 SEM images of **MeDBP-TFB** COF at 3k and 10k magnification

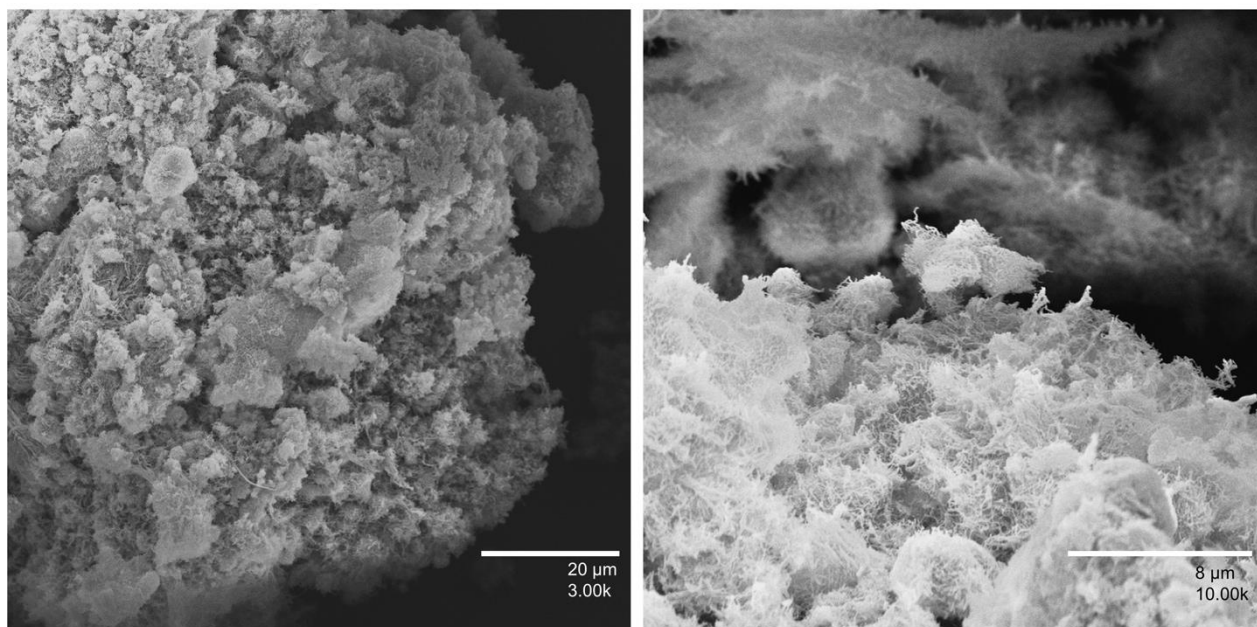


Figure S41 SEM images of **PhDBP-TFP** COF at 3k and 10k magnification

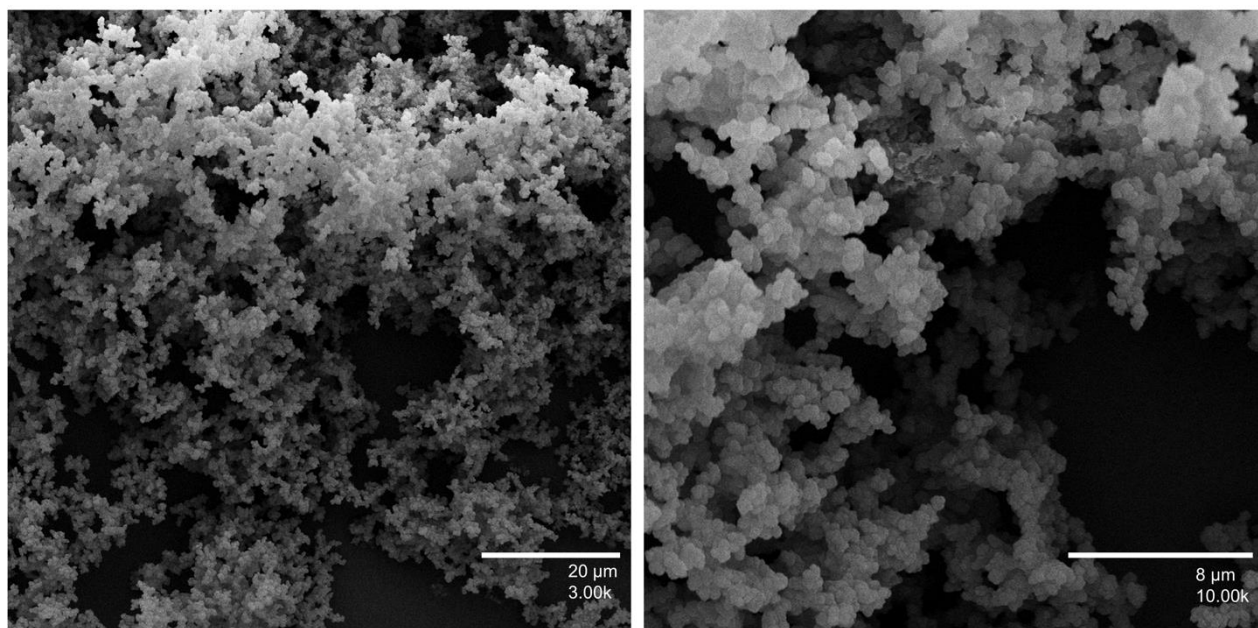


Figure S42 SEM images of **PhDBP-TFB** POP at 3k and 10k magnification

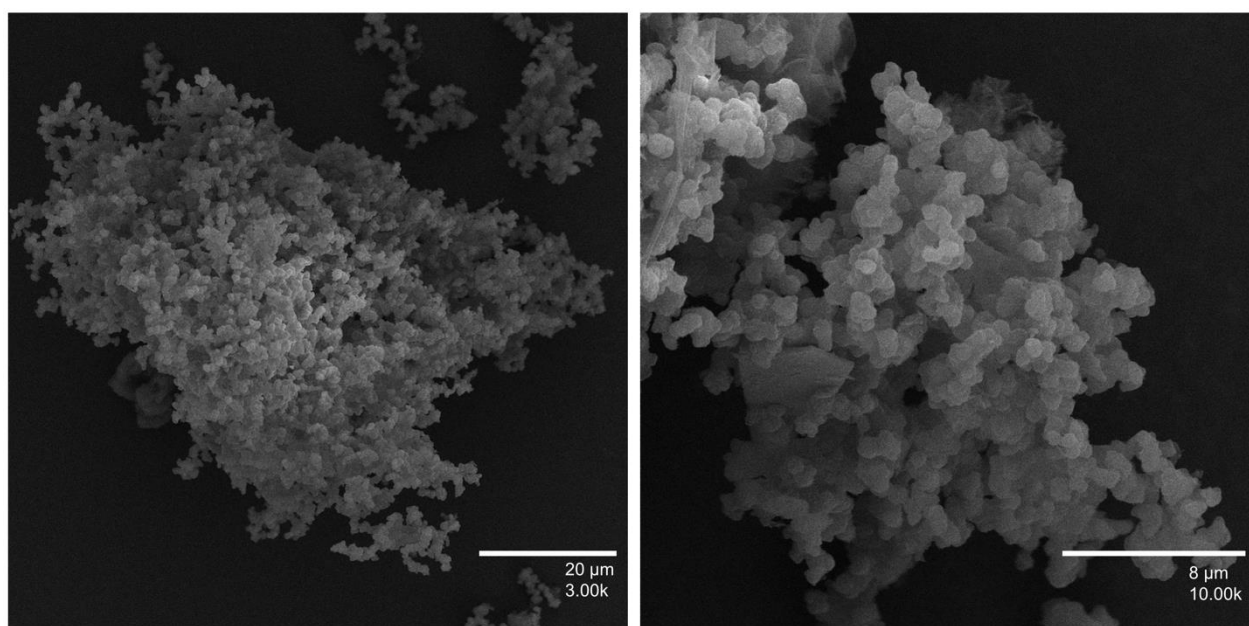


Figure S43 SEM images of **DBP-Ph-TFB** COF at 3k and 10k magnification

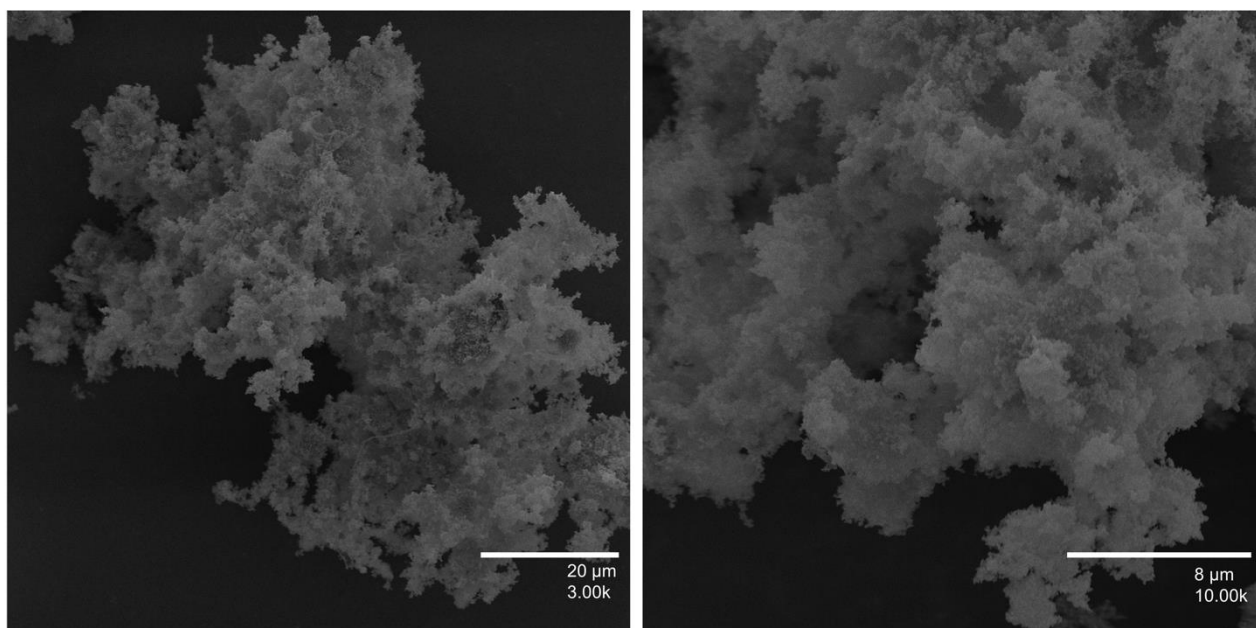


Figure S44 SEM images of **DBP-Ph-TFP** POP at 3k and 10k magnification

S11. Elemental Analysis of COFs

A significant discrepancy between calculated and measured elemental composition can be observed especially for the carbon content, while proton and nitrogen content are replicated well. This is in line with previously reported elemental analyses of COFs.^{15–17} Heating under vacuum (120 °C) immediately before measurement brings the measured values closer to the calculated values, indicating that adsorbents in the pores play a role in the atomic composition. All samples have been subjected to this treatment prior to measurement. All measurements were conducted in triplicates.

MeDBP-TFB COF

Calculated: C: 86.72%, H: 4.85%, N: 8.43%

Measured: C: 76.11%, H: 4.50%, N: 5.03%

PhDBP-TFB POP

Calculated: C: 89.54%, H: 4.42%, N: 6.14%

Measured: C: 78.45%, H: 3.92%, N: 4.65%

PhDBP-TFP COF

Calculated: C: 83.59%, H: 4.13%, N: 5.73%, O: 6.55%

Measured: C: 76.93%, H: 4.15%, N: 4.184%

DBP-Ph-TFB COF

Calculated: C: 89.54%, H: 4.42%, N: 6.14%

Measured: C: 82.63%, H: 3.42%, N: 4.65%

DBP-Ph-TFP POP

Calculated: C: 83.59%, H: 4.13%, N: 5.73%, O: 6.55%

Measured: C: 79.31%, H: 4.28%, N: 3.46%

S12. Thermogravimetric Analysis

Approximately 1 mg of material was used for each measurement. Onset temperatures for all materials are listed below:

Table 8 TGA onset temperature of COFs and POPs

Material	Onset Temperature [°C]
MeDBP-TFB COF	325
PhDBP-TFB POP	402
PhDBP-TFP COF	336
DBP-Ph-TFB COF	343
DBP-Ph-TFP POP	410

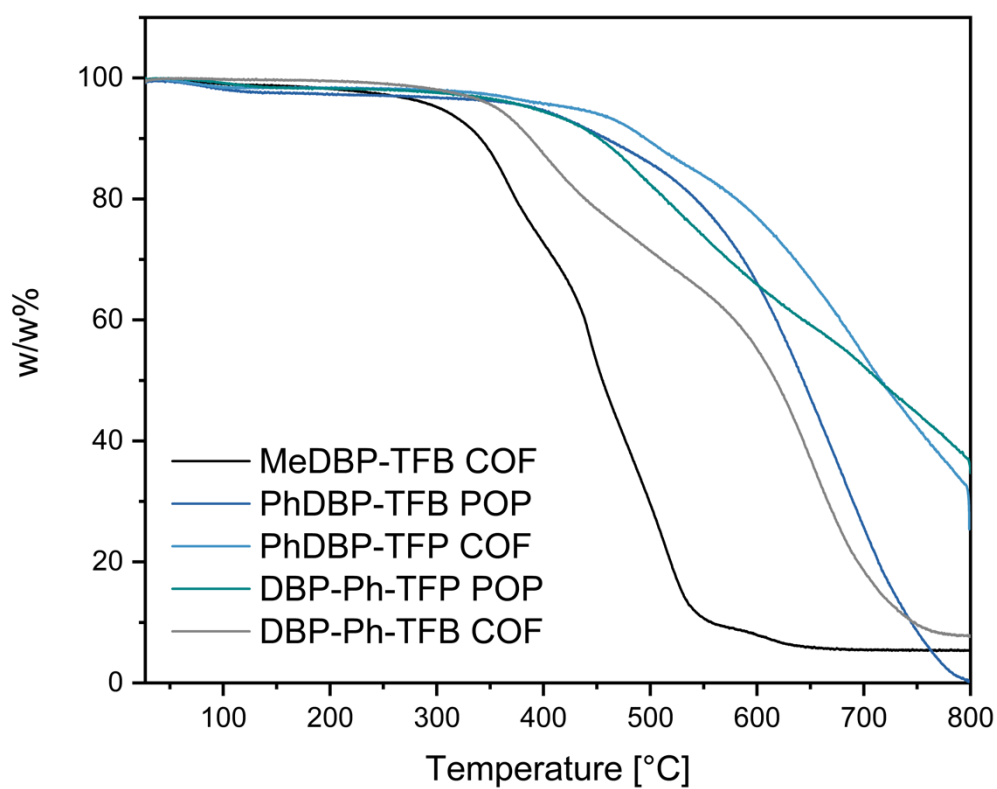


Figure S45 Thermogravimetric analysis of all COFs and POPs.

S13. COF Stability Tests

To confirm the chemical stability of our COFs we subjected them to various conditions. 3 mg of COF materials was suspended in 2 mL of solvent (water, 1 M HCl, 1 M NaOH, or EC/DMC 1:1) for 48 h. The materials were transferred to a PTFE-lined fritted funnel, washed with acetone and methanol and subsequently dried via $scCO_2$. PXRD shows minimal to no deviation to the pristine COF and confirms excellent stability under those conditions.

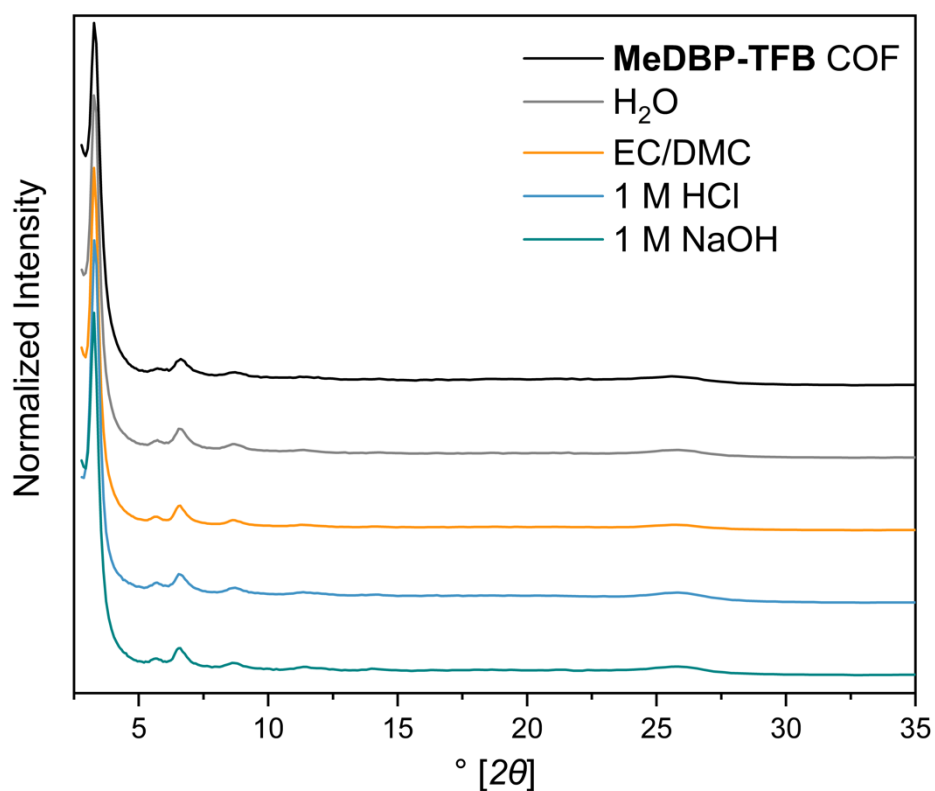


Figure S46 PXRD of **MeDBP-TFB** COF after treatment with various conditions.

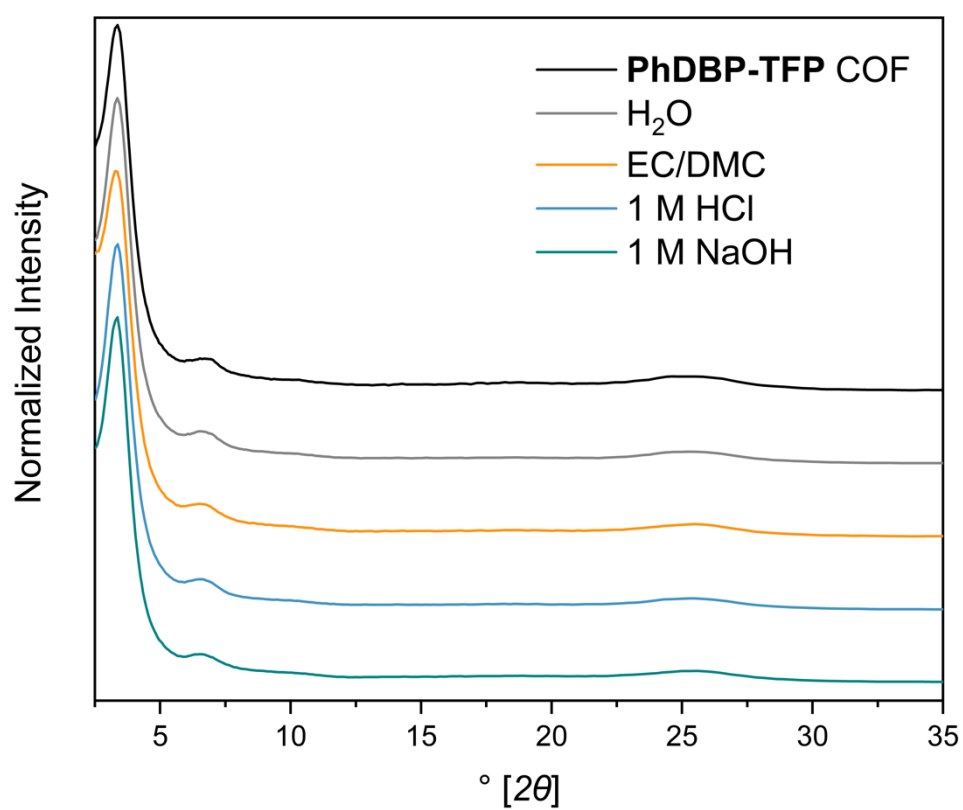


Figure S47 PXRD of **PhDBP-TFB COF** after treatment with various conditions.

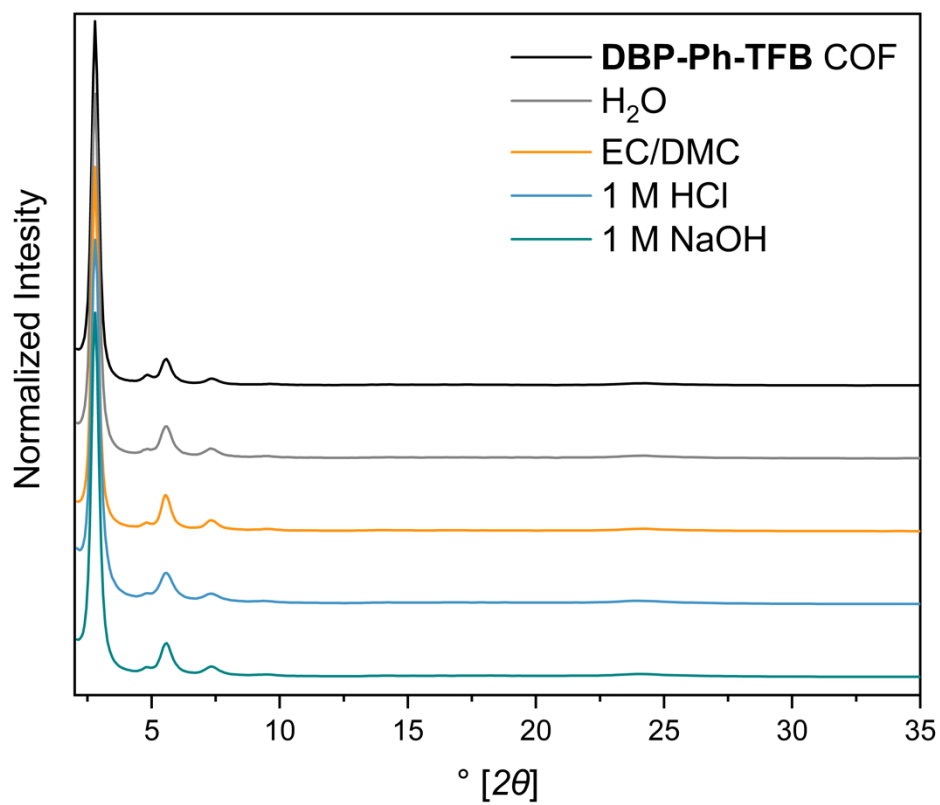


Figure S48 PXRD of **DBP-Ph-TFB COF** after treatment with various conditions.

S14. UV-vis Spectroscopy

For diffuse reflectance UV-vis spectroscopy, the solid COF or POP samples were mixed with BaSO₄ for dilution before measurement to avoid saturation of the detector.

For the preparation of COF dispersions, 5 mg of material was stirred in MeCN/THF/TFA 7:3:2 (1 mg mL⁻¹) for 18 h. The dispersion was filtered through folded filter paper (pore size ~ 7 µm) to remove larger particles. The dispersion was diluted tenfold before the UV-vis measurement. The concentration of the dispersions could not be determined. Therefore, the spectra were normalized from 0 to 1 between 280 nm and 800 nm. It cannot be excluded that monomers or oligomers are present in the dispersion. Optical band gaps of the materials were determined with Tauc plots from the diffuse reflectance data and are as follows: 1.3 eV (**PhDBP-TFP** COF), 1.5 eV (**MeDBP-TFB** COF), 1.6 eV (**PhDBP-TFB** POP), and 1.7 eV (**DBP-Ph-TFB** COF and **DBP-Ph-TFP** POP).

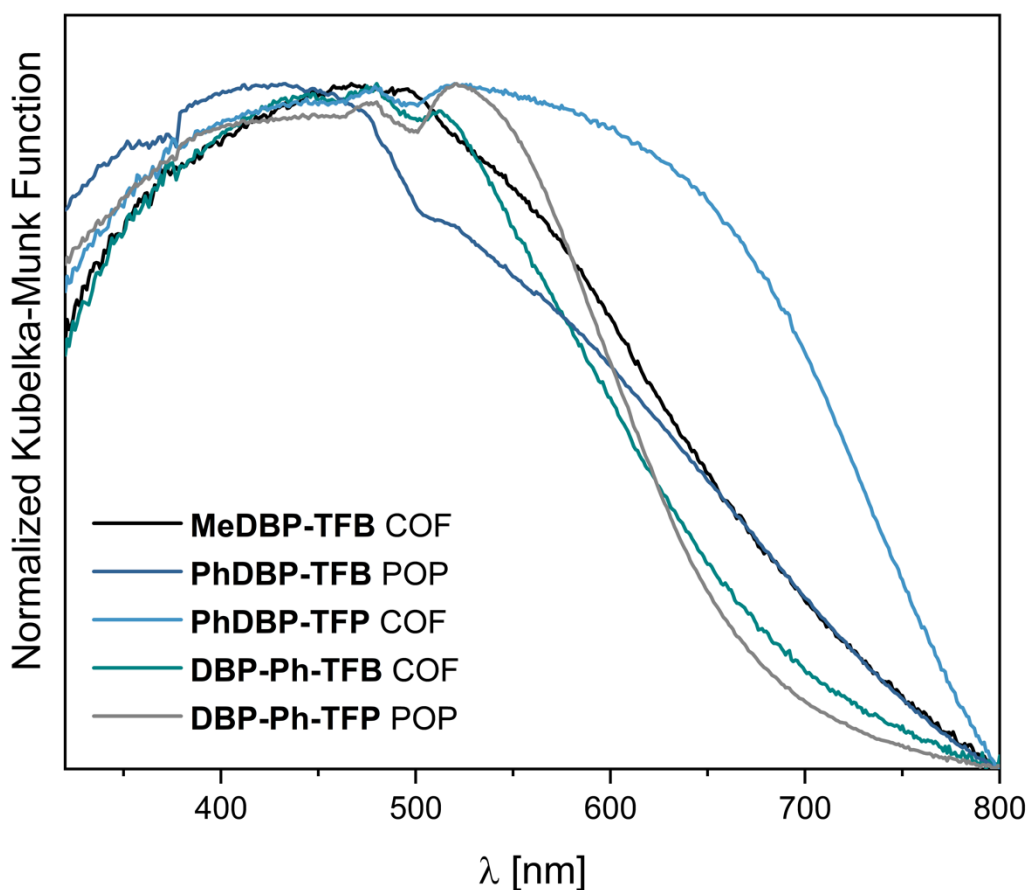


Figure S49 Normalized absorption spectra of solid COF samples recorded by diffuse-reflectance UV-vis spectroscopy and converted using the Kubelka–Munk function. **MeDBP-TFB** COF (black), **PhDBP-TFB** POP (dark blue), **PhDBP-TFP** COF (green), **DBP-Ph-TFB** COF (light blue), and **DBP-Ph-TFP** POP (grey).

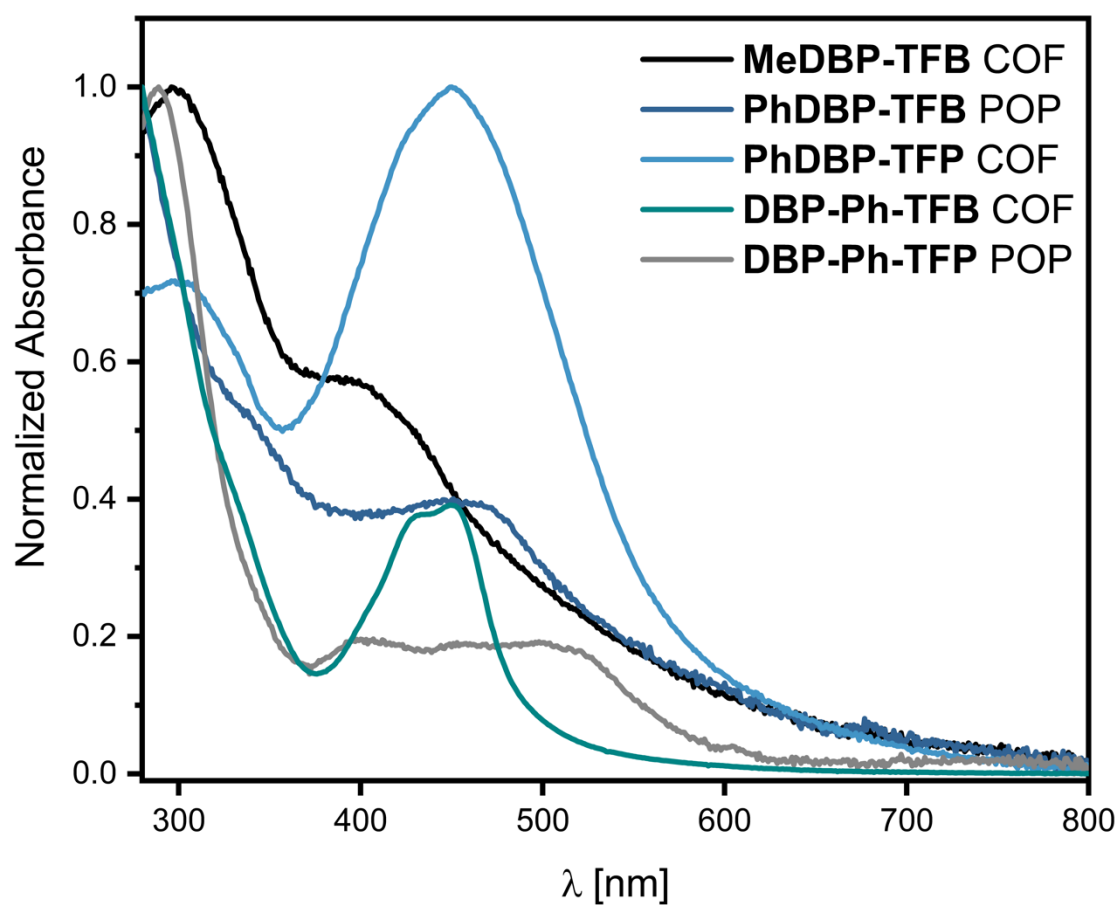


Figure S50 UV-vis spectra of COF and POP dispersions (through TFA-assisted exfoliation); **MeDBP-TFB** COF (black), **PhDBP-TFB** POP (dark blue), **PhDBP-TFP** COF (green), **DBP-Ph-TFB** COF (light blue), and **DBP-Ph-TFP** POP (grey).

S15. Thin Film Preparation and Conductivity Measurements

For the preparation of COF thin films, dispersions were produced by two different exfoliation protocols:

Protocol 1, acid-free exfoliation: 1 mg mL⁻¹ of COF material was sonicated in MeOH for 1 h using a sonotrode.

Protocol 2, acid-mediated exfoliation:¹⁸ 1 mg mL⁻¹ of **PhDBP-TFP** COF was sonicated in MeCN/THF/TFA (7:3:2) for 30 min using a sonication bath and subsequently stirred for 18 h at room temperature.

Ossila “Interdigitated ITO Substrates for OFET and Sensing” S161: w × L: 30 mm × 50 μm were coated by dropcasting. Multiple rounds of dropcasting and evaporation were necessary to cover the surface with material. The devices were dried in vacuum for 18 h prior to measurements. Laser scanning microscopy was used to determine the film thickness. (Figure 45 and 46) The microscopy images revealed patchy films with many holes. The determination of film thickness is therefore merely a rough estimation. We believe that an optimized film preparation will lead to significantly improved properties of our antiaromatic COF devices.

For each measurement, 5 devices were measured simultaneously to improve the signal to noise ratio and to average out inhomogeneities in the film quality at a 20 V bias. Photoconductivity was observed during irradiation with an LED for wavelengths down to 660 nm. Doping with I₂ lead to a conductivity increase of 3–4 orders of magnitude. For the I₂-doped devices a decrease of conductivity over time can be observed that is ascribed to the gradual evaporation of I₂ especially well observed at the long measurements (Figure S56)

Both exfoliation protocols 1 and 2 led to similar results in terms of conductivity and photoconductivity. The main difference is an overall slower response time for the devices prepared by acid-mediated protocol 1. This can especially be seen in the long measurements with 100 on/off cycles (10 s on, 10 s off) gradually increasing conductivity in the irradiated and non-irradiated state.

For comparison, devices were fabricated from the well-known literature **BND-TFB** COF.¹⁹ Film-fabrication by Protocol 1 led to slightly better results than for DBP COFs (as expected, as Protocol 1 was optimized for this COF structure by Dichtel and coworkers)¹⁸ The devices show only a very low conductivity and no photoresponse. (Figure S60) This encourages that the observed activity of DPB-based COFs indeed stems from the DBP functionality.

The conductivity was calculated using the following equation:

$$\sigma = \frac{L}{U \frac{I}{wt}}$$

σ = electrical conductivity; L = channel length; I = current; w = channel width; t = film thickness

We list recent functional framework materials and their conductivity in the table below the following graphs.

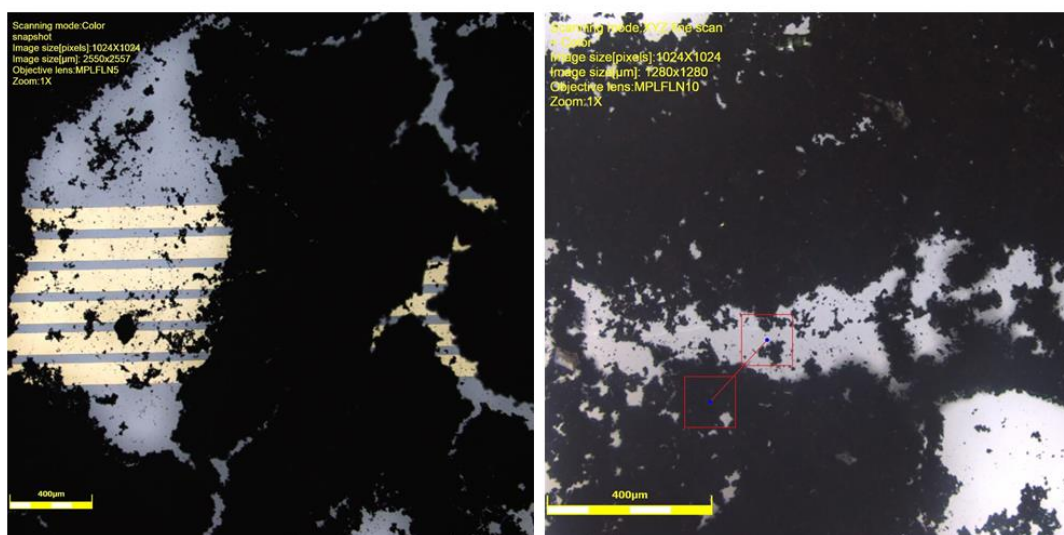


Figure S51 Laser scanning microscopy images of **PhDBP-TFP** COF film on interdigitated ITO substrate prepared by acid-mediated exfoliation protocol 1 with a film thickness of 30 μm .

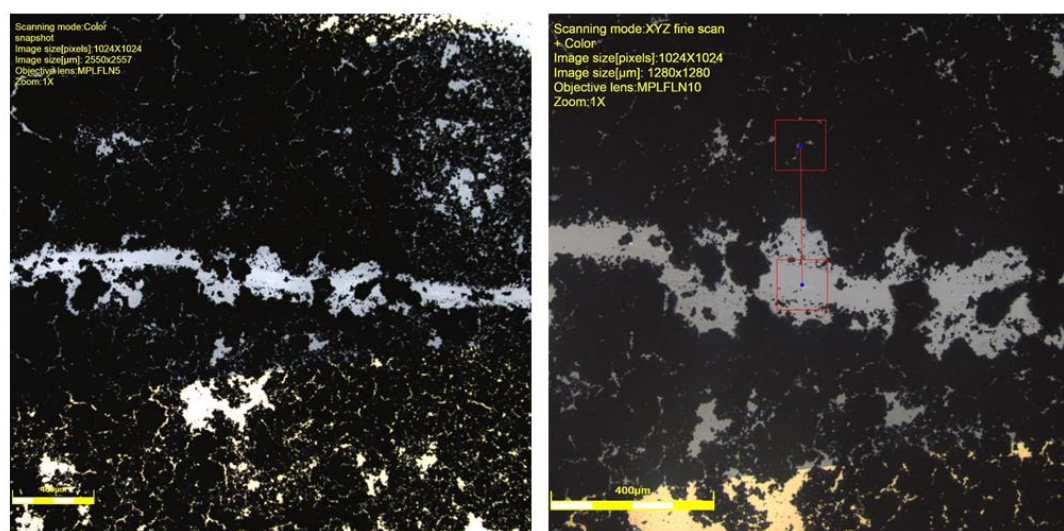


Figure S52 Laser scanning microscopy images of **PhDBP-TFP** COF film on interdigitated ITO substrate prepared by acid-free exfoliation protocol 2 with a film thickness of 17 μm .

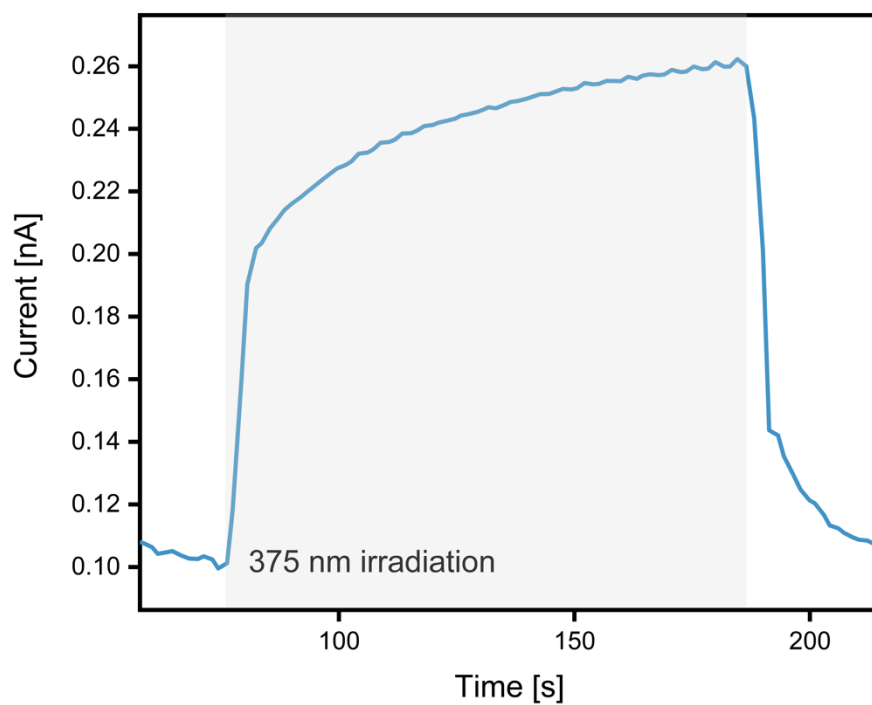


Figure S53 Conductivity of acid-free exfoliated **PhDBP-TFP** COF film at 20 V bias, with (grey) and without (white) irradiation with a 375 nm LED; On/off ratio 2.6

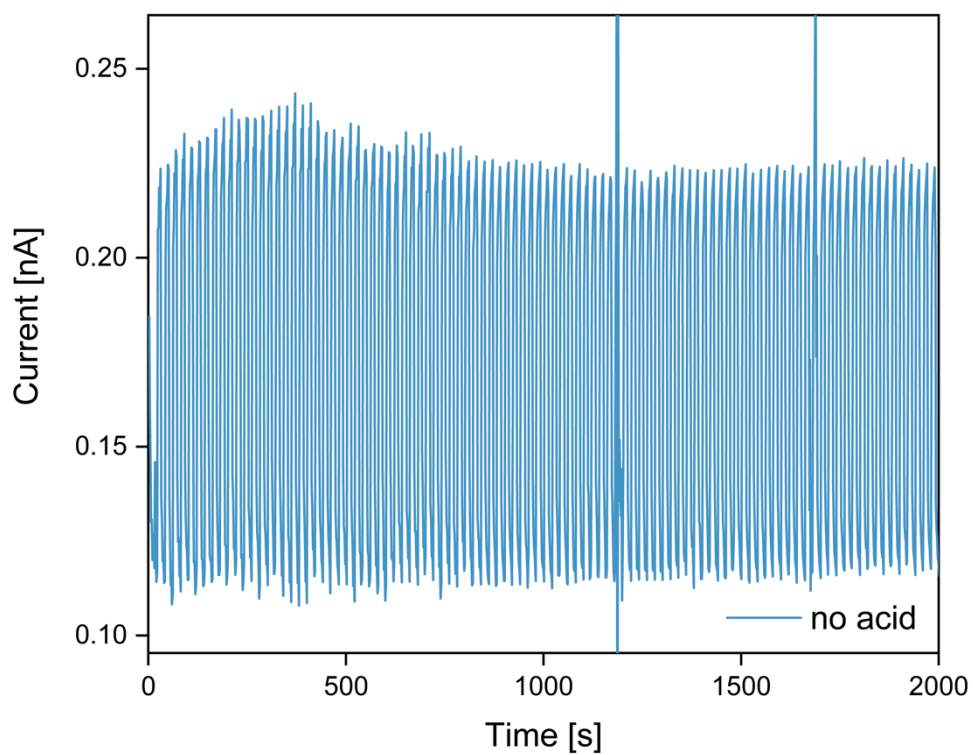


Figure S54 Conductivity of acid-free exfoliated **PhDBP-TFP** COF film at 20 V bias, through 100 cycles of irradiation with a 375 nm LED (10 s on, 10 s off).

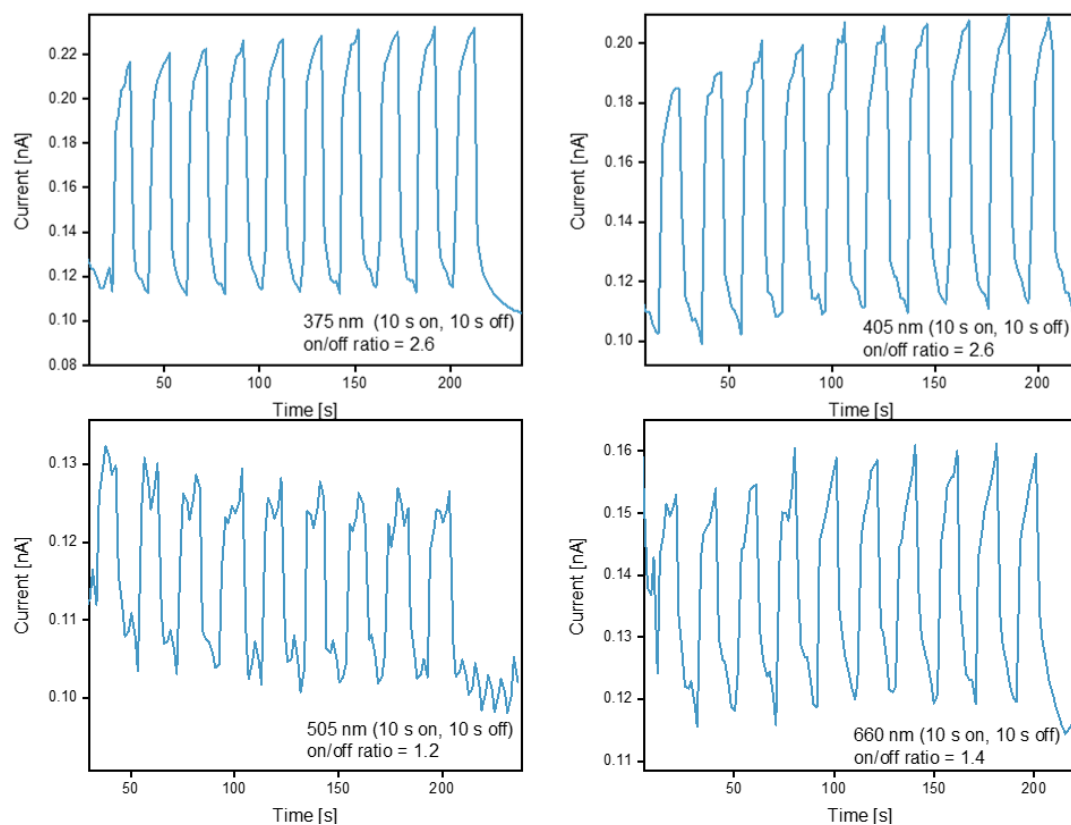


Figure S55 Conductivity of acid-free exfoliated **PhDBP-TFP** COF film at 20 V bias, through 10 cycles of irradiation with different wavelength LEDs (10 s on, 10 s off)

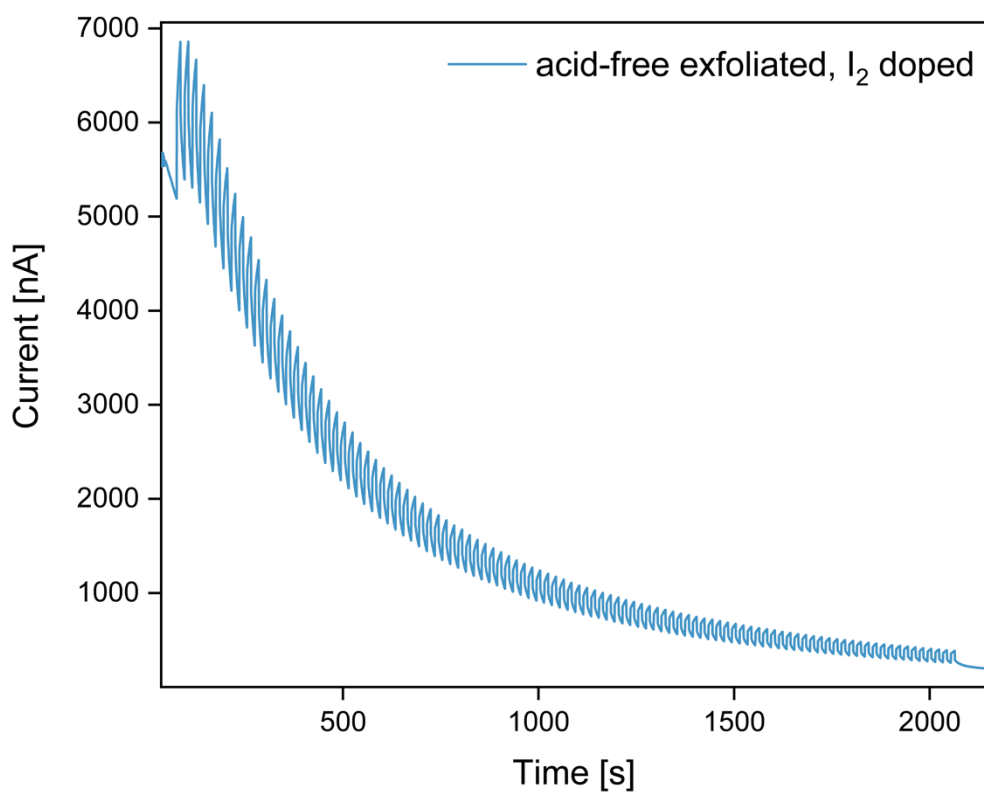


Figure S56 Conductivity of acid-free exfoliated **PhDBP-TFP** COF film at 20 V bias, through 100 cycles of irradiation with a 375 nm LED (10 s on, 10 s off).

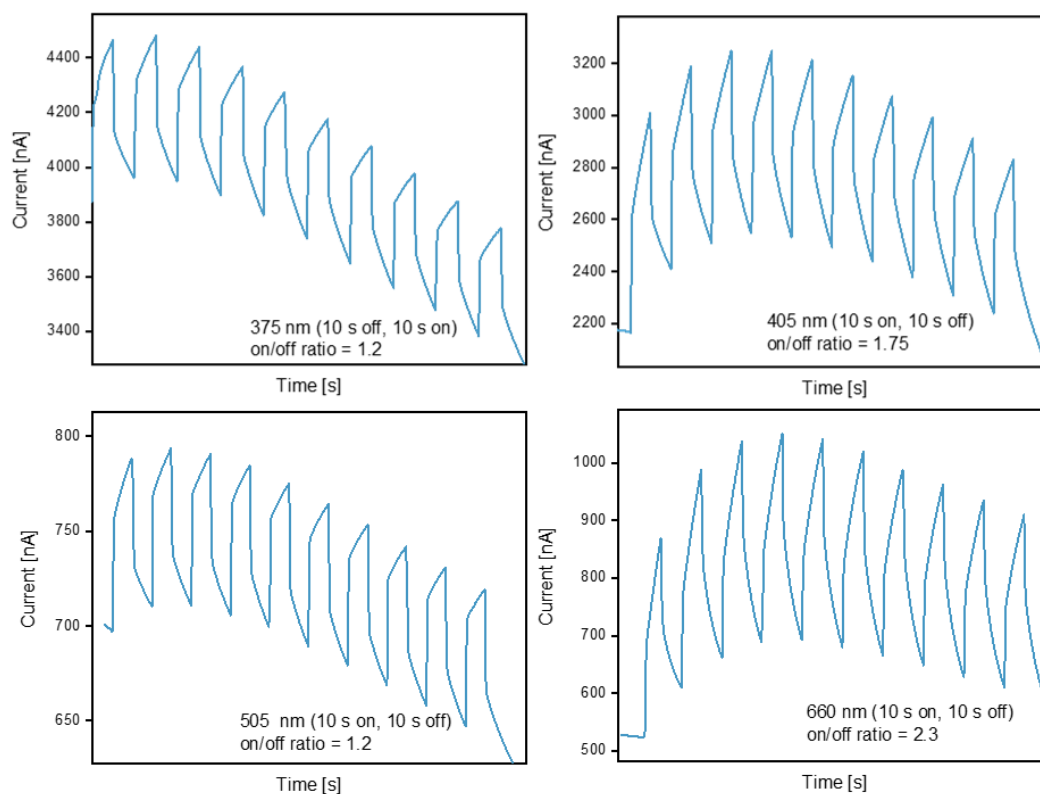


Figure S57 Conductivity of acid-free exfoliated I₂-doped **PhDBP-TFP** COF film at 20 V bias, through 10 cycles of irradiation with different wavelength LEDs (10 s on, 10 s off)

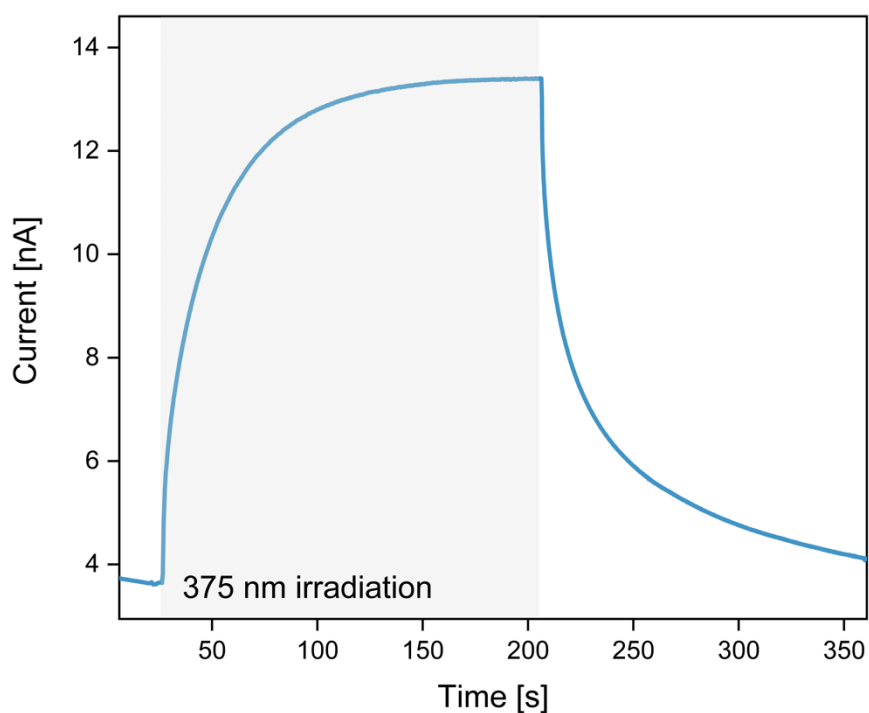


Figure S58 Conductivity of acid-exfoliated **PhDBP-TFP** COF film at 20 V bias, with (grey) and without (white) irradiation with a 375 nm LED; On/off ratio 3.6.

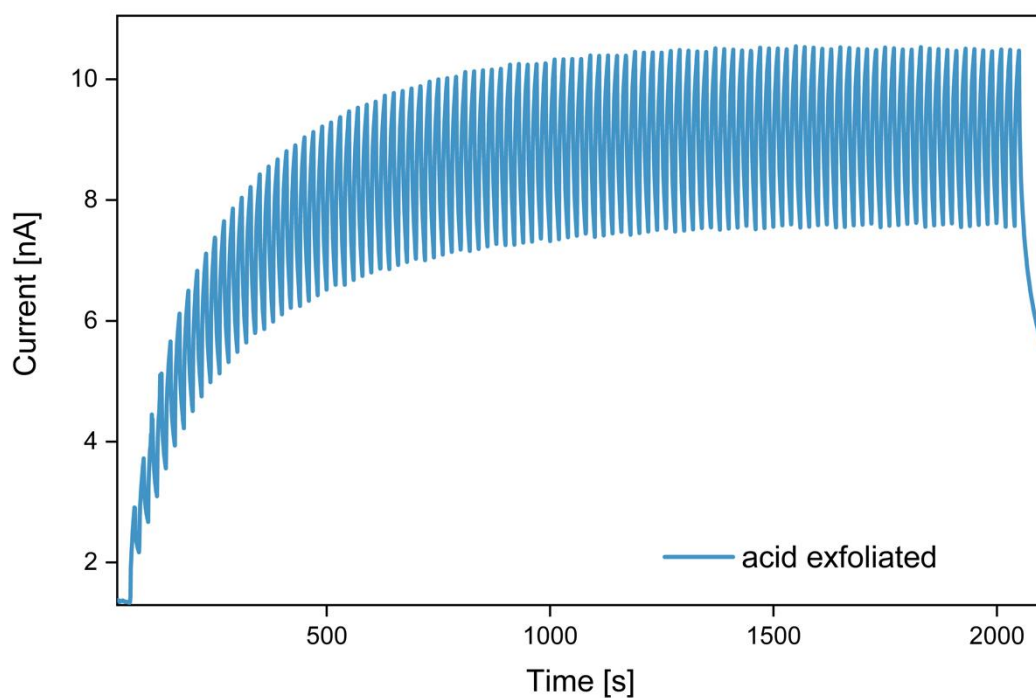


Figure S59 Conductivity of acid exfoliated **PhDBP-TFP** COF film at 20 V bias, through 100 cycles of irradiation with a 375 nm LED (10 s on, 10 s off).

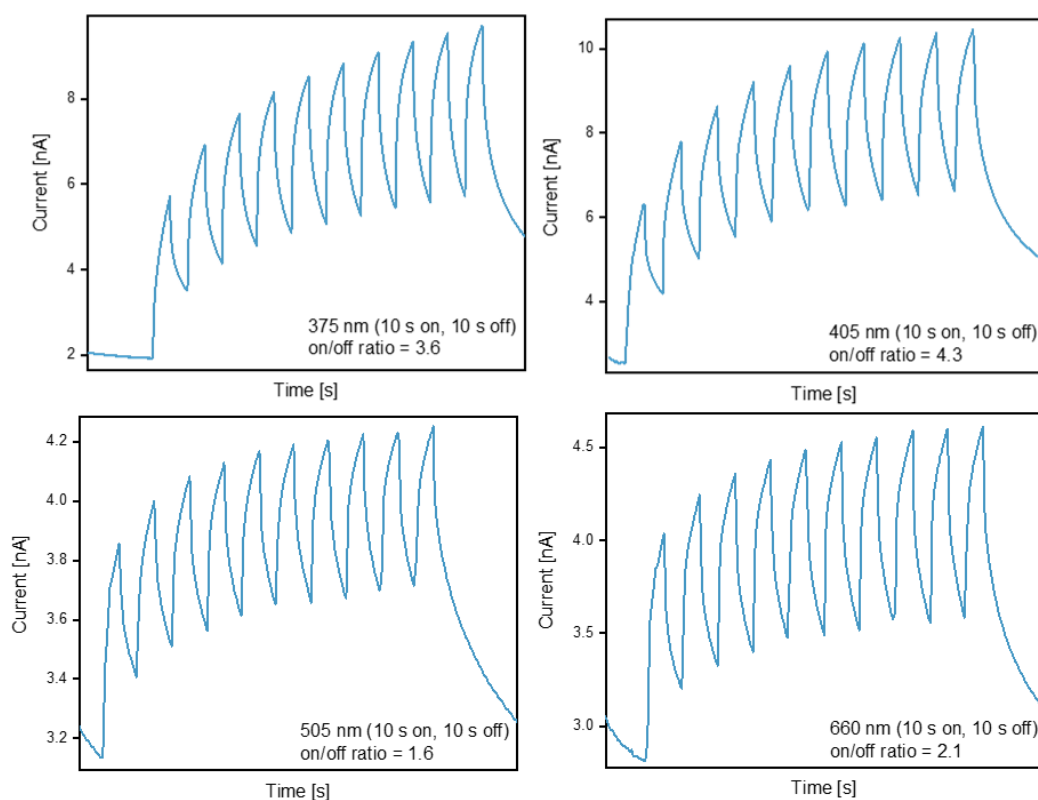


Figure S60 Conductivity of acid exfoliated **PhDBP-TFP** COF film at 20 V bias, through 10 cycles of irradiation with different wavelength LEDs (10 s on, 10 s off)

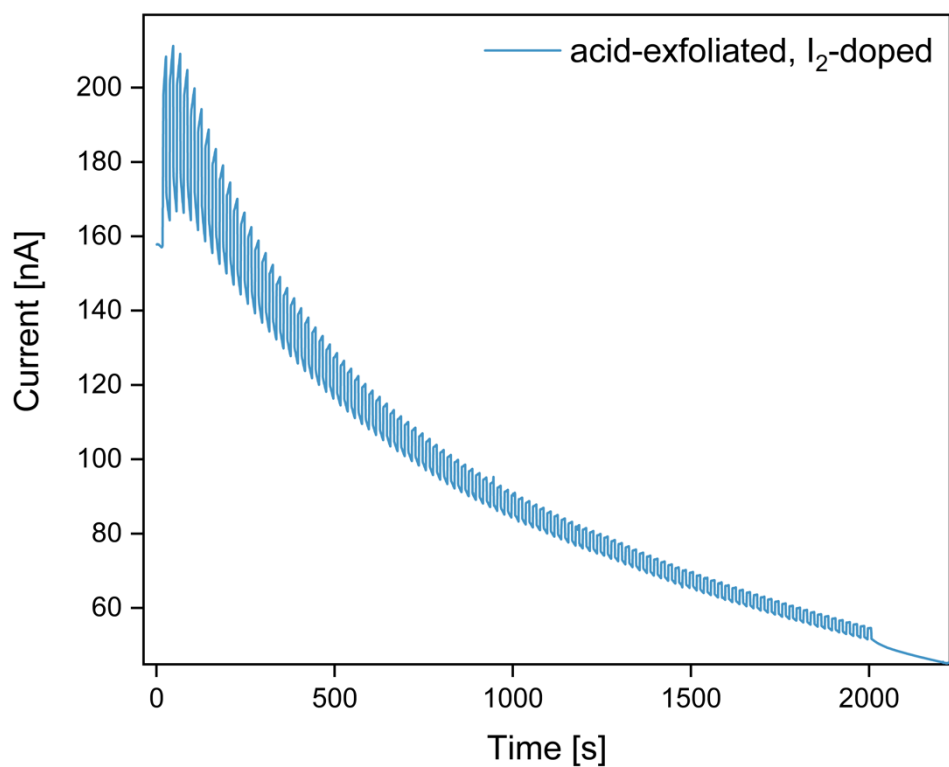


Figure S61 Conductivity of acid exfoliated I_2 -doped **PhDBP-TFP** COF film at 20 V bias, through 100 cycles of irradiation with a 375 nm LED (10 s on, 10 s off).

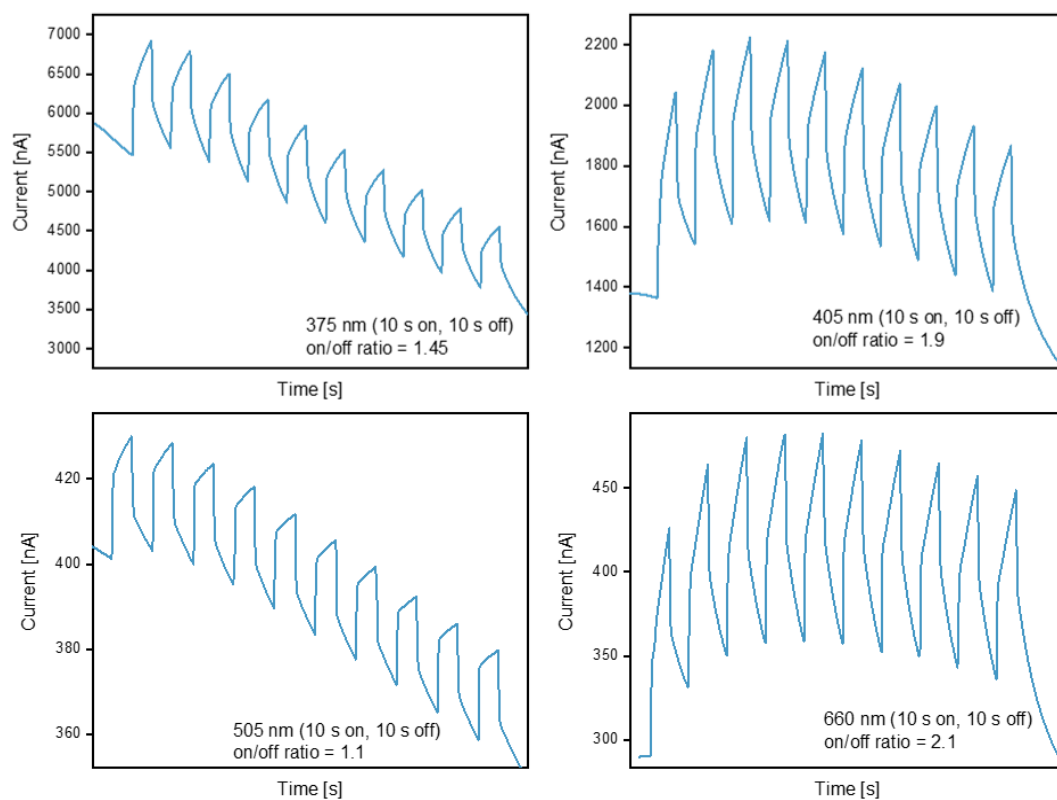


Figure S62 Conductivity of acid exfoliated I_2 -doped **PhDBP-TFP** COF film at 20 V bias, through 10 cycles of irradiation with different wavelength LEDs (10 s on, 10 s off)

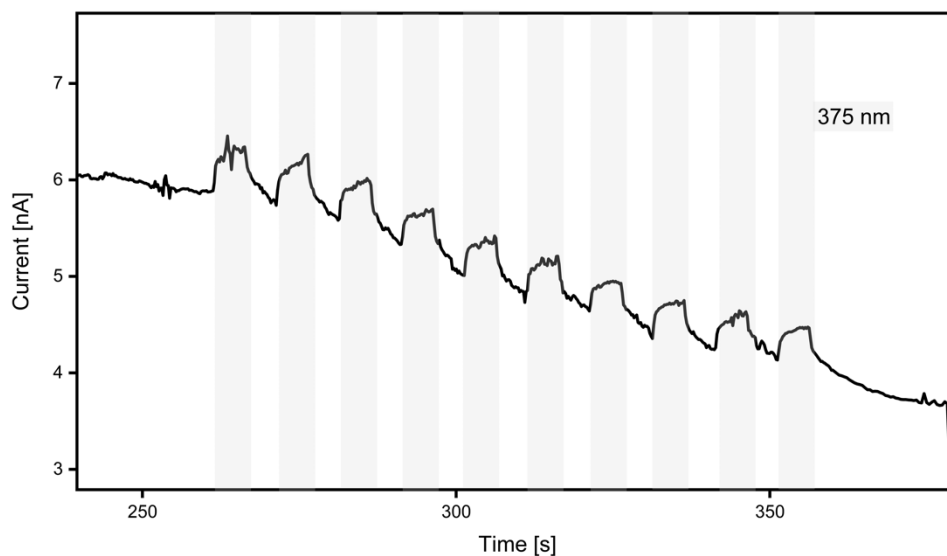


Figure S63 Conductivity of acid-exfoliated **BND-TFB** COF film at 20 V bias through 10 cycles of irradiation with a 375 nm LED (10 s on, 10 s off). On/off ratio: 1.07

Table S9 Recent all-organic functional framework materials and their conductivity

Material	Conductivity [S cm^{-1}]	Reference
PhDBP-TFP COF	1×10^{-12}	This work
I2@PhDBP-TFP COF	4×10^{-8}	This work
AntTTH	3×10^{-8}	20
PTI-LiBr	4×10^{-10}	21
3D ep-POP	8×10^{-10}	22
3D p-POP	5×10^{-8}	22
I2@3D p-POP	6×10^{-4}	22
TANG-COF	$5 \times 10^{-12} / 2 \times 10^{-5}$	23
I2@TANG-COF	2×10^{-3}	23
I2@JUC-518	3×10^{-4}	24
sp ² c-COF	6×10^{-16}	25
I2@sp ² c-COF	7×10^{-4}	25
TTF-COF	1×10^{-6}	26
I2@TTF-COF	3×10^{-3}	26
I2@TTF-COF	2×10^{-6}	27

S16. Cyclic Voltammetry of Dibenzopentalene Monomers

Cyclic Voltammetry of the DBP monomers **2**, **3**, **4**, **5**, and **S4** show reversible and non-reversible oxidation and reduction waves. Substitution of the 5,10-positions of the DBP core with phenyl groups generally leads to more reversible redox events compared to methyl-substitution. This is in line with the increased steric and electronic stabilization that phenyl substituents offer over methyl substituents to the reactive 5- and 10-position of the DBP. Due to low solubility CV measurements of **7** were not achieved.

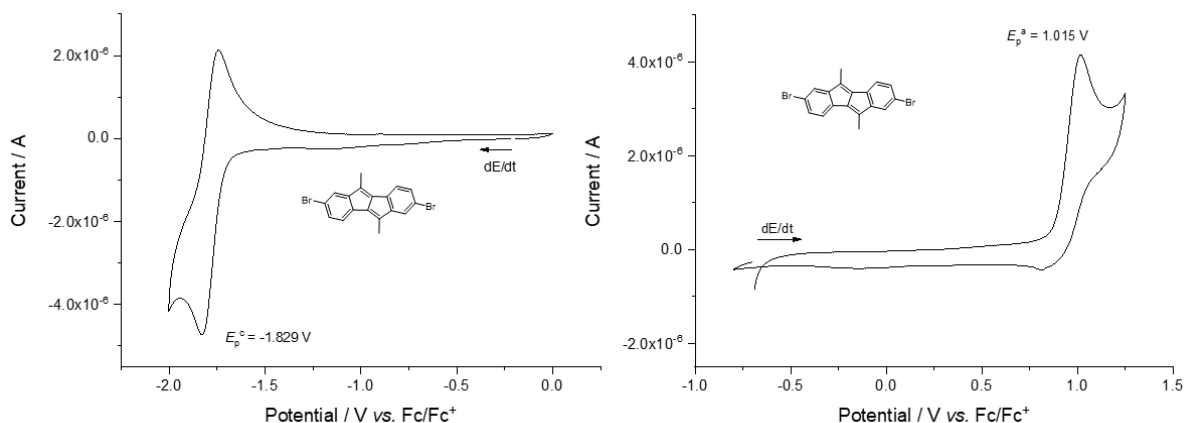


Figure S64 Cyclic Voltammograms of **2** in CH_2Cl_2 with 1 M Bu_4NPF_6 , scan rate $dE/dt = 100 \text{ mV s}^{-1}$, $T = 295 \text{ K}$. The reduction wave is reversible, while the oxidation wave shows nearly no reversibility.

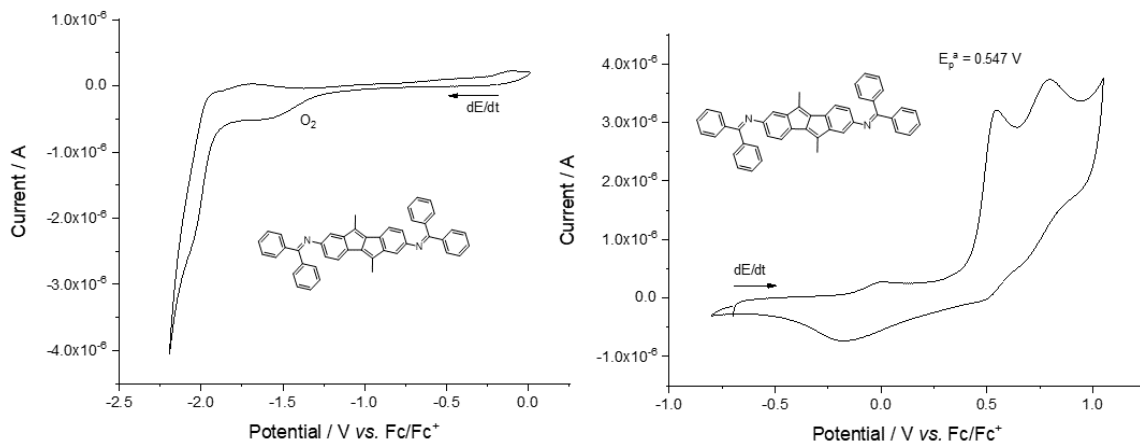


Figure S65 Cyclic Voltammograms of **3** in CH_2Cl_2 with 1 M Bu_4NPF_6 , scan rate $dE/dt = 100 \text{ mV s}^{-1}$, $T = 295 \text{ K}$. The two oxidation waves show low reversibility, no clear reduction wave is visible.

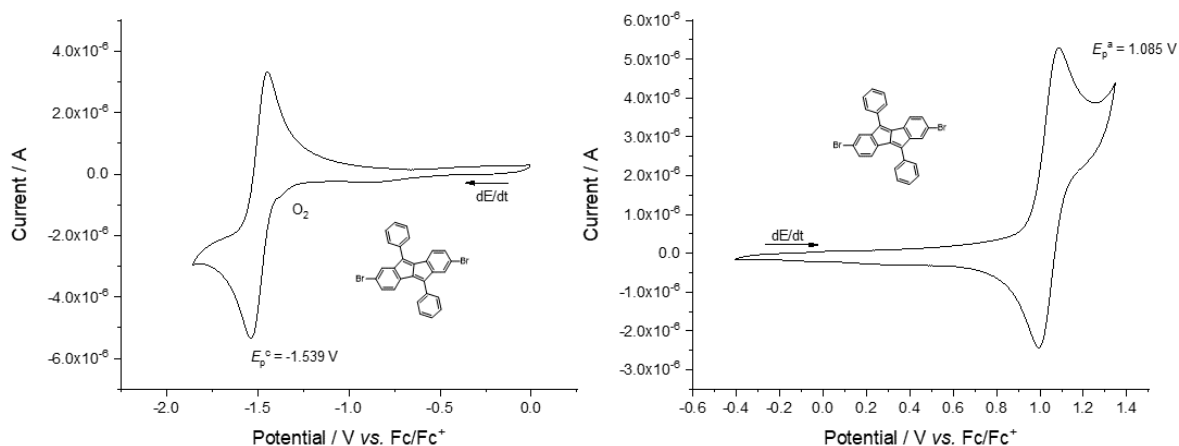


Figure S66 Cyclic Voltammograms of **4** in CH_2Cl_2 with 1 M Bu_4NPF_6 , scan rate $dE/dt = 100 \text{ mV s}^{-1}$, $T = 295 \text{ K}$. Reduction and oxidation wave are reversible in contrast to the CVs of Me-substituted **2**.

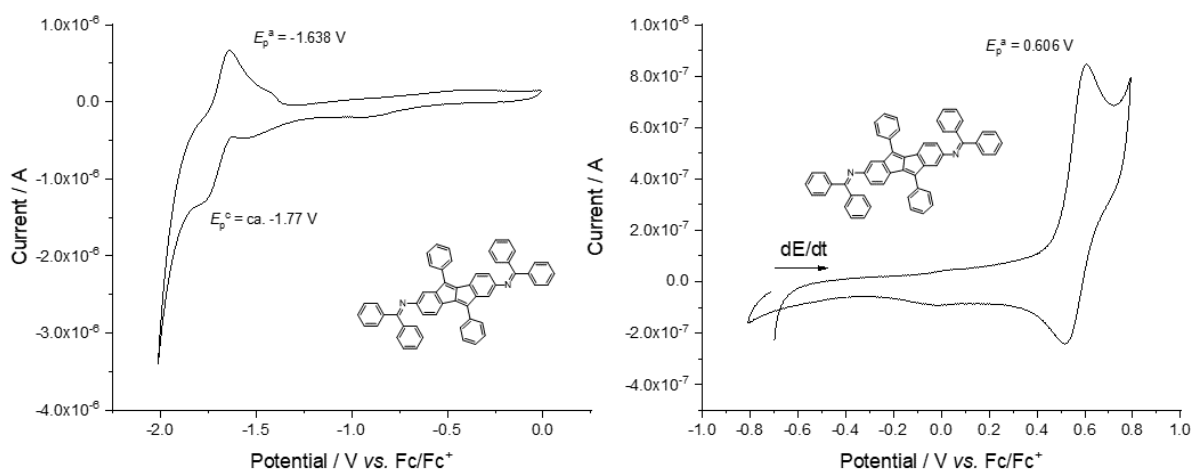


Figure S67 Cyclic Voltammograms of **5** in CH_2Cl_2 with 1 M Bu_4NPF_6 , scan rate $dE/dt = 100 \text{ mV s}^{-1}$, $T = 295 \text{ K}$. Reduction and oxidation wave are quasi-reversible in contrast to the CVs of Me-substituted **3**.

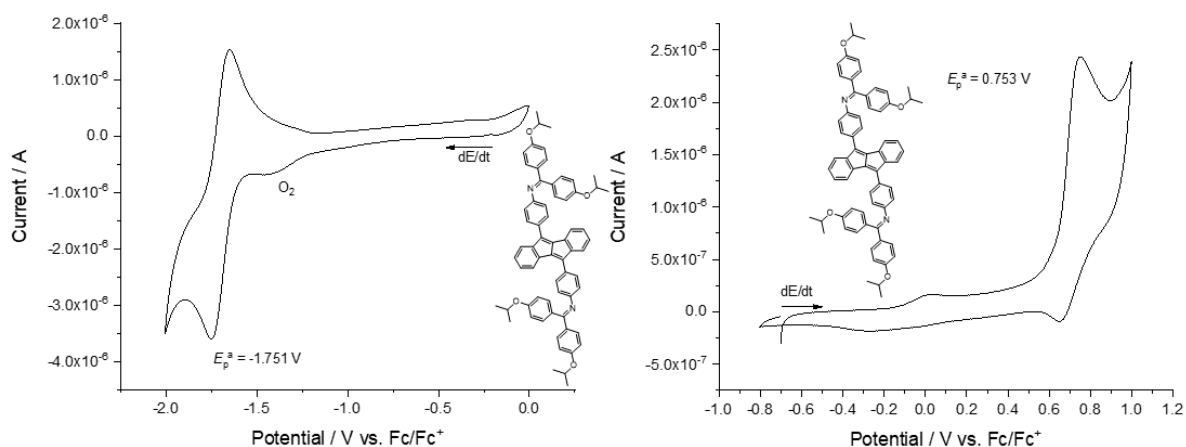


Figure S68 Cyclic Voltammograms of **5** in CH_2Cl_2 with 1 M Bu_4NPF_6 , scan rate $dE/dt = 100 \text{ mV s}^{-1}$, $T = 295 \text{ K}$. Reduction and oxidation wave are quasi-reversible.

S17. Single-Crystal X-Ray Data

Single-Crystal X-Ray Data for 2,7-Dibromo-5,10-dimethylindeno[2,1-a]indene (**2**)

Red needles were grown by slow evaporation from a CH₂Cl₂ solution containing Bu₄NPF₆ (cyclic voltammetry solution).

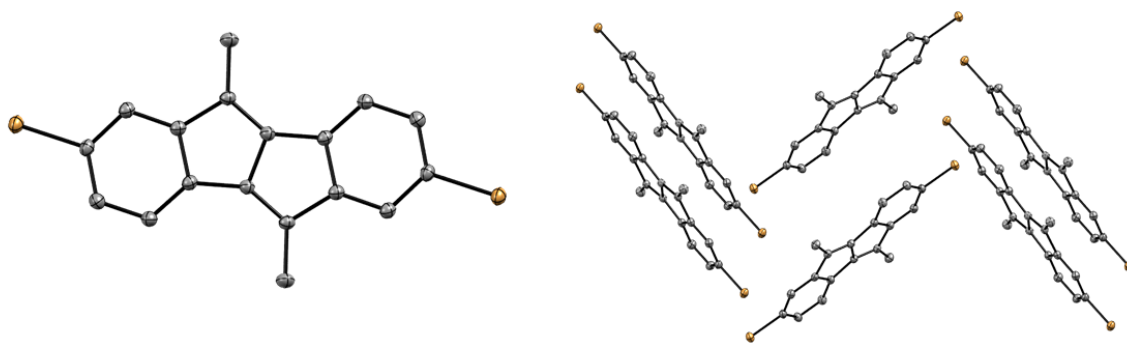


Figure S69 X-Ray structure of **2**. Ellipsoids are shown at 50% probability level, hydrogens are omitted for clarity. Color code: Carbon: grey, bromine: orange.

Table S10 Crystal Data and structure refinement for 2,7-dibromo-5,10-dimethylindeno[2,1-a]indene (**2**)

CCDC number	2082679
Empirical formula	C ₁₈ H ₁₂ Br ₂
Formula weight	388.10
Temperature/K	100.0
Crystal system	monoclinic
Space group	<i>P</i> 2 ₁ / <i>n</i>
<i>a</i> /Å	5.3259(6)
<i>b</i> /Å	17.2628(19)
<i>c</i> /Å	7.6698(8)
α /°	90
β /°	96.048(4)
γ /°	90
Volume/Å ³	701.24(13)
<i>Z</i>	2
ρ_{calc} /g/cm ³	1.838
μ /mm ⁻¹	5.765
<i>F</i> (000)	380.0
Crystal size/mm ³	0.668 × 0.33 × 0.27
Radiation	MoK α (λ = 0.71073)
2 θ range for data collection/°	5.84 to 63.316
Index ranges	−7 ≤ <i>h</i> ≤ 7, −25 ≤ <i>k</i> ≤ 25, −11 ≤ <i>l</i> ≤ 11
Reflections collected	31463
Independent reflections	2341 [<i>R</i> _{int} = 0.0655, <i>R</i> _{sigma} = 0.0222]
Data/restraints/parameters	2341/0/92
Goodness-of-fit on <i>F</i> ²	1.119
Final <i>R</i> indexes [<i>I</i> ≥ 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.0259, <i>wR</i> ₂ = 0.0717
Final <i>R</i> indexes [all data]	<i>R</i> ₁ = 0.0280, <i>wR</i> ₂ = 0.0731
Largest diff. peak/hole / e Å ⁻³	1.01/−0.83

Single-Crystal X-Ray Data for 2,7-Dibromo-5,10-diphenylindeno[2,1-a]indene (4)

Red needles were grown by slow evaporation from a CH₂Cl₂ solution containing Bu₄NPF₆ (cyclic voltammetry solution).

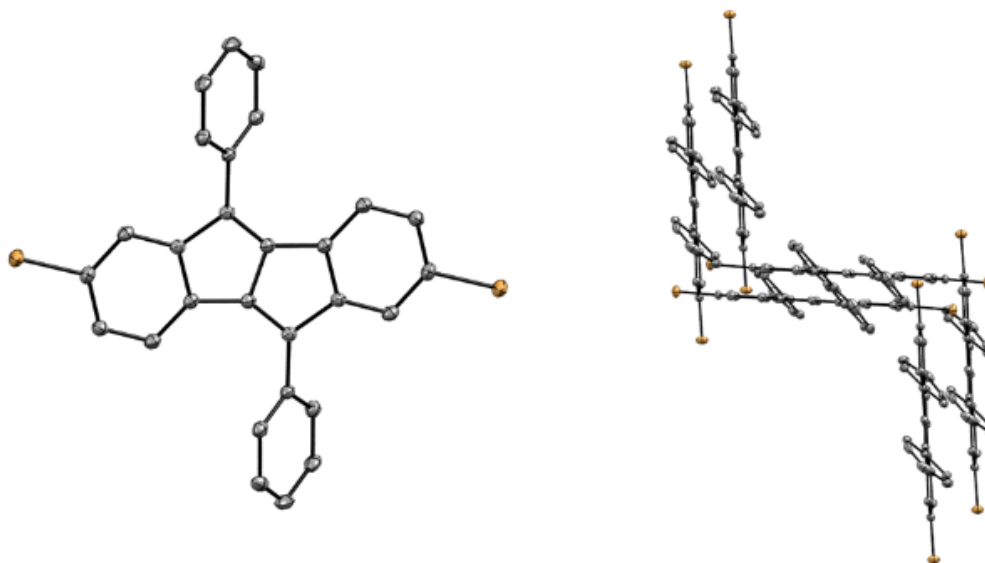


Figure S70 X-Ray structure of **4**. Ellipsoids are shown at 50% probability level, hydrogens and disorder are omitted for clarity. Color code: Carbon: grey, bromine: orange.

Table S11 Crystal Data and structure refinement for 2,7-dibromo-5,10-diphenylindeno[2,1-a]indene (**4**)

CCDC Number	2098056
Empirical formula	C ₂₈ H ₁₆ Br ₂
Formula weight	512.23
Temperature/K	100.0
Crystal system	monoclinic
Space group	C2/c
<i>a</i> /Å	28.786(3)
<i>b</i> /Å	4.9066(6)
<i>c</i> /Å	19.352(2)
α /°	90
β /°	131.807(3)
γ /°	90
Volume/Å ³	2037.4(4)
<i>Z</i>	4
ρ_{calc} / g/cm ³	1.670
μ /mm ⁻¹	3.992
<i>F</i> (000)	1016.0
Crystal size/mm ³	0.453 × 0.443 × 0.146
Radiation	MoK α (λ = 0.71073)
2 θ range for data collection/°	3.796 to 56.678
Index ranges	−38 ≤ <i>h</i> ≤ 38, −6 ≤ <i>k</i> ≤ 6, −25 ≤ <i>l</i> ≤ 25
Reflections collected	46465
Independent reflections	2552 [<i>R</i> _{int} = 0.0588, <i>R</i> _{sigma} = 0.0226]
Data/restraints/parameters	2552/3/173
Goodness-of-fit on <i>F</i> ²	1.071
Final <i>R</i> indexes [<i>I</i> ≥ 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.0302, <i>wR</i> ₂ = 0.0789
Final <i>R</i> indexes [all data]	<i>R</i> ₁ = 0.0331, <i>wR</i> ₂ = 0.0804
Largest diff. peak/hole / e Å ⁻³	0.73/−0.61

Single-Crystal X-Ray Data for *N,N'*-(Indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-diphenylmethanimine) (7)

Red crystals were grown by slow evaporation of a saturated solution in chlorobenzene at 130°C.

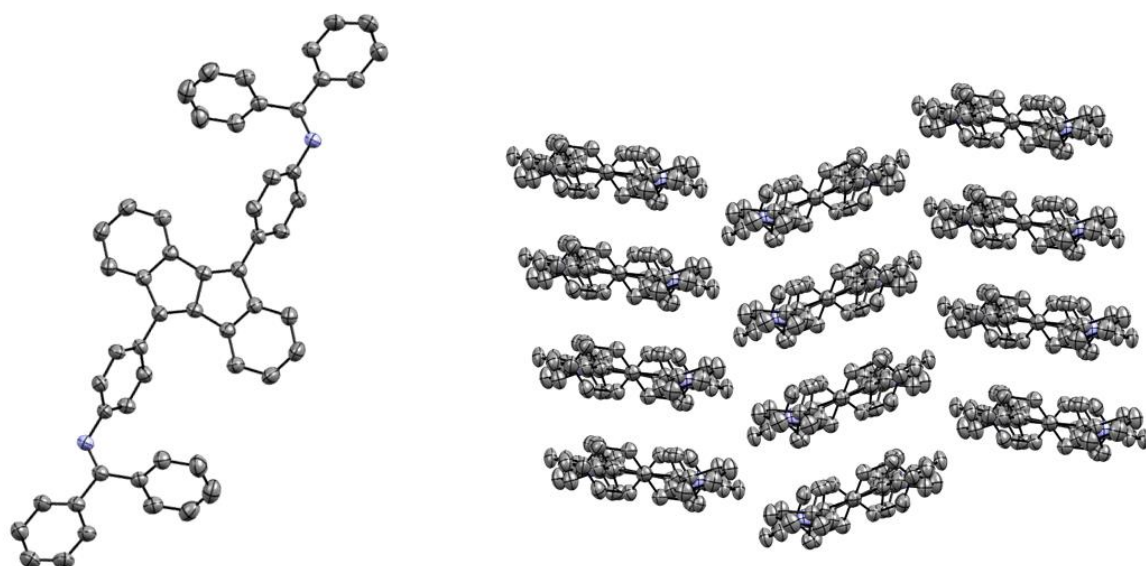


Figure S71 X-Ray structure of **7**; Ellipsoids are shown at 50% probability level, hydrogens and disorder are omitted for clarity. Color code: Carbon: grey, nitrogen: blue;

Table S12 Crystal Data and structure refinement for *N,N'*-(Indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-diphenylmethanimine) (**7**)

CCDC number	2219119
Empirical formula	C ₅₄ H ₃₆ N ₂
Formula weight	712.85
Temperature/K	100.0
Crystal system	monoclinic
Space group	<i>P</i> 2 ₁ / <i>c</i>
<i>a</i> /Å	10.896(4)
<i>b</i> /Å	18.162(6)
<i>c</i> /Å	9.738(4)
α /°	90
β /°	96.613(15)
γ /°	90
Volume/Å ³	1914.4(13)
<i>Z</i>	2
ρ_{calc} / g/cm ³	1.237
μ /mm ⁻¹	0.071
<i>F</i> (000)	748.0
Crystal size/mm ³	0.393 × 0.17 × 0.1
Radiation	MoK α (λ = 0.71073)
2 θ range for data collection/°	4.38 to 50.858
Index ranges	−13 ≤ <i>h</i> ≤ 13, −21 ≤ <i>k</i> ≤ 21, −11 ≤ <i>l</i> ≤ 11
Reflections collected	98524
Independent reflections	3500 [<i>R</i> _{int} = 0.1247, <i>R</i> _{sigma} = 0.0346]
Data/restraints/parameters	3500/0/253
Goodness-of-fit on <i>F</i> ²	1.056
Final <i>R</i> indexes [<i>I</i> ≥ 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.0639, <i>wR</i> ₂ = 0.1738
Final <i>R</i> indexes [all data]	<i>R</i> ₁ = 0.0908, <i>wR</i> ₂ = 0.1977
Largest diff. peak/hole / e Å ⁻³	0.28/−0.20

S18. DFT Calculations

NICS_{iso}(0) and NICS_{iso}(1) values were calculated using Gaussian 16 Revision A.03 on the chccs-cluster of the Department of Chemistry at Humboldt University of Berlin and at the Justus High Performance Computing Cluster (JUSTUS 2).

For molecules **3**, **5**, and **7** the structures are confirmed ground-state minima according to the analysis of their analytical frequencies computed at the same level, showing no imaginary frequencies. Geometry optimizations were conducted at the DFT²⁸/B3LYP^{29,30}/6-31G(d,p)³¹ level of theory. Isotropic NICS_{iso}(0) and NICS_{iso}(1) values³² were calculated at the DFT³³/mPW1PW91/6-311G+(2d,p)³¹ level of theory applying the *GIAO* (Gauge Including Atomic Orbitals) method. In the centers of the 5-membered rings of the dibenzopentalenes, NICS values indicate paratropicity, while the NICS values in the 6-membered rings of dibenzopentalenes indicate diatropicity. Cartesian Coordinates of the optimized structures including ghost atoms for NICS calculation can be found in section S19.

Table S13 Isotropic NICS_{iso}(0) and NICS_{iso}(1) values of DBPs **3**, **5**, and **7**.

Molecule	NICS _{iso} (0) _{5-Ring} [ppm]	NICS _{iso} (1) _{5-Ring} [ppm]	NICS _{iso} (0) _{6-Ring} [ppm]	NICS _{iso} (1) _{6-Ring} [ppm]
3	10.89	5.48	-3.19	-4.88
5	11.16	6.26	-2.52	-4.61
7	11.88	6.67	-2.85	-5.46

All further DFT calculations were performed on the Justus High Performance Computing Cluster (JUSTUS 2) with the TURBOMOLE v7.5.0 program package.³⁴ The resolution-of-identity³⁵ (RI, RIJDX for SP) approximation for the Coulomb integrals was used in all DFT calculations employing matching auxiliary basis set def2-XVP/J.²⁸ Further, the D3 dispersion correction scheme²⁸ with the Becke-Johnson damping function was applied.^{36,37} The geometries were first optimized without symmetry restrictions (in neutral oxidation state) using the PBEh-3c³⁸ composite scheme followed by harmonic vibrational frequency analysis to confirm minima as stationary points. This geometry was used as an input for further geometry optimizations in the neutral and oxidized radical cation states. They were each optimized on the B3LYP-D3/def2-TZVP level of theory.^{39,40} In all further calculations the COSMO solvation model ($\epsilon = 37.5$, acetonitrile) was used, as this models the polarity of the battery electrolyte employed. Structures were optimized with restricted orbitals for the neutral molecules and with unrestricted orbitals for the radical cation states.

PhDBP-TFP* and **DBP-Ph-TFP*** were chosen as model compounds for **PhDBP-TFP** COF and **DBP-Ph-TFP** POP, containing one DBP-unit and one third of the TFP-node. For both, the neutral and radical cation state was investigated.

NICS values of COF model fragments **PhDBP-TFP*** and **DBP-Ph-TFP*** were calculated applying the standard *GIAO* (Gauge Including Atomic Orbitals) method on the B3LYP/6-31G(d) level of theory. NICS_{zz}(1.7) values: To minimize the influence of the ring current from the sigma scaffold, the dummy atoms were placed 1.7 Å above the centers of the five- and six-membered rings, and the zz-component of the shielding tensor was used and the values for both rings of the same size were averaged.⁴¹

Negative NICS values indicate a diatropic ring current (associated with an aromatic character) while positive NICS values indicate a paratropic ring current (associated with antiaromatic character).

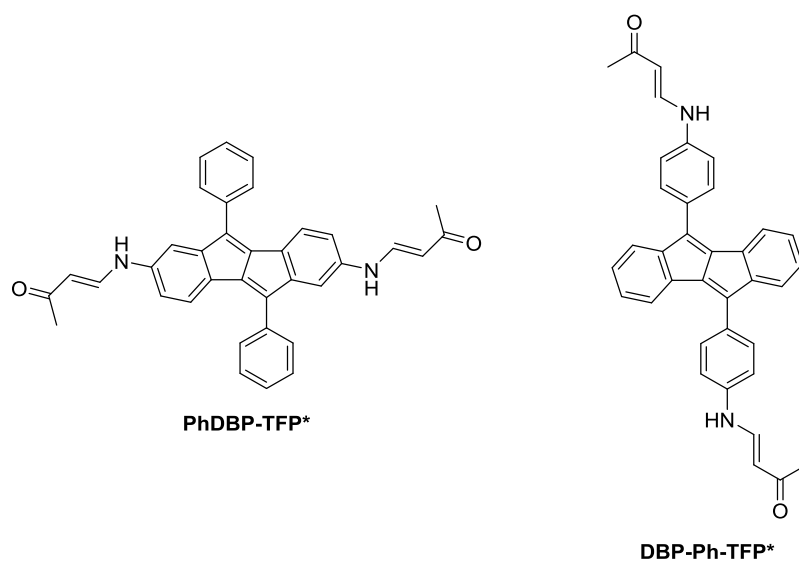


Figure S72 Subunits of the **PhDBP-TFP** COF and **DBP-Ph-TFP** POP used for the DFT calculations.

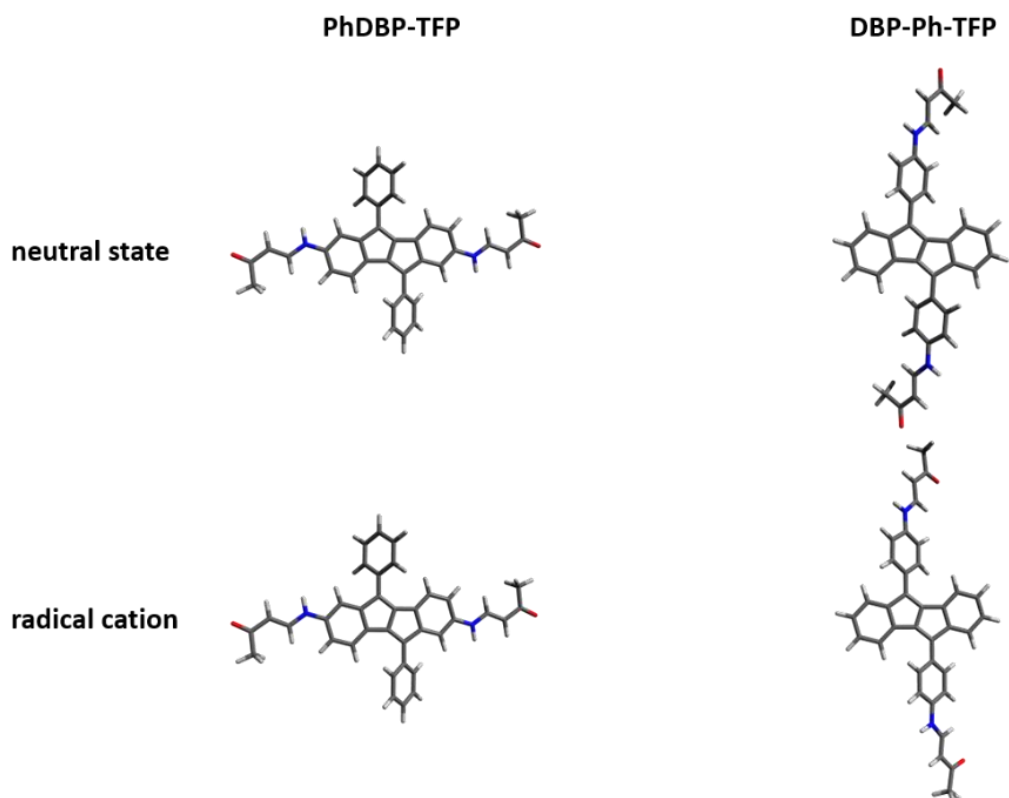


Figure S73 Optimized structures of **PhDBP-TFP*** and **DBP-Ph-TFP*** in the neutral or radical cation state (B3LYP-D3/def2-TZVP+COSMO).

Calculations of Frontier Molecular Orbitals, Spin Densities, and Electrostatic Potentials Maps

Single point energies were calculated on the PW6B95-D3/def2-QZVP+COSMO⁴² level of theory on the optimized structures for the different oxidation states. Molecular orbitals were plotted using ChimeraX,⁴³ spin densities and electrostatic potential maps were plotted using gOpenMol.

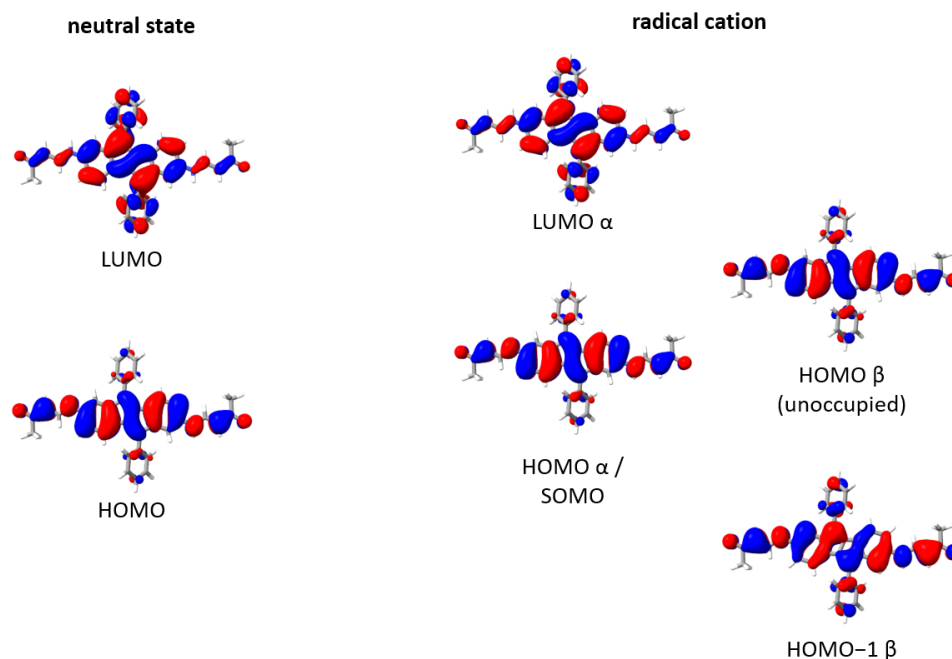


Figure S74 Frontier molecular orbitals of **PhDBP-TFP*** in the neutral and radical cation state (PW6B95-D3/def2-QZVP+COSMO).

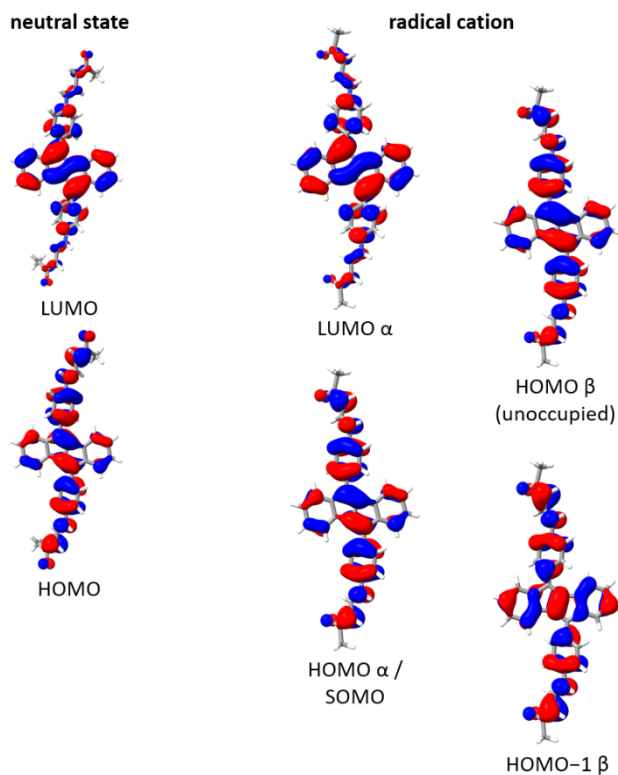


Figure S75 Frontier molecular orbitals of **DBP-Ph-TFP*** in the neutral and radical cation state (PW6B95-D3/def2-QZVP+COSMO).

S19. Electrode Preparation and Electrochemical Measurements

The electrodes were prepared by mixing 30 wt% of active material with 50 wt% of acetylene black and 20 wt% of polyvinylidene fluoride (PVdF), using *N*-methyl-2-pyrrolidone (NMP) as solvent. The acetylene black electrodes were prepared by mixing 50 wt% of acetylene black and 50 wt% of PVdF binder and NMP used as a solvent. Then the slurry was coated on Al foil or Cu foil using a doctor blade and electrodes were dried overnight under vacuum at 80 °C. 2032 Coin-type cells were assembled in Ar-filled glove box with water and oxygen level under 0.5 ppm. Lithium was used as the counter electrode, a glass fiber membrane (Whatman, GF/D) was used as separator and 1 M LiPF₆ in ethylene carbonate and dimethyl carbonate (1:1) was used as electrolyte. Cyclic voltammetry and galvanostatic experiments were performed at different current densities with a BioLogic MPG 2 potentiostat. Electrochemical Impedance Spectroscopy (EIS) measurements were carried out on a BioLogic MPG 2 potentiostat applying a signal amplitude of 10 mV in the frequency range of 0.02 MHz to 10 MHz.

Investigations as positive electrode materials are described in the main text. Using **PhDBP-TFP** COF and **DBP-Ph-TFP** POP as negative electrode materials in Li-based half cells, the CVs in the potential range of 3.0–0.05 V vs. Li/Li⁺ showed a peak in cathodic scan direction with a peak potential of ca. 1.0 V, a sharp current increase towards 0.05 V, and two peak potentials in cathodic scan direction at ca. 0.3 and 1.0 V in the second cycle (Figure S76a). In the first cycle, both materials showed irreversible peaks in cathodic scan direction at 1.66 and 0.64 V vs. Li/Li⁺, the latter related to the formation of a solid–electrolyte interphase.⁴⁴ The charge/discharge curves from constant current cycling measurements confirm that the Li-ion intercalation occurs between 0.05–1.2 V and furnishes specific capacities of 309 mAh g⁻¹ for the second cycle for both materials (Figure S76b). Cycling at 100 mA g⁻¹ for 100 cycles interestingly showed a higher cycling stability for the **DBP-Ph-TFP** POP than for the **PhDBP-TFP** COF with a capacity retention of 80% vs. 62%, respectively (Figure S76c). Since the reduction of the DBP unit is expected to occur at ca. 1.6 V vs. Li/Li⁺ (based on the reported reduction potential of 1,5-diphenyl-DBP of –1.68 V vs. Fc/Fc⁺⁴⁵ and assuming 3.25 V for Fc/Fc⁺ vs. Li/Li⁺⁴⁶), the electrochemical signature of the DBP-COF/POP resembles more that of carbon materials. Indeed, electrodes only consisting of acetylene black (and binder) showed a CV profile similar to those in Figure S76b (Figure S77). However, the specific capacities were higher with **PhDBP-TFP** COF and **DBP-Ph-TFP** POP than those with acetylene black alone (177 mAh g⁻¹ for the second cycle).

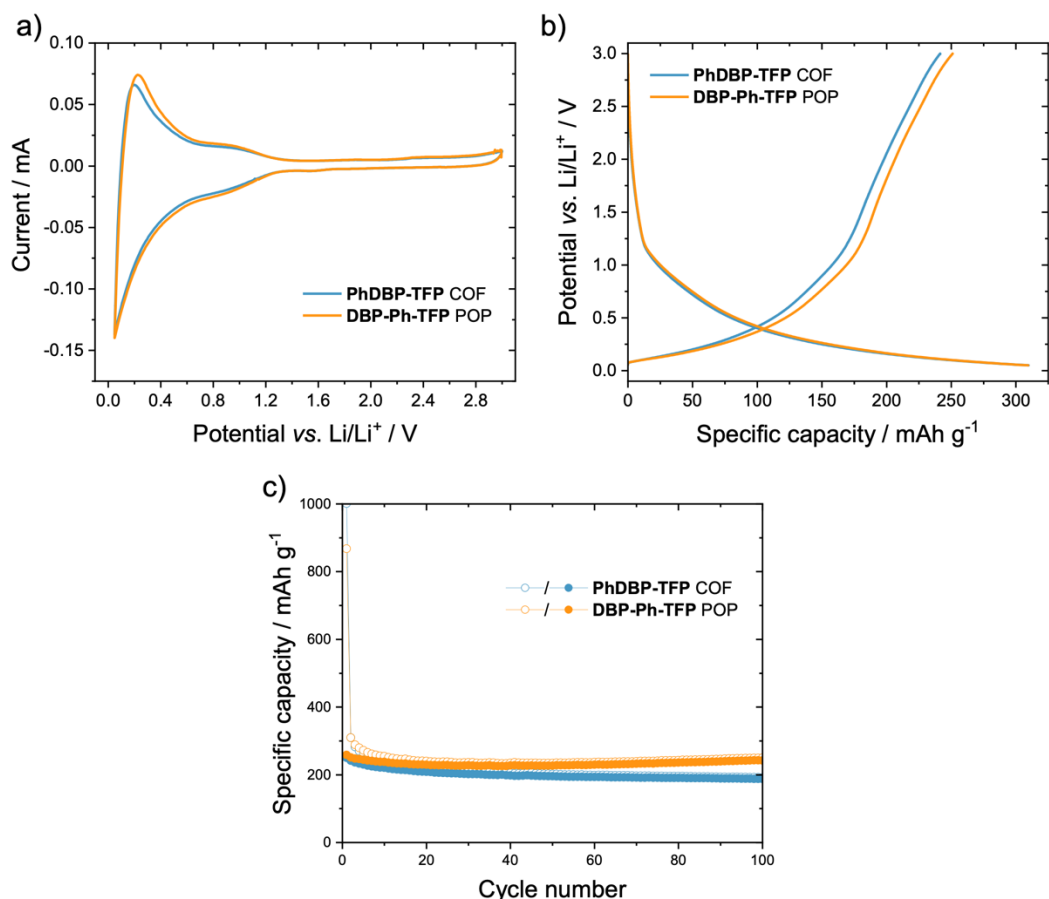


Figure S76 Electrochemical performance of **PhDBP-TFP COF** and **DBP-Ph-TFP POP** in composite electrodes (COF or POP/acetylene black/PVdF binder (30:50:20 wt%), 1 M LiPF₆ in EC/DMC (1:1), counter electrode: Li foil), as negative electrode materials on Cu foil. a) Cyclic voltammograms (2nd cycle, 0.2 mV s⁻¹); b) Charge/discharge curves during constant current cycling at 100 mA g⁻¹, 2nd cycle; c) Cycling performance at 100 mA g⁻¹ (solid symbols represent the charge capacity, and open symbols represent the discharge capacity).

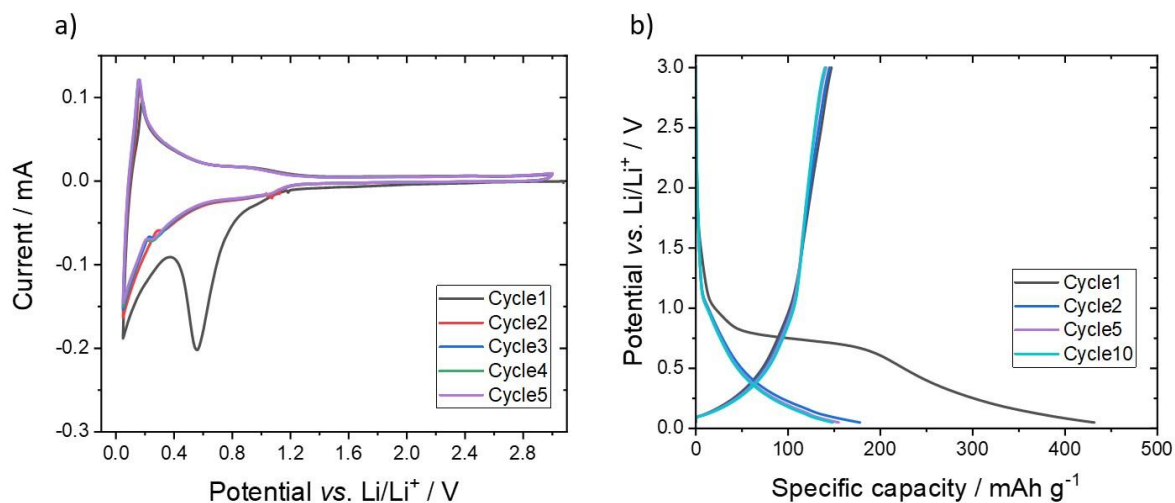


Figure S77 a) Cyclic Voltammogram, scan rate 0.2 mV s⁻¹, and b) charge/discharge curves during constant current cycling at 100 mA g⁻¹ of **acetylene black electrodes** between 3 and 0.05 V vs. Li/Li⁺ (acetylene black/PVdF binder (50:50 wt%) on Cu foil, 1 M LiPF₆ in EC/DMC (1:1), counter electrode: Li foil).

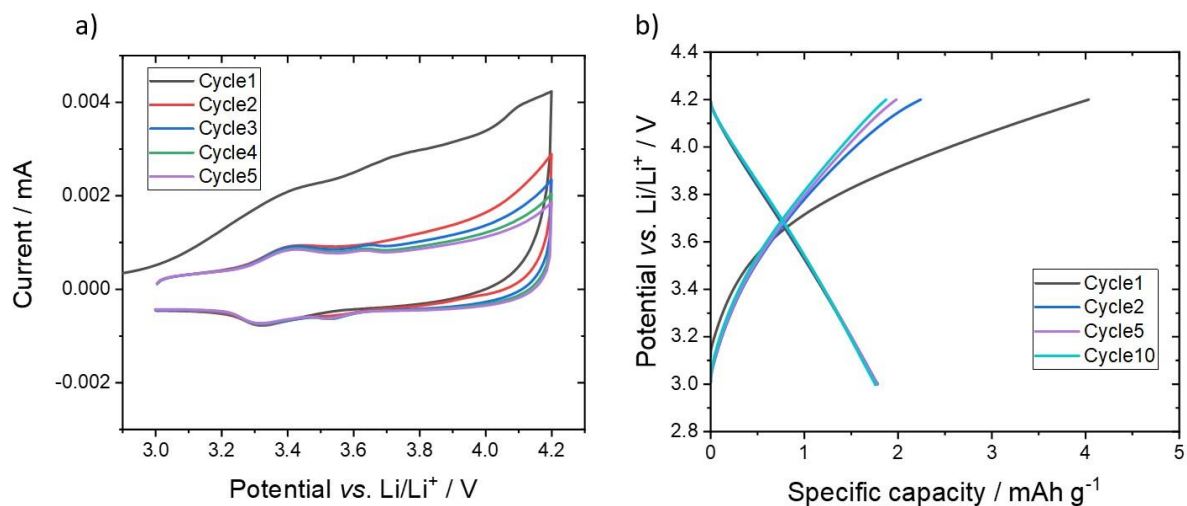


Figure S78 a) Cyclic Voltammogram, scan rate 0.2 mV s^{-1} , and b) charge/discharge curves during constant current cycling at 100 mA g^{-1} of **acetylene black electrodes** between 3 and 4.2 V vs. Li/Li^+ (acetylene black/PVdF binder (50:50 wt%) on Al foil, 1 M LiPF_6 in EC/DMC (1:1), counter electrode: Li foil).

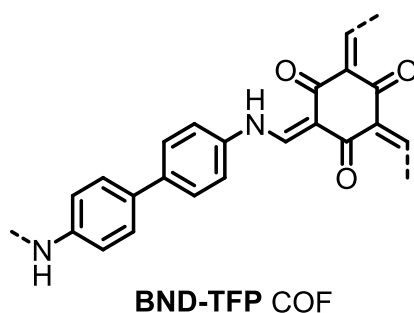


Figure S79 Structure of **BND-TFP COF** (also referred to as TpBD COF)⁴⁷ used as reference in electrochemical measurements, prepared according to literature.^{12,19}

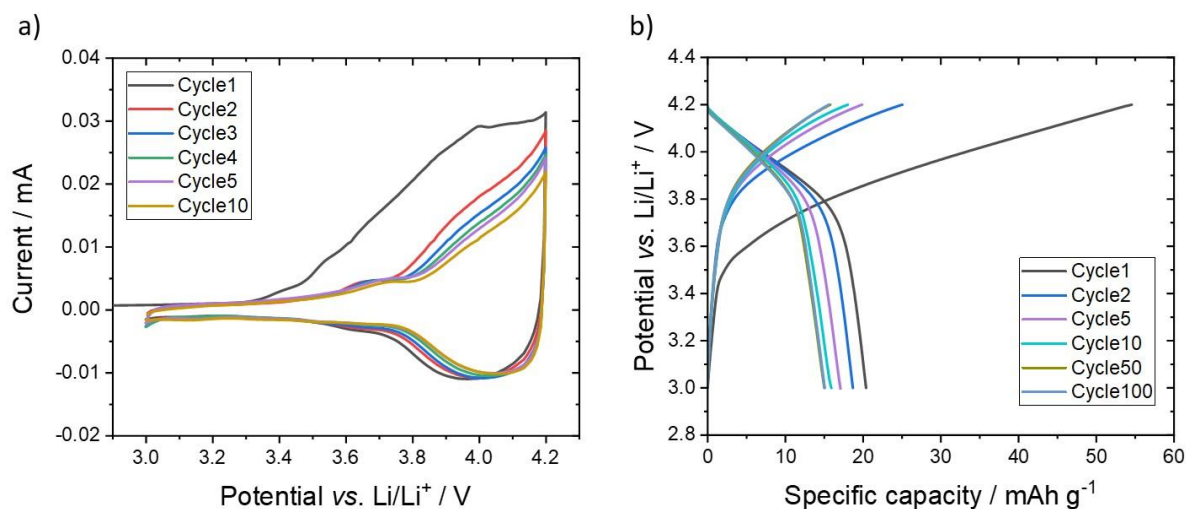
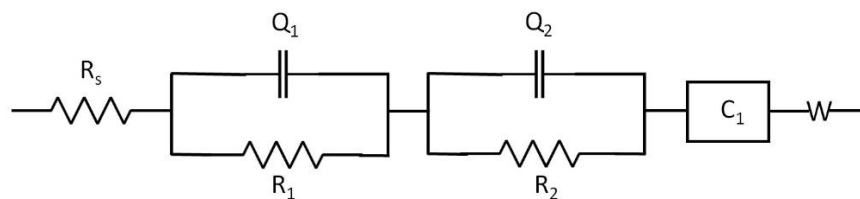


Figure S80 a) Cyclic Voltammogram, scan rate 0.2 mV s^{-1} , and b) charge/discharge curves during constant current cycling at 100 mA g^{-1} of **BND-TFP COF** electrodes between 3 and 4.2 V vs. Li/Li^+ (COF/acetylene black/PVdF binder (30:50:20 wt%), 1 M LiPF_6 in EC/DMC (1:1), counter electrode: Li foil).

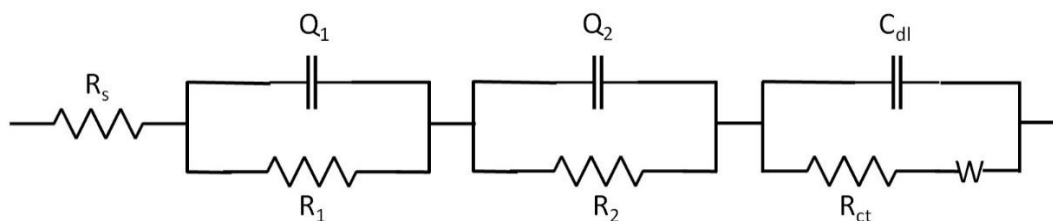
Electrochemical impedance spectroscopy (EIS) measurements were performed on the **PhDBP-TFP** COF and **DBP-Ph-TFP** POP in different states of charge.

The equivalent circuits of cells at OCP and charged/discharged states and the simulation result after Z-fitting are shown below. R_s is the electrolyte resistance, R_1 and R_2 are the resistances arising from mesoporous structure of the COF/POP electrode and Q_1 and Q_2 are the constant phase elements. C_1 is the double layer capacitance from the as-prepared mesoporous structure, R_{ct} and C_{dl} are the charge-transfer resistance and the double layer capacitance after the charge/discharge process, respectively, and the W indicates the Warburg ion diffusion inside the porous COF and POP. The unit of all resistances is Ohm.

At OCP



At Charged/discharged state



At the open circuit potential, the Nyquist plots show a single semicircle at a high frequency region and a linear slope at low frequency region, indicating the resistances of the COF and the diffusion of ions inside the mesopore (Warburg impedance) respectively. After the 6th charge cycle, a new semicircle is formed in the low frequency region due to the charge transfer resistance, amounting to 21.82 Ohm for **PhDBP-TFP** COF and 16.31 Ohm for the **DBP-Ph-TFP** POP. This could be related to a decomposition of part of the COF and POP during the charging process.⁴⁸ The charge transfer resistances decrease to 14.72 Ohm for **PhDBP-TFP** COF and 9.66 Ohm for the **DBP-Ph-TFP** POP during the discharge.

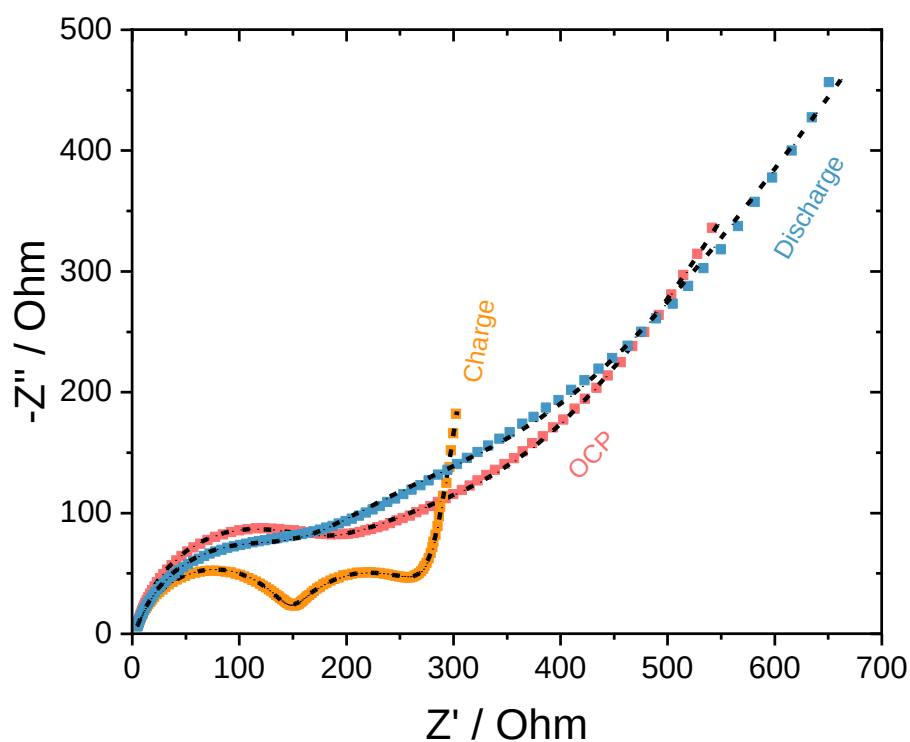


Figure S81 Nyquist plot of **PhDBP-TFP** COF at open-circuit potential (red), after the 6th cycled charged state (orange), and discharged state (blue). The dot and dashed lines indicate the experimental and simulation data based on the equivalent circuits, respectively.

Table S14 Simulation results of **PhDBP-TFP** COF after Z-fitting

State of the PhDBP-TFP COF electrode	R_s	R_1 (Q_1)	R_2 (Q_2)	R_{ct} (C_{dl})
OCP	2.018	156.5 (0.2894×10^{-3})	138.3 (4.722×10^{-6})	NA
After 6 th charge	2.165	145 (13.52×10^{-6})	131.6 (1.865×10^{-3})	21.82 (0.04021)
After 6 th discharge	2.571	128 (11.85×10^{-6})	244 (0.2905×10^{-3})	14.72 (0.5689×10^{-3})

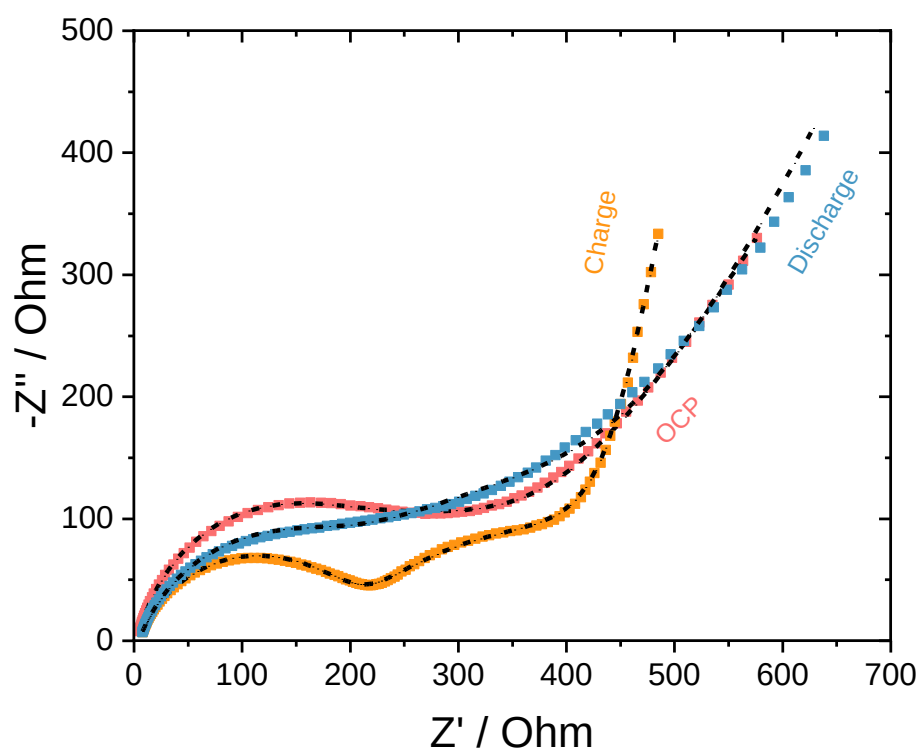


Figure S82 Nyquist plot of **DBP-Ph-TFP** POP at open-circuit potential (red), after the 6th cycled charged state (orange), and discharged state (blue). The dot and dashed lines indicate the experimental and simulation data based on the equivalent circuits, respectively.

Table S15 Simulation results of **DBP-Ph-TFP** POP after Z-fitting.

State of the DBP-Ph-TFP POP electrode	R_s	R_1 (Q_1)	R_2 (Q_2)	R_{ct} (C_{dl})
OCP	1.823	151.8 (0.1097×10^{-3})	163.3 (4.619×10^{-6})	NA
After 6 th charge	2.769	204.3 (23.62×10^{-6})	249.7 (1.645×10^{-3})	16.31 (0.01723)
After 6 th discharge	4.591	181.6 (18×10^{-6})	259.6 (0.3159×10^{-3})	9.66 (1.293×10^{-3})

S20. Selected NMR Spectra

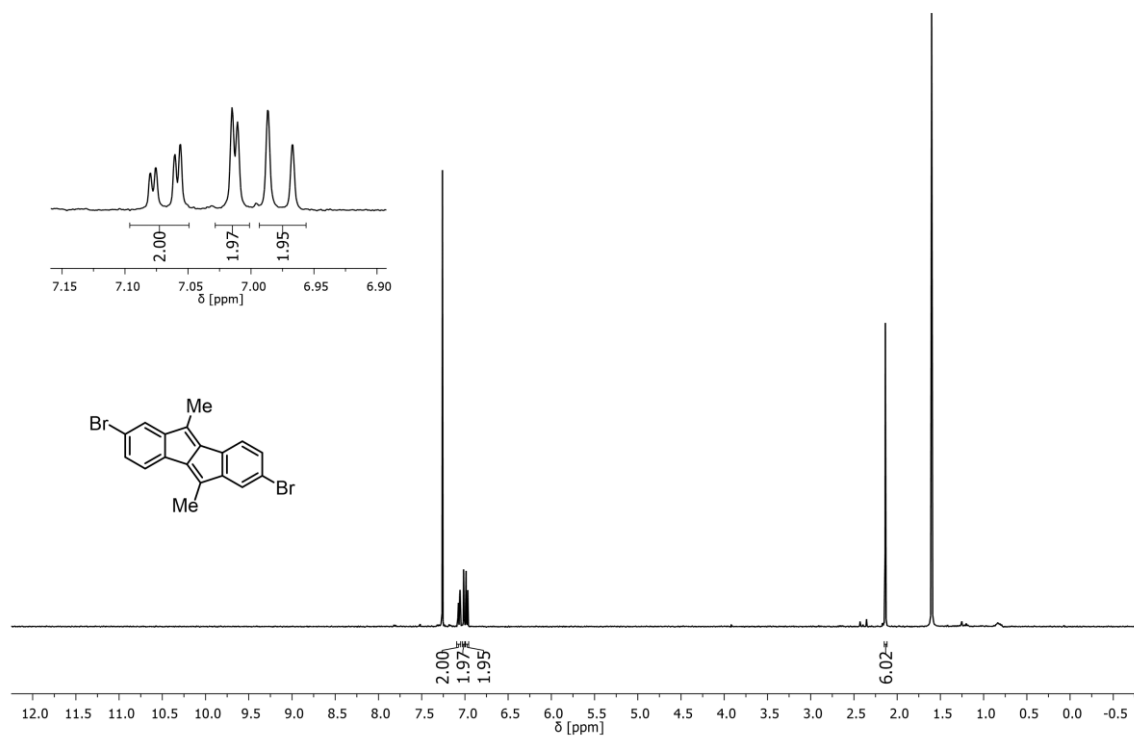


Figure S83 ¹H NMR spectrum (500 MHz) of 2,7-dibromo-5,10-dimethylindeno[2,1-a]indene **2** in CDCl₃

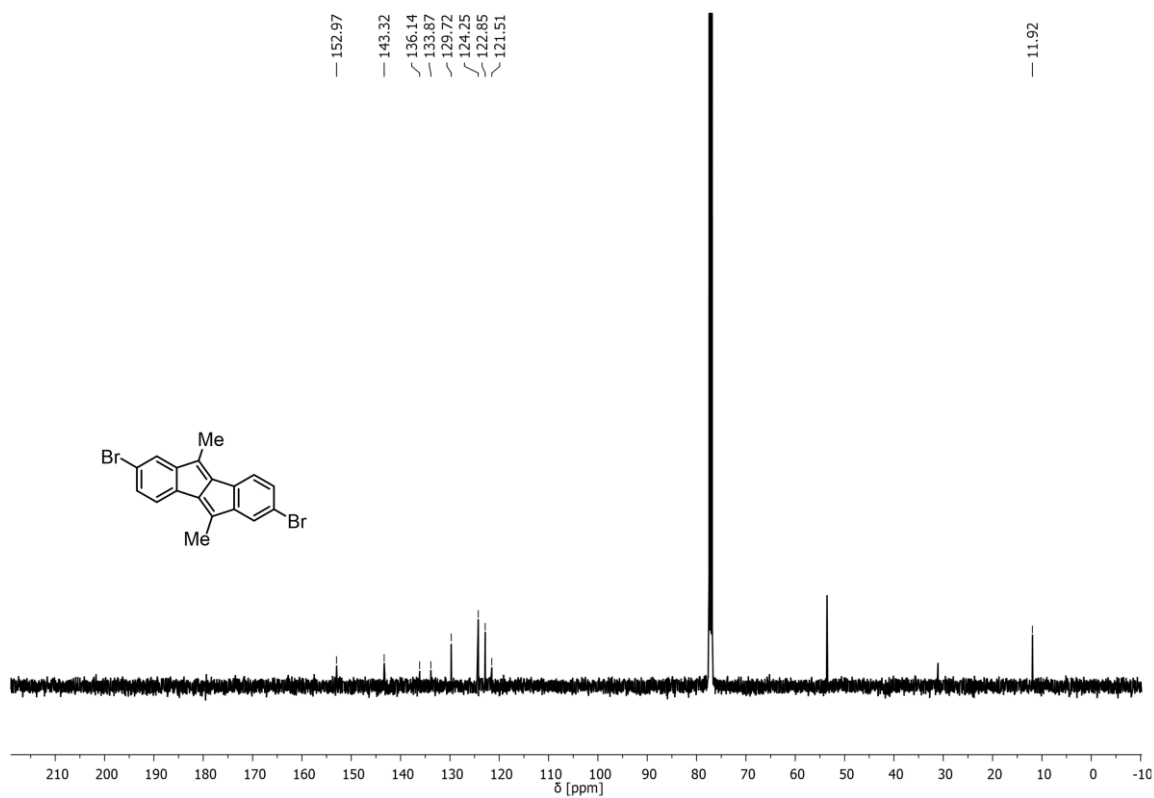


Figure S84 ¹³C NMR spectrum (126 MHz) of 2,7-dibromo-5,10-dimethylindeno[2,1-a]indene **2** in CDCl₃

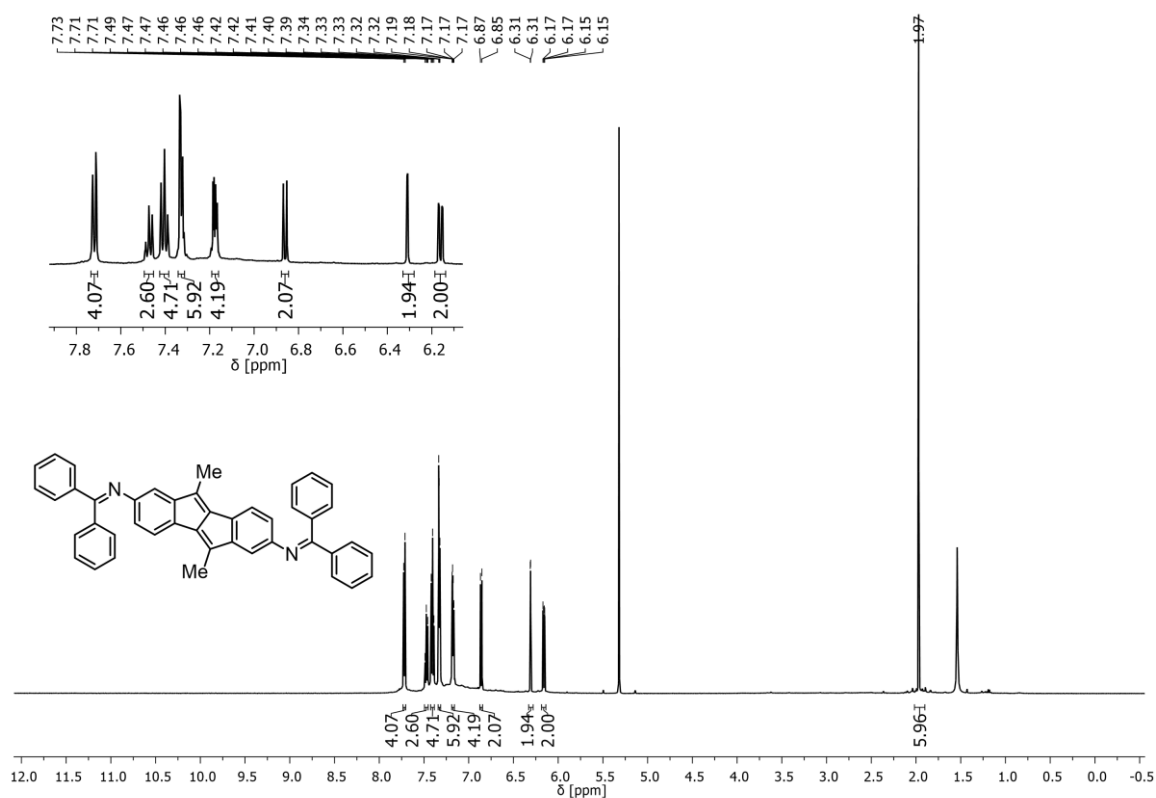


Figure S85 ¹H NMR spectrum (500 MHz) of *N,N'*-(5,10-dimethylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) **3** in CD₂Cl₂

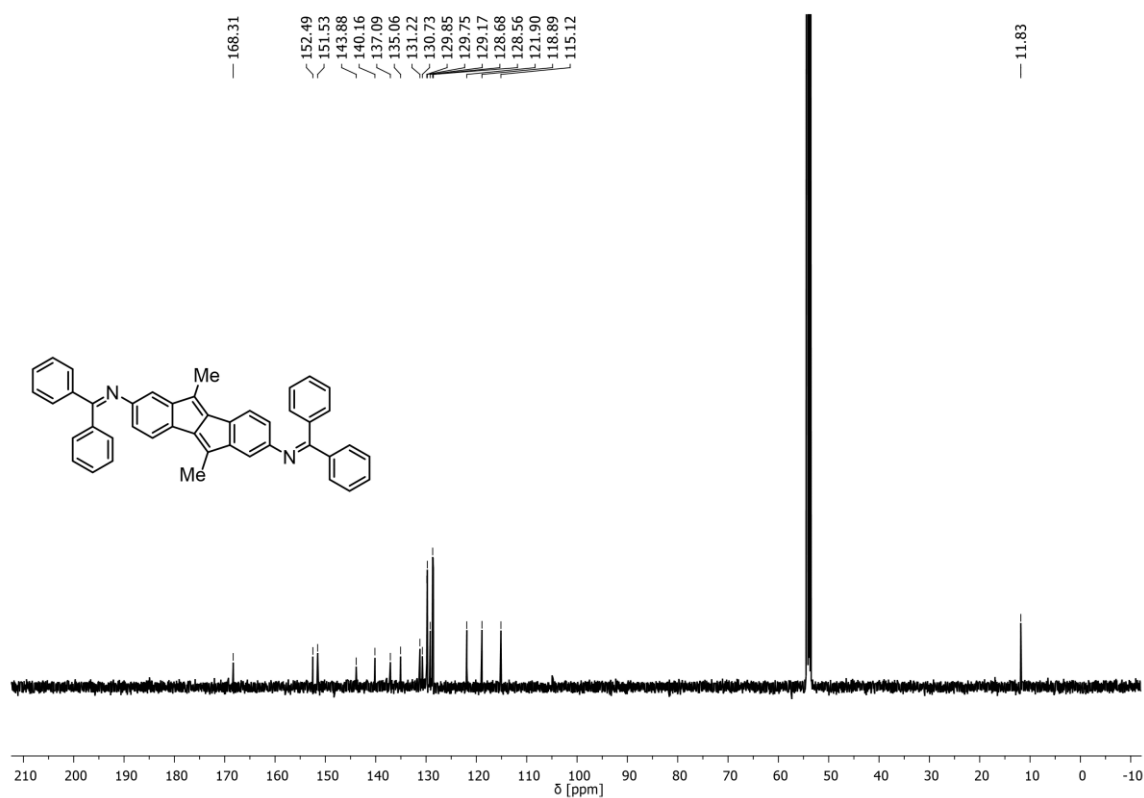


Figure S86 ¹³C NMR spectrum (500 MHz) of *N,N'*-(5,10-Dimethylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) **3** in CD₂Cl₂

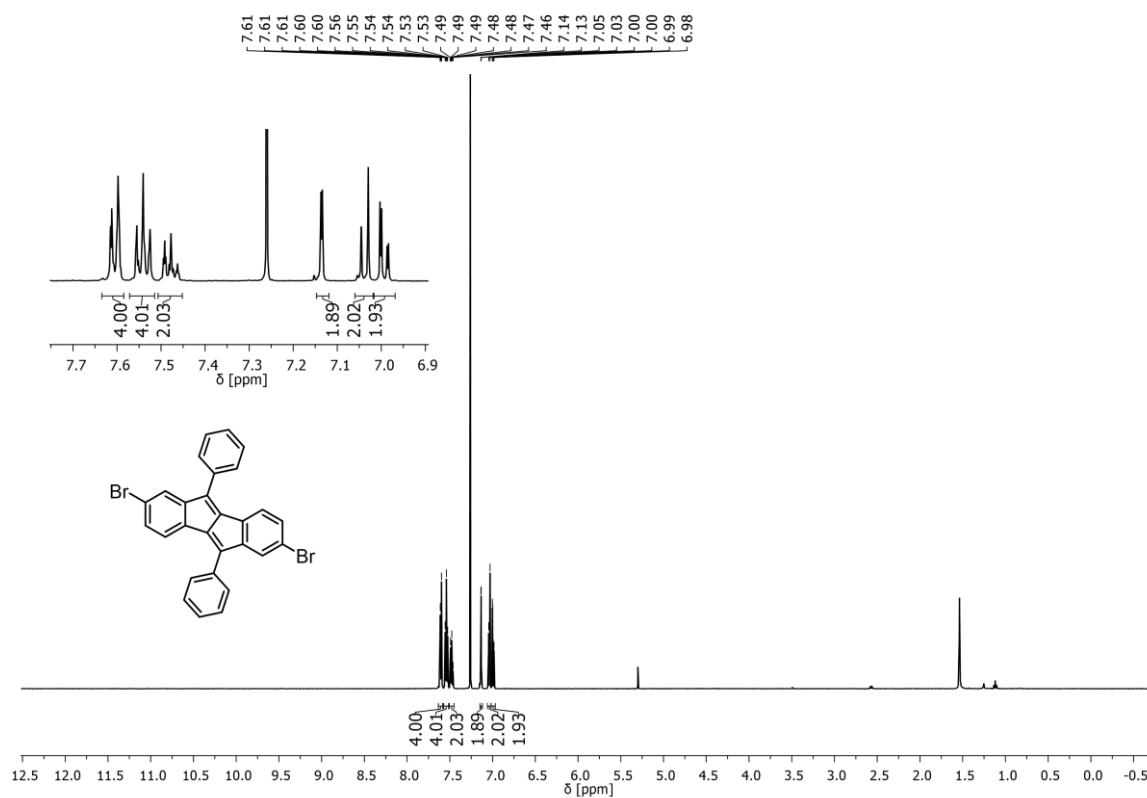


Figure S87 ¹H NMR spectrum (500 MHz) of 2,7-dibromo-5,10-diphenylindeno[2,1-a]indene **4** in CDCl₃

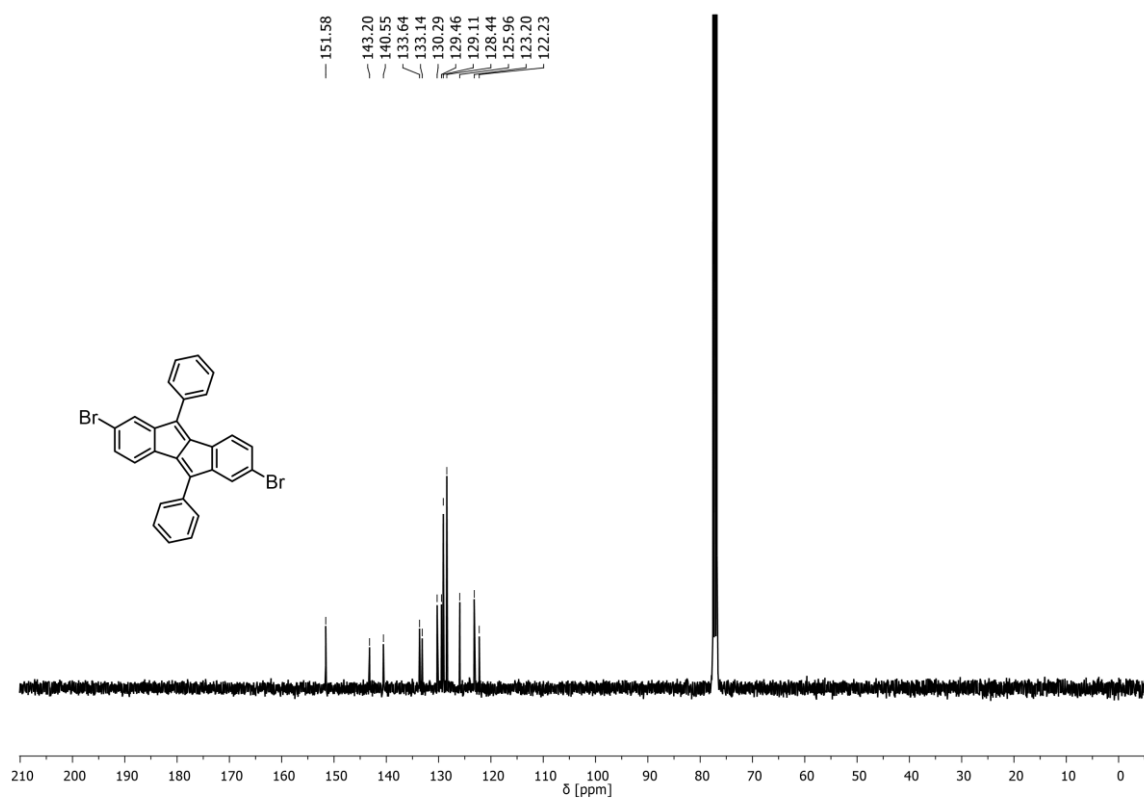


Figure S88 ¹³C NMR spectrum (100 MHz) of 2,7-dibromo-5,10-diphenylindeno[2,1-a]indene **4** in CDCl₃

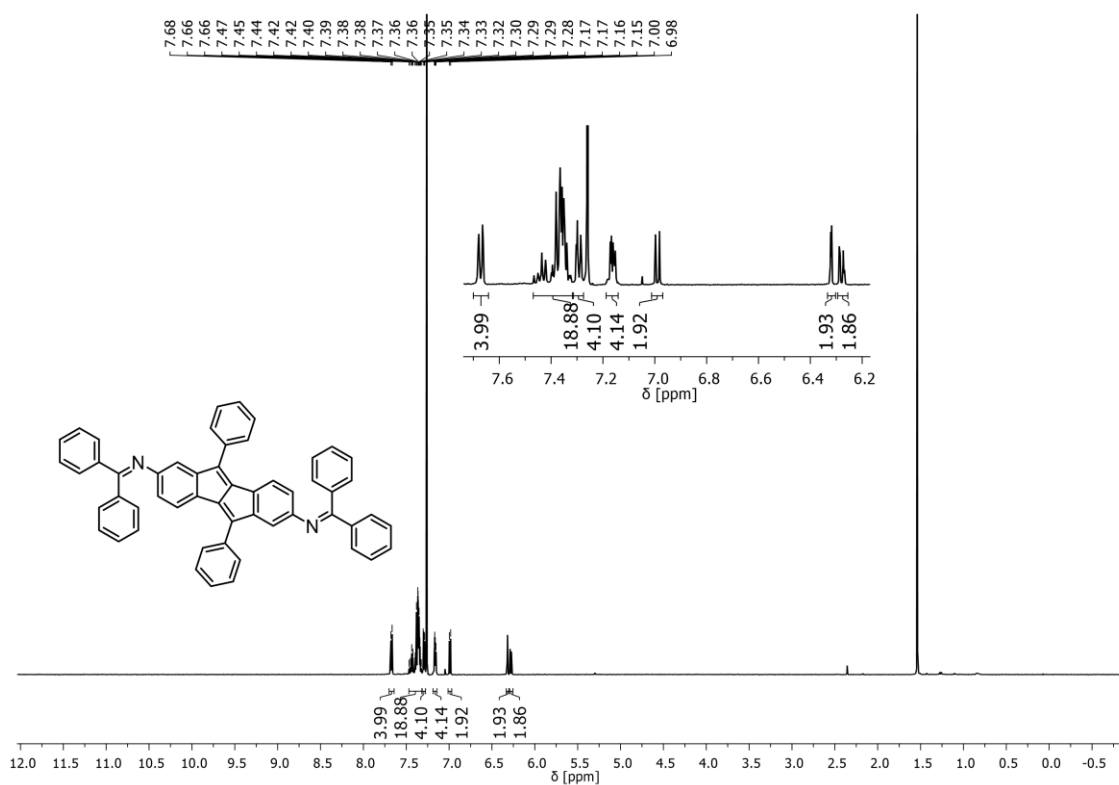


Figure S89 ¹H NMR spectrum (500 MHz) of *N,N'*-(5,10-diphenylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) **5** in CDCl₃

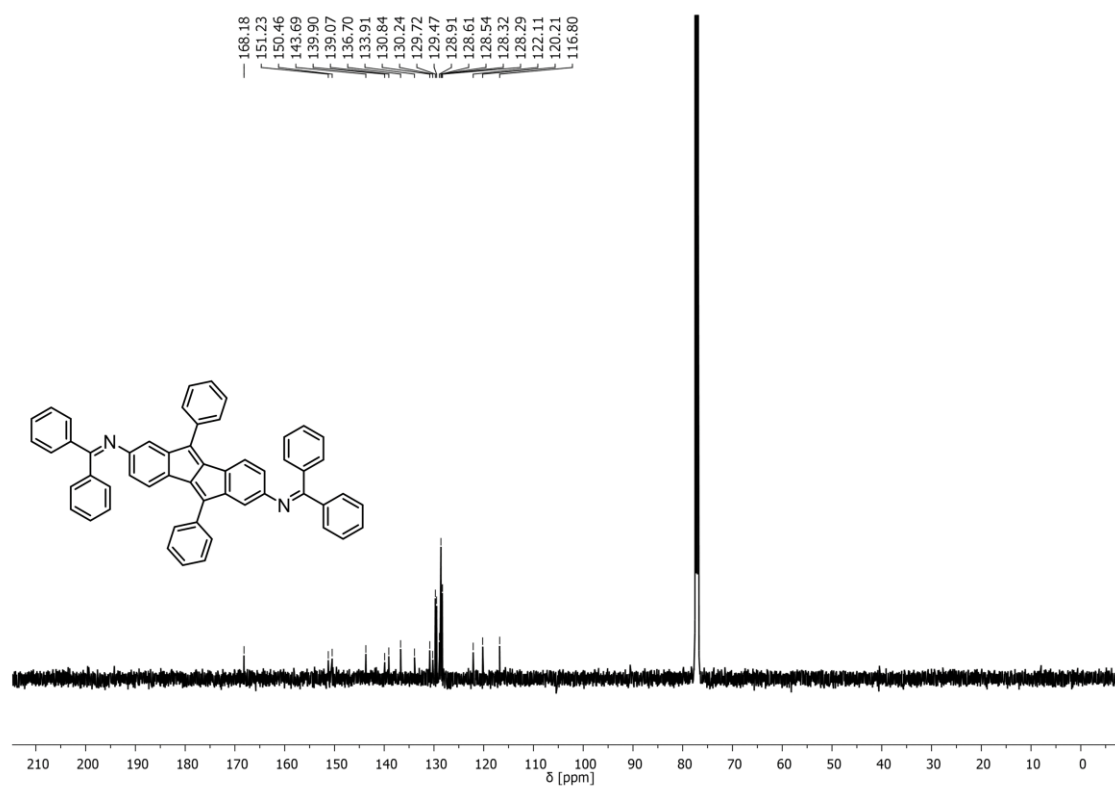


Figure S90 ¹³C NMR spectrum (126 MHz) of *N,N'*-(5,10-diphenylindeno[2,1-*a*]indene-2,7-diyl)bis(1,1-diphenylmethanimine) **5** in CDCl₃

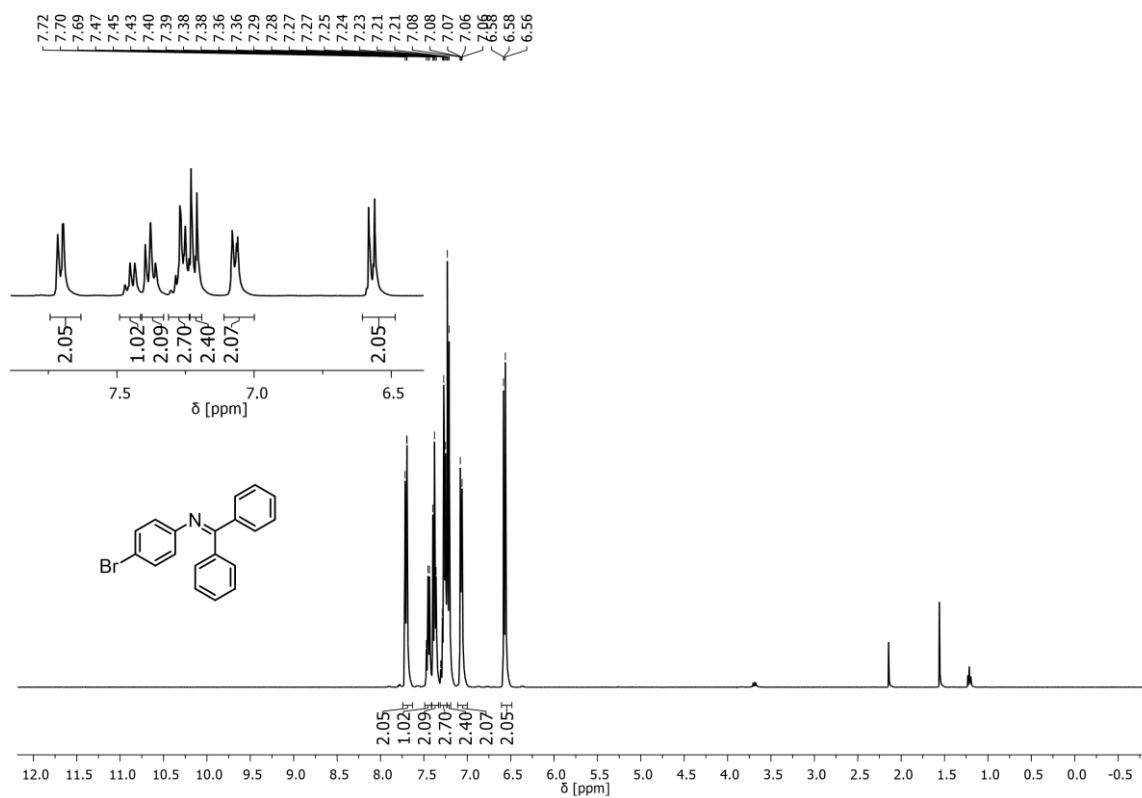


Figure S91 ¹H NMR spectrum (400 MHz) of *N*-(4-bromophenyl)-1,1-diphenylmethanimine **A** in CDCl₃

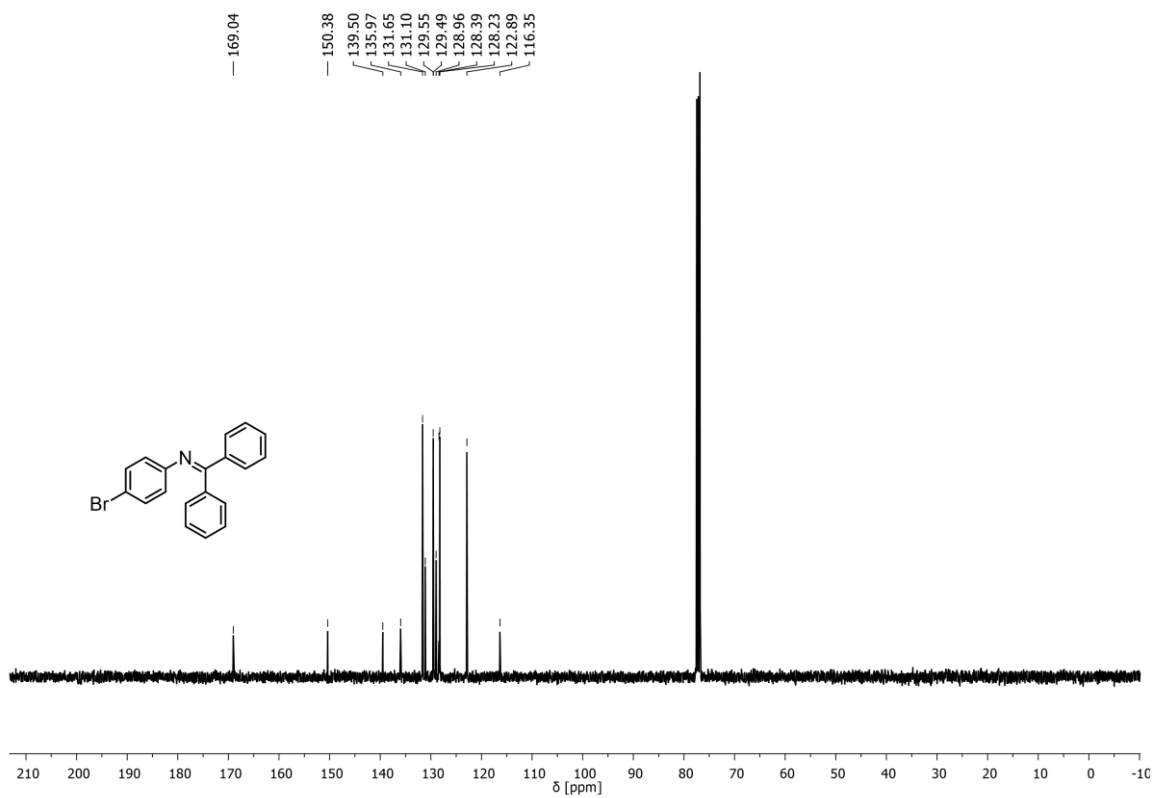


Figure S92 ¹³C NMR spectrum (100 MHz) of *N*-(4-bromophenyl)-1,1-diphenylmethanimine **A** in CDCl₃

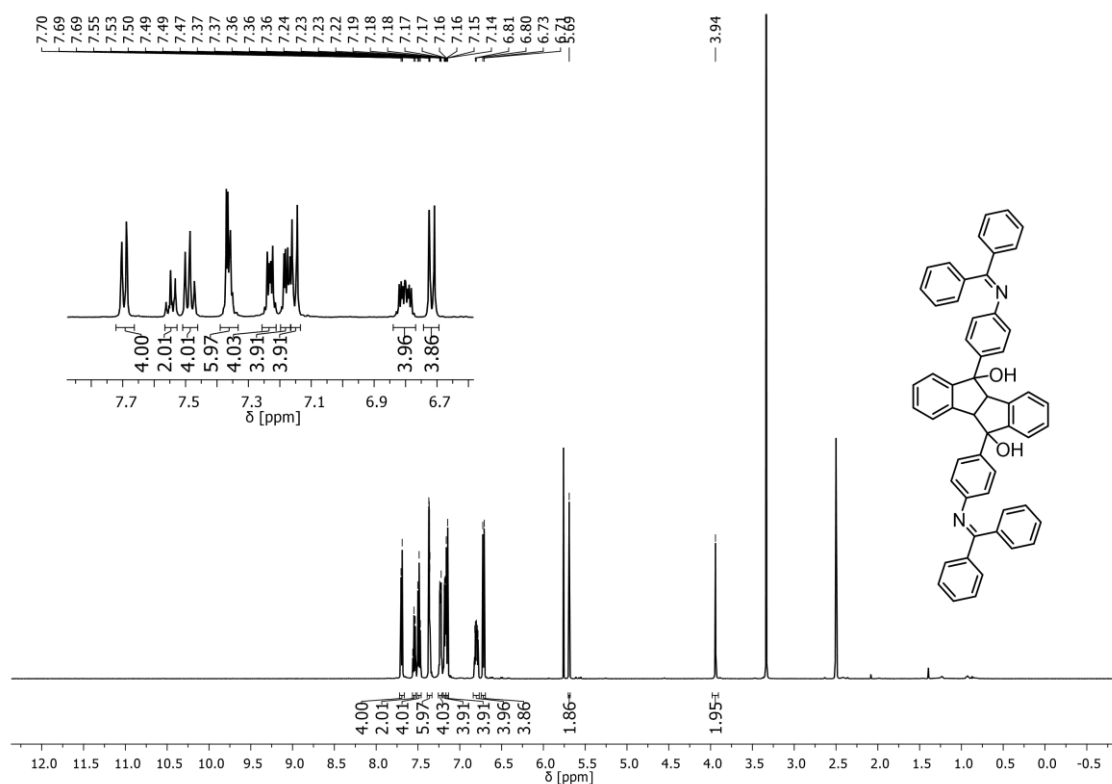


Figure S93 ¹H NMR spectrum (500 MHz) of 5,10-bis(4-((diphenylmethylene)amino)phenyl)-4b,5,9b,10-tetrahydroindeno[2,1-a]indene-5,10-diol **6** in (CD₃)₂SO

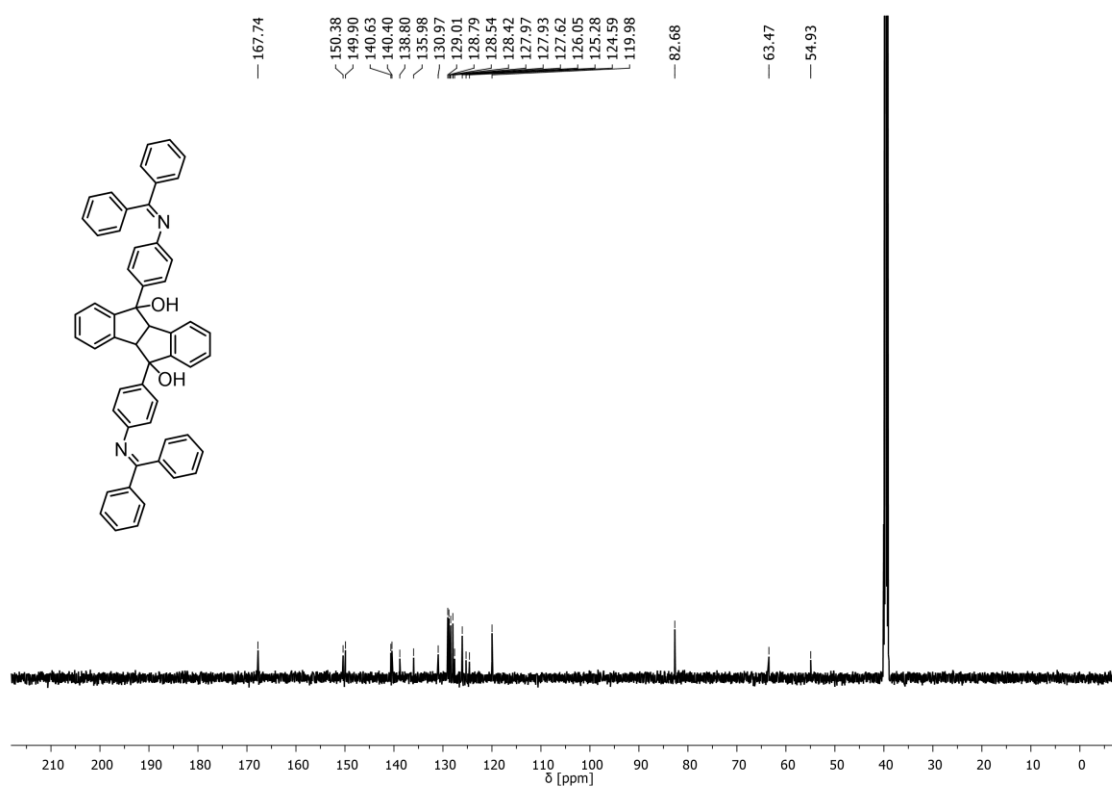


Figure S94 ¹³C NMR spectrum (126 MHz) of 5,10-bis(4-((diphenylmethylene)amino)phenyl)-4b,5,9b,10-tetrahydroindeno[2,1-a]indene-5,10-diol **6** in (CD₃)₂SO

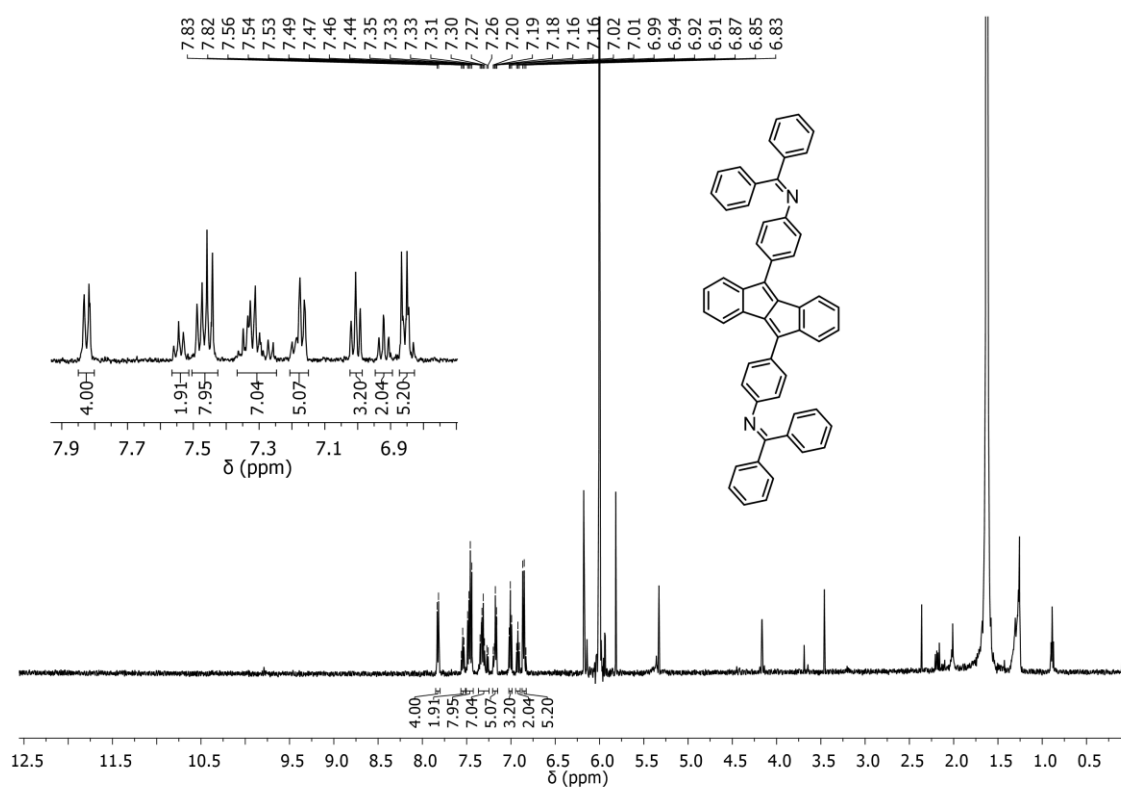


Figure S95 ^1H NMR spectrum (500 MHz) of *N,N'*-(Indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-diphenylmethanimine) **7** in $\text{C}_2\text{D}_2\text{Cl}_4$.

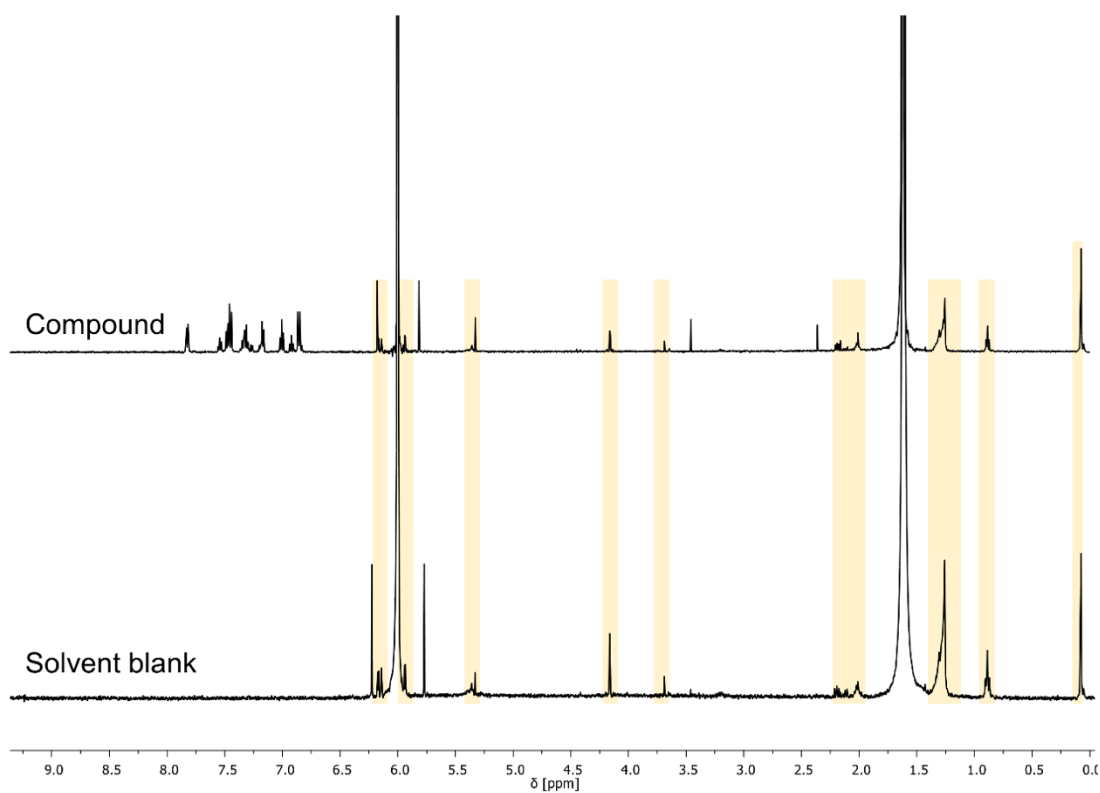


Figure S96 Comparison sample ^1H NMR spectrum of **7** and blank deuterated solvent scan, both at 1024 scans in $\text{C}_2\text{D}_2\text{Cl}_4$. Impurities that stem from the solvent are marked in yellow.

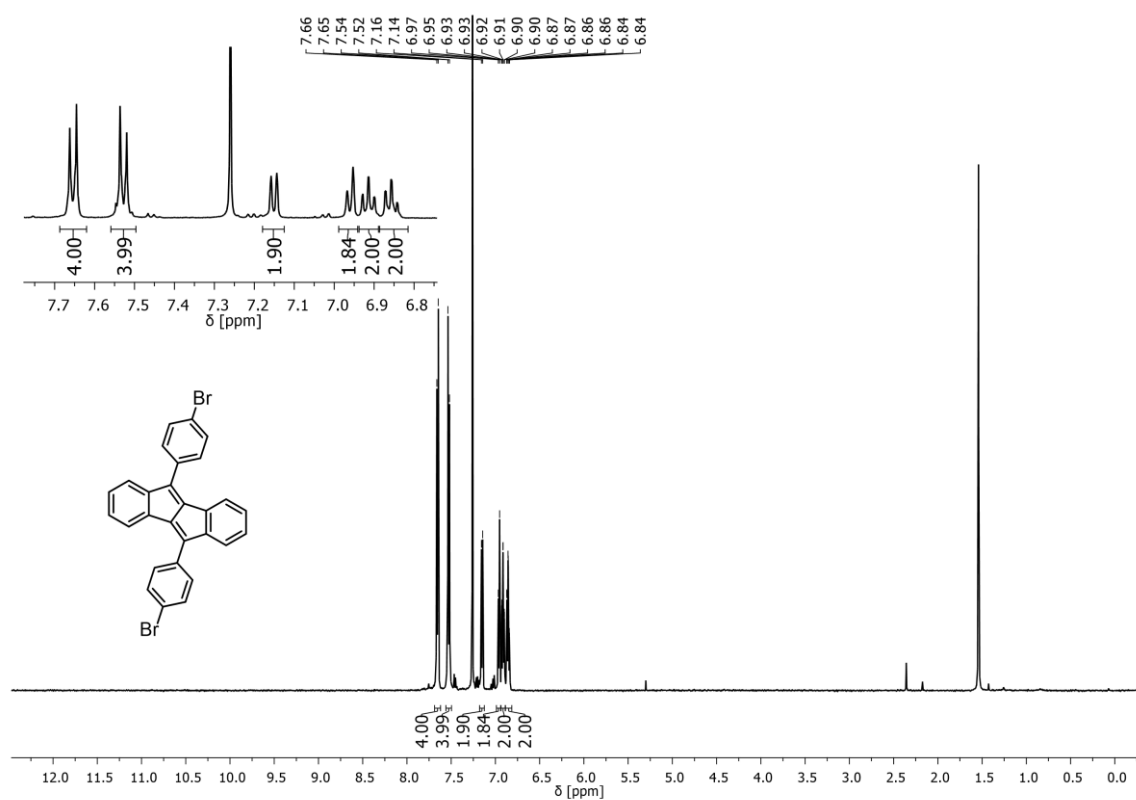


Figure S97 ¹H NMR spectrum (500 MHz) of 5,10-bis(4-bromophenyl)indeno[2,1-a]indene **S1** in CDCl₃

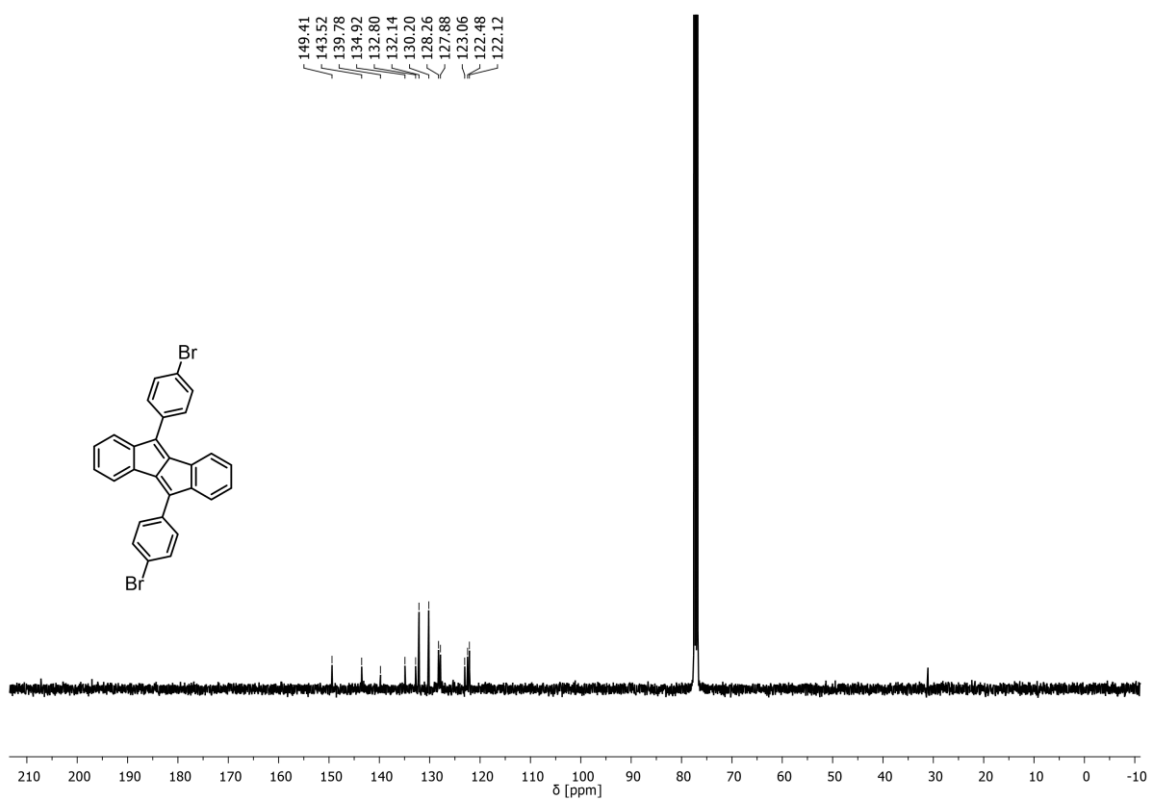


Figure S98 ¹³C NMR spectrum (100 MHz) of 5,10-bis(4-bromophenyl)indeno[2,1-a]indene **S1** in CDCl₃

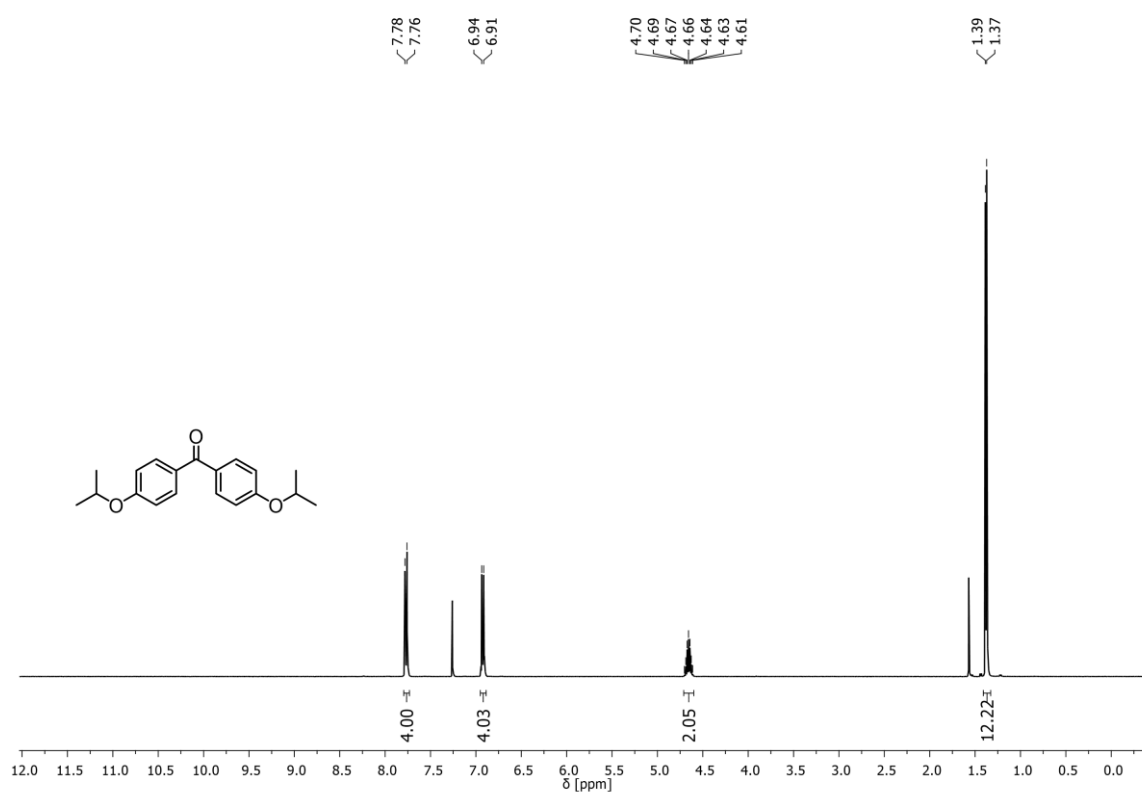


Figure S99 ¹H NMR spectrum (400 MHz) of bis(4-isopropoxyphenyl)methanone **S2** in CDCl₃

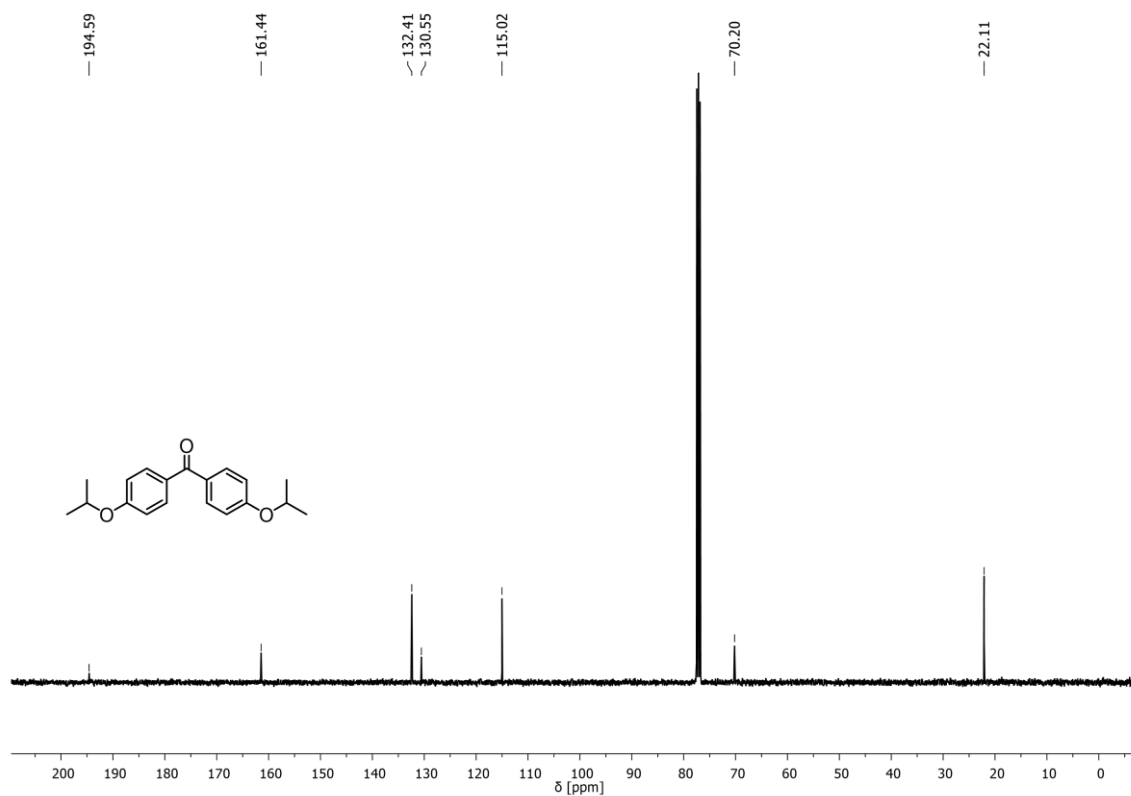


Figure S100 ¹³C NMR spectrum (100 MHz) of bis(4-isopropoxyphenyl)methanone **S2** in CDCl₃

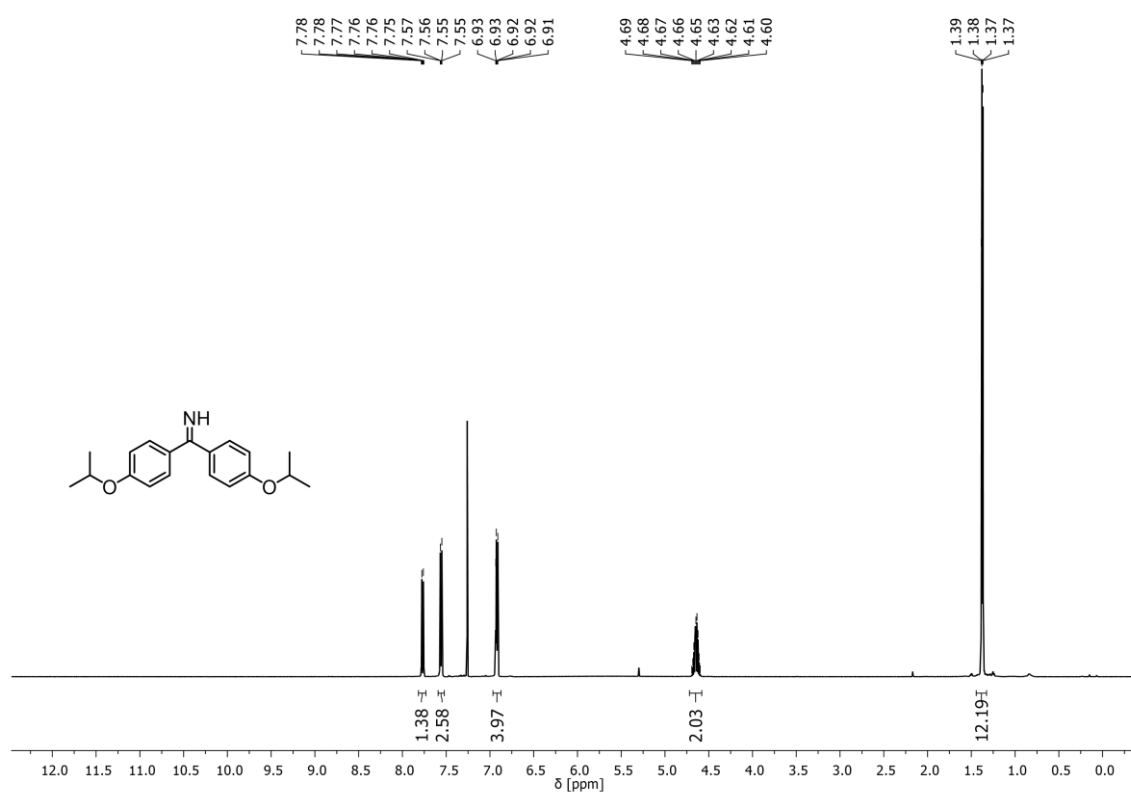


Figure S101 ¹H NMR spectrum (500 MHz) of bis(4-isopropoxyphenyl)methanimine **S3** in CDCl₃

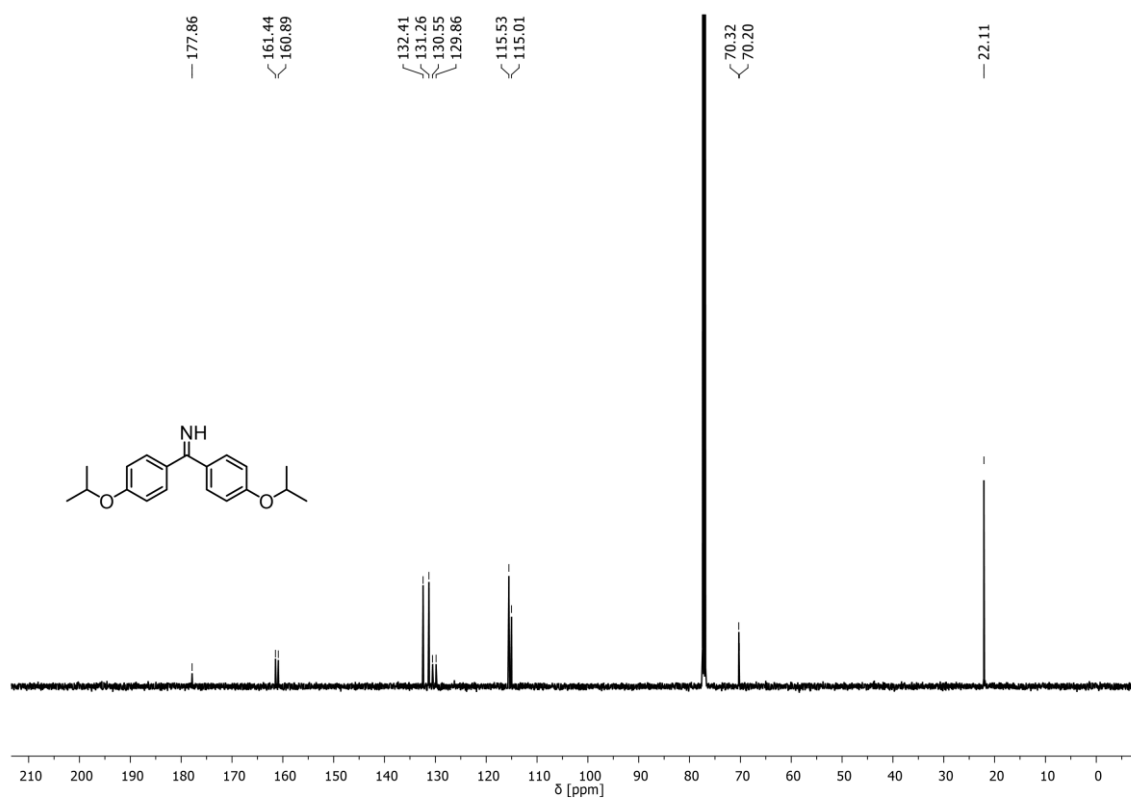


Figure S102 ¹³C NMR spectrum (126 MHz) of bis(4-isopropoxyphenyl)methanimine **S3** in CDCl₃

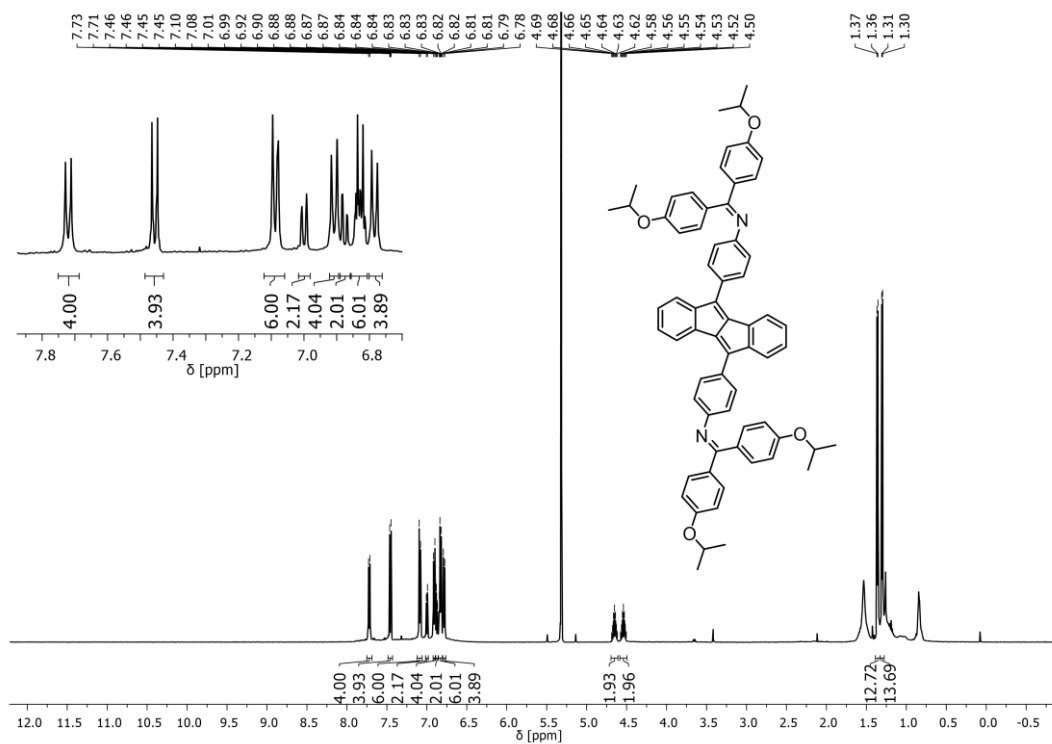


Figure S103 ¹H NMR spectrum (500 MHz) of *N,N'*-(indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-bis(4-isopropoxyphenyl)methanimine) **S4** in CD₂Cl₂

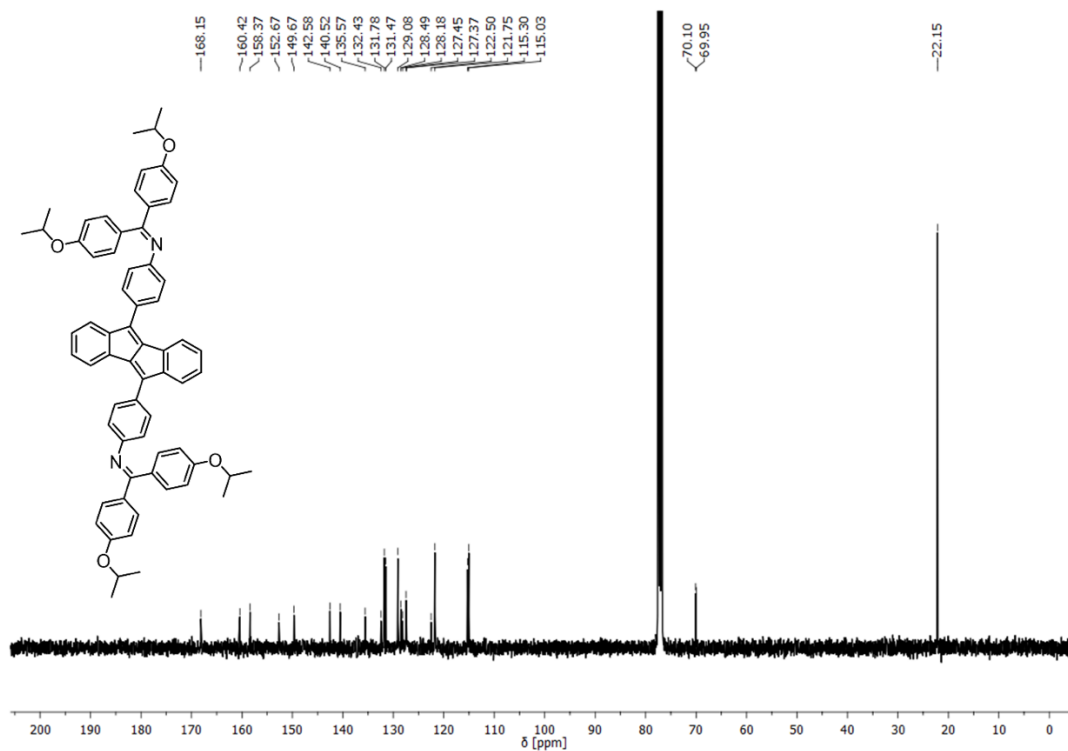


Figure S104 ¹³C NMR spectrum (126 MHz) of *N,N'*-(indeno[2,1-*a*]indene-5,10-diylbis(4,1-phenylene))bis(1,1-bis(4-isopropoxyphenyl)methanimine) **S4** in CDCl₃

S21. Cartesian Coordinates

MeDBP Imine 3

C	-4.42993100	-0.72096700	0.49124000	H	-2.47183300	1.53591800	2.19156700
C	-4.07782900	0.34779400	1.35859800	H	-3.67738500	-2.28534600	-0.81256900
C	-2.73821300	0.70504300	1.55228600	H	3.40976200	1.78036600	1.31295500
C	-1.77146800	-0.02108800	0.87153000	H	4.65134100	-1.33764700	-1.42828400
C	-2.08895900	-1.09365200	0.04429400	H	2.32537300	-2.18124400	-1.54314500
C	-3.42099700	-1.46923200	-0.15027600	H	-0.03397200	-3.55892000	-1.05505900
C	-0.33720100	0.13747800	0.79710200	H	-0.23000100	-2.42030500	-2.44888400
C	0.15512900	-0.87135700	-0.08244000	H	-1.67354100	-3.23175700	-1.73247100
C	-0.83507200	-1.63816900	-0.54845600	H	0.03930600	1.73014600	3.14298700
C	0.63882500	0.92078100	1.25334000	H	-0.20814800	2.83120100	1.72719200
C	1.88626400	0.41959600	0.62344600	H	1.44856800	2.56517300	2.38978400
C	1.59137900	-0.67869800	-0.17806700	H	-8.16752400	2.04290600	0.78003500
C	3.18686500	0.90995900	0.71000900	H	-7.67450000	4.40221000	0.31335400
C	4.20523900	0.26840800	-0.01803300	H	-5.68692500	5.02898400	-1.02500500
C	3.88938400	-0.85444400	-0.83251200	H	-4.18990200	3.26913800	-1.91514900
C	2.57168200	-1.33855400	-0.91102100	H	-4.66111400	0.90888100	-1.49073800
C	-0.68668000	-2.78118700	-1.50379000	H	-7.66766600	-2.61649700	0.86914800
C	0.47231400	2.08250100	2.18338500	H	-9.94316500	-3.51738400	0.75185300
N	-5.79676300	-1.03522300	0.24366800	H	-11.76848100	-2.19514800	-0.27301600
N	5.50914700	0.84103000	0.02053700	H	-11.29805300	0.04935000	-1.20185800
C	6.68075700	0.28603100	-0.05470500	H	-9.05391100	0.97904100	-1.12254100
C	6.87765800	-1.20471900	0.03317200	H	6.61215400	2.93957800	-0.75173800
C	7.84645900	1.23125300	-0.19023200	H	8.47247000	4.50784000	-0.97181400
C	-6.74904300	-0.18680300	-0.02558900	H	10.79638000	3.77154400	-0.54523800
C	-6.45843200	1.26795200	-0.31375500	H	11.24917900	1.43276600	0.12534000
C	-8.15232800	-0.72488700	-0.09356800	H	9.42797900	-0.16425200	0.37370100
C	-7.30128400	2.28531900	0.17828200	H	8.48814500	-1.29049700	-1.42405500
C	-7.02240400	3.63178600	-0.07697900	H	8.81051900	-3.71416600	-1.25593900
C	-5.90178600	3.98645400	-0.82897900	H	7.40093300	-5.06582800	0.26713100
C	-5.05776700	2.99559300	-1.32857700	H	5.65853800	-3.96670800	1.64181800
C	-5.33059100	1.64940500	-1.07489200	H	5.32320400	-1.54389700	1.51723800
C	-8.44105900	-2.01080100	0.41435100	Bq	-0.87135321	-1.30425969	1.00432753
C	-9.73708400	-2.53276800	0.35257800	Bq	-0.78843508	-1.75075710	1.53752329
C	-10.76636000	-1.78903700	-0.22388900	Bq	-0.97551420	-0.69735760	0.21640600
C	-10.50131300	-0.52369600	-0.74470000	Bq	-3.00192227	-1.42950632	2.02553764
C	-9.20935500	0.00515800	-0.68453000	PhDBP Imine 5			
C	7.61375100	2.57844300	-0.55492400	C	-4.38659800	-0.51487200	0.31013700
C	8.67068900	3.48286100	-0.68494500	C	-4.03945900	0.50920000	1.20550300
C	9.97945400	3.06864600	-0.44560600	C	-2.70650100	0.91047600	1.35924300
C	10.23308000	1.74967500	-0.07044600	C	-1.71229200	0.30490500	0.59948100
C	9.18038600	0.83785700	0.05935900	C	-2.05696500	-0.74350100	-0.30770700
C	7.86584100	-1.85017900	-0.73812900	C	-3.37041200	-1.16871700	-0.42589300
C	8.05056100	-3.23310000	-0.65384100	C	-0.25991700	0.44814600	0.52625900
C	7.25551100	-3.99536700	0.20225000	C	0.23277900	-0.55931300	-0.41735900
C	6.27294000	-3.37590200	0.97474000	C	-0.81519800	-1.27315600	-0.93812200
C	6.08299800	-1.99369200	0.89243900	C	0.78812700	1.16207200	1.04676300
H	-4.84647400	0.91571100	1.86683700	C	2.02982200	0.63245900	0.41622200

C	1.68516200	-0.41609300	-0.49074900	H	11.38688900	2.92067800	-0.08104500
C	3.34322600	1.05789600	0.53433300	H	11.27875700	0.71174900	1.05726500
C	4.35945800	0.40405200	-0.20163400	H	9.15602200	-0.54382800	1.13022900
C	4.01234600	-0.62021600	-1.09680300	H	8.10795000	-1.74538900	-1.20721400
C	2.67944400	-1.02164100	-1.25045700	H	8.10858600	-4.17475700	-0.72920100
N	-5.68660200	-1.02939100	0.20060700	H	6.77107500	-5.07232700	1.16664200
N	5.65942000	0.91869700	-0.09219200	H	5.42035900	-3.52176600	2.56625800
C	6.72403000	0.21764900	0.09105400	H	5.38769500	-1.09961200	2.05740600
C	6.74183600	-1.25496700	0.38833400	Bq	-0.82464200	-1.00859400	0.64487900
C	8.02948300	0.94033600	0.03760500	Bq	-0.74531600	-1.48407900	1.15311900
C	-6.75114700	-0.32827700	0.01726200	Bq	-2.97651700	-0.77187900	1.22082000
C	-6.76881000	1.14434600	-0.27999200	Bq	-0.92359600	-0.36025900	-0.11090800
C	-8.05664700	-1.05087000	0.07057700	Bq	-2.94979100	-1.16679700	1.68412200
C	-7.54072900	2.02478200	0.49366300	Bq	-3.03155100	-0.04499000	0.41626100
C	-7.54245300	3.39319000	0.22169300	C	0.74802100	2.26713400	2.01633500
C	-6.79030500	3.89699300	-0.84093100	C	-0.20126800	3.29806400	1.89050400
C	-6.03056800	3.02655700	-1.62568400	C	1.65519800	2.32467200	3.09059200
C	-6.01443600	1.66163400	-1.34365000	C	-0.24647000	4.34438400	2.80977100
C	-8.12841700	-2.30662800	0.70005300	H	-0.88926300	3.28154900	1.05054600
C	-9.32684000	-3.01121000	0.74368900	C	1.60437300	3.36890800	4.01266900
C	-10.47811500	-2.47979900	0.15357800	H	2.38489500	1.53032000	3.21405900
C	-10.41843900	-1.23884800	-0.48046300	C	0.65506100	4.38352700	3.87595100
C	-9.21881100	-0.52719600	-0.51916500	H	-0.98249400	5.13500800	2.68902700
C	8.10111400	2.19604600	-0.59198200	H	2.30707900	3.38840300	4.84157700
C	9.29947600	2.90072600	-0.63574800	H	0.61973700	5.19963000	4.59257600
C	10.45082700	2.36946700	-0.04565000	C	-0.77510800	-2.37799500	-1.90788000
C	10.39128900	1.12857200	0.58851200	C	-1.68214000	-2.43516400	-2.98231000
C	9.19172500	0.41682000	0.62733900	C	0.17392300	-3.40916300	-1.78211000
C	7.51363200	-2.13538400	-0.38547200	C	-1.63139300	-3.47923600	-3.90454900
C	7.51548600	-3.50378200	-0.11346600	H	-2.41165800	-1.64063900	-3.10575700
C	6.76360100	-4.00759800	0.94934200	C	0.21908400	-4.45530900	-2.70158700
C	6.00398300	-3.13718500	1.73423300	H	0.86177400	-3.39301200	-0.94201800
C	5.98771300	-1.77226900	1.45216100	C	-0.68228500	-4.49408000	-3.76790700
H	-4.81519200	0.97963500	1.80049000	H	-2.33402300	-3.49846900	-4.73352700
H	-2.46141700	1.68759500	2.07650600	H	0.95499100	-5.24604900	-2.58089200
H	-3.64761700	-2.00234600	-1.06282500	H	-0.64702500	-5.31003700	-4.48469900
H	3.62026700	1.89170900	1.17107300	DBP-Ph Imine 7			
H	4.78807600	-1.09064800	-1.69179600	C	-1.04526600	4.47724900	0.04004800
H	2.43447000	-1.79885700	-1.96766500	C	0.35708500	4.46688900	-0.07720400
H	-8.13525600	1.63480100	1.31526100	C	1.05147700	3.27248000	-0.28799000
H	-8.13565600	4.06417500	0.83731700	C	0.31422900	2.10809300	-0.37319500
H	-6.79767600	4.96172800	-1.05820600	C	-1.09690600	2.10781800	-0.26557700
H	-5.44674500	3.41112900	-2.45757300	C	-1.78372900	3.29285700	-0.05507700
H	-5.41433300	0.98896000	-1.94879000	C	0.66384200	0.71936200	-0.58160400
H	-7.22698100	-2.71146100	1.14726100	C	-0.56679000	-0.05630600	-0.61888500
H	-9.36648400	-3.97693100	1.24077500	C	-1.63707200	0.73129100	-0.41346200
H	-11.41422600	-3.03093200	0.18887400	C	1.73526200	-0.08508400	-0.71212000
H	-11.30585000	-0.82190200	-0.94921500	C	1.18241900	-1.46052300	-0.88008200
H	-9.18299700	0.43350100	-1.02195600	C	-0.22894700	-1.44653900	-0.81925400
H	7.19961900	2.60076500	-1.03917200	C	1.85934000	-2.65854500	-1.05170100
H	9.33901300	3.86640400	-1.13292500				

C	1.10981300	-3.83343900	-1.17616500	C	9.96659600	1.82522300	0.00420200
C	-0.29558800	-3.80351000	-1.12177900	H	-1.56182400	5.42054600	0.20249700
C	-0.97993700	-2.59947500	-0.93880900	H	0.90792000	5.40223900	-0.00570400
C	-3.00683100	0.24527000	-0.29893100	H	2.13135500	3.26813300	-0.38659100
C	3.13429100	0.33058900	-0.63821900	H	-2.86555200	3.30626000	0.02995300
C	-3.50511000	-0.12750200	0.95408200	H	2.94352500	-2.69128400	-1.08412900
C	-4.79806200	-0.64038400	1.06989500	H	1.61997900	-4.78409000	-1.31348000
C	-5.61162700	-0.77041300	-0.05728300	H	-0.85676200	-4.73020200	-1.21924300
C	-5.10648900	-0.41326300	-1.31003700	H	-2.06324300	-2.57758800	-0.89070000
C	-3.81203700	0.09998000	-1.43266700	H	-2.88376600	-0.03821000	1.84199000
C	3.97850600	0.17599400	-1.74283700	H	-5.17426900	-0.94959200	2.04057100
C	5.33804500	0.48122800	-1.63624600	H	-5.72387900	-0.54434800	-2.19363300
C	5.87072800	0.92317400	-0.42339800	H	-3.43234000	0.36872200	-2.41529500
C	5.01766600	1.13034100	0.66235800	H	3.58739200	-0.19575800	-2.68681300
C	3.65911200	0.82474600	0.56182400	H	5.99301800	0.35109200	-2.49255800
N	-6.89125000	-1.32642500	0.06738100	H	5.42216900	1.50257700	1.59878000
N	7.24577500	1.16904200	-0.30039500	H	3.01935100	0.95153800	1.43209000
C	-7.93844500	-0.56559200	0.23007600	H	-7.43775300	0.97709900	2.39381900
C	8.03998100	0.23117300	0.14311000	H	-7.34200000	3.45966100	2.46728100
C	-7.87200300	0.91754200	0.27810800	H	-7.71675600	4.77648800	0.39769200
C	-7.60720100	1.55753400	1.49127400	H	-8.18247400	3.62461500	-1.75003400
C	-7.55299700	2.95015400	1.53049700	H	-8.27793200	1.14259700	-1.83266200
C	-7.76336600	3.69041700	0.36459500	H	-10.43134800	0.59753200	0.73484600
C	-8.02623900	3.04299100	-0.84502900	H	-12.57931000	-0.55644700	0.97967200
C	-8.08128100	1.65081700	-0.89271300	H	-12.71384000	-3.02548900	0.78193200
C	-9.28541400	-1.21961000	0.37988400	H	-10.65996000	-4.34506400	0.33012000
C	-10.45107800	-0.48391000	0.63623900	H	-8.48650800	-3.21088900	0.07691000
C	-11.68064400	-1.13460900	0.77921800	H	7.71609900	-1.89736700	-1.49173200
C	-11.75730800	-2.52169000	0.66870200	H	6.91056000	-4.16182700	-0.85771500
C	-10.60582400	-3.26275400	0.41518600	H	6.30414600	-4.64672400	1.49922800
C	-9.37626400	-2.61552500	0.27198300	H	6.50058600	-2.87989600	3.22937600
C	7.56719700	-1.12768000	0.52161100	H	7.30617900	-0.61363700	2.60458900
C	7.45949500	-2.11700200	-0.45924300	H	10.12615900	-1.45792700	0.92439700
C	7.00574600	-3.38608400	-0.10233200	H	12.49594300	-0.87873100	1.14418300
C	6.66470100	-3.65802800	1.22471100	H	13.28353000	1.41948100	0.64235700
C	6.77576600	-2.66414200	2.20002700	H	11.66217700	3.15151500	-0.08965800
C	7.22924600	-1.39267400	1.85134100	H	9.27043500	2.59752100	-0.31681900
C	9.50726600	0.53072400	0.28707000	Bq	-0.28902875	3.15898863	0.81878264
C	10.43183500	-0.44065500	0.69753400	Bq	-0.36718500	3.28756433	-0.16983250
C	11.78667100	-0.11916100	0.82454100	Bq	-0.39599021	0.99033695	0.53837009
C	12.22990000	1.17141000	0.54268600	Bq	-0.46453940	1.12205160	-0.45054460
C	11.32079500	2.14360900	0.13196600				

Cartesian coordinates of **PhDBP-TFP*** in the neutral state

C	-2.2503154	-3.4725113	-1.4488227	C	5.1311409	6.8843226	3.5452759
C	-2.7794274	-2.3640827	-2.1115266	C	5.8376607	5.853819	4.4066364
C	-2.2455677	-1.0910631	-1.8963914	O	5.446521	8.0749054	3.6384614
C	-1.1956942	-0.9191145	-1.0118375	C	-5.2581198	-6.8591112	-3.4048605
C	-0.6730475	-2.0497566	-0.3182699	C	-6.0108332	-5.8192635	-4.214375
C	-1.1823421	-3.3105285	-0.543205	O	-5.5912359	-8.0461992	-3.47876
C	-0.4191927	0.2308765	-0.5664335	H	-3.5998812	-2.4727094	-2.8054946
C	0.5670282	-0.2595542	0.4023589	H	-2.6649615	-0.2509925	-2.432403
C	0.436095	-1.6083687	0.569293	H	-0.7787502	-4.1809163	-0.0423733
C	-0.3408156	1.5896518	-0.6748804	H	0.838015	4.1689078	-0.0243572
C	0.7373176	2.0370265	0.2470922	H	3.5994563	2.4732132	2.8060149
C	1.2902977	0.900704	0.9070354	H	2.7473262	0.2355825	2.3416407
C	1.2313524	3.3008067	0.4884509	H	3.1064288	-1.6392159	0.8830573
C	2.2756271	3.4676744	1.4204496	H	4.4311387	-3.0890796	2.3719088
C	2.8021564	2.3599154	2.0863183	H	3.3014073	-4.792434	3.7761685
C	2.3127052	1.0782858	1.8221529	H	0.8353956	-5.0224922	3.6800824
C	1.2138331	-2.4896973	1.4408879	H	-0.4878448	-3.5645153	2.2063352
C	-1.1419622	2.4755	-1.5203311	H	-1.5050428	3.9402688	0.0157051
C	2.6081417	-2.3692397	1.5071877	H	-2.8678513	5.4050947	-1.4148115
C	3.3538458	-3.1925084	2.3396802	H	-3.3007374	4.7910689	-3.7767219
C	2.7202985	-4.1511632	3.1257051	H	-2.3461359	2.6946643	-4.6950449
C	1.3359577	-4.2827092	3.0681548	H	-0.9677523	1.2346705	-3.2657129
C	0.5889783	-3.4653905	2.2296323	H	-2.2574177	-5.4909705	-1.1027812
C	-1.6758322	3.6699878	-1.0176405	H	2.2786606	5.4867878	1.0784301
C	-2.4524083	4.4925924	-1.8234852	H	-4.237881	-4.3770839	-2.9792441
C	-2.699376	4.1461489	-3.148633	H	4.1690368	4.3908732	3.0557283
C	-2.1660727	2.9673996	-3.6628146	H	3.6272812	7.2494638	2.044091
C	-1.3967803	2.1397758	-2.8564926	H	-3.6676263	-7.2415479	-2.0003678
N	-2.7285656	-4.7717974	-1.635153	H	5.1311515	5.3445832	5.0655803
N	2.7359107	4.7704463	1.6264883	H	6.3234223	5.0924552	3.7930466
C	-3.7472061	-5.1655248	-2.4277011	H	6.5888595	6.3570159	5.0115568
C	3.7027806	5.1743897	2.4766473	H	-6.4797988	-5.0798737	-3.5616612
C	4.0968892	6.4654145	2.6286024	H	-6.7815319	-6.3184505	-4.7976873
C	-4.1634619	-6.4522245	-2.555379	H	-5.3406481	-5.2842027	-4.8902039

Cartesian coordinates of **PhDBP-TFP*** in the radical cation state

C	-2.2453898	-3.455322	-1.4152749	C	0.4538531	-1.6179548	0.5692912
C	-2.7889493	-2.3266909	-2.0760159	C	-0.3423044	1.5958399	-0.6936504
C	-2.2579811	-1.0696234	-1.8742574	C	0.7174565	2.0376632	0.2264433
C	-1.1832216	-0.9047696	-0.9973708	C	1.2708752	0.8872839	0.8995958
C	-0.6503495	-2.0513172	-0.3014532	C	1.207481	3.2976752	0.4737693
C	-1.1646518	-3.3065493	-0.5223654	C	2.2383224	3.4562789	1.422234
C	-0.4216789	0.2266381	-0.5735637	C	2.7591189	2.3321032	2.1084971
C	0.5747727	-0.2566183	0.4028636	C	2.285312	1.0639669	1.8435061

C	1.2478651	-2.4912544	1.4269386	H	-0.7614315	-4.1814551	-0.0317856
C	-1.1333663	2.4691522	-1.5541133	H	0.8261787	4.1680127	-0.0418681
C	2.6401047	-2.3391357	1.4893249	H	3.5406758	2.4558175	2.8421915
C	3.4039447	-3.1594556	2.3064998	H	2.7047405	0.2195962	2.370565
C	2.7904366	-4.1399109	3.0818918	H	3.1220951	-1.5945781	0.8698967
C	1.4085836	-4.2995074	3.0292366	H	4.4790113	-3.0371367	2.3355708
C	0.6418778	-3.4892993	2.2033262	H	3.3864883	-4.7775581	3.7221044
C	-1.6549372	3.6786908	-1.0734296	H	0.9261842	-5.0555944	3.6352378
C	-2.4229885	4.4895508	-1.8974987	H	-0.4327282	-3.6071341	2.1848903
C	-2.6681343	4.1175826	-3.2163362	H	-1.4867884	3.9660153	-0.0446876
C	-2.144929	2.9243499	-3.707645	H	-2.8332025	5.4127153	-1.5088191
C	-1.3880026	2.1044544	-2.8835578	H	-3.2619884	4.7552248	-3.8586267
N	-2.7358294	-4.7219386	-1.6079582	H	-2.323627	2.6347876	-4.7351757
N	2.7032358	4.7278646	1.6434411	H	-0.9630042	1.1898365	-3.2749544
C	-3.768171	-5.1056077	-2.4089287	H	-2.2740127	-5.4564983	-1.085539
C	3.6778763	5.1226311	2.5087321	H	2.2618801	5.458499	1.098335
C	4.058508	6.4114557	2.6472882	H	-4.2566028	-4.3164643	-2.9591135
C	-4.1726068	-6.3897784	-2.5206183	H	4.139286	4.3386412	3.0888057
C	5.0967734	6.8505773	3.5695332	H	3.5862571	7.1877694	2.0553587
C	5.8037001	5.8372509	4.4418623	H	-3.6708131	-7.1717479	-1.9613368
O	5.3913039	8.0425135	3.6358993	H	5.0929989	5.3267649	5.0952928
C	-5.2771076	-6.8160359	-3.3689869	H	6.297471	5.0763888	3.8339335
C	-6.0344548	-5.7929827	-4.185917	H	6.5463203	6.349253	5.049595
O	-5.5887551	-8.0045518	-3.4156775	H	-6.4941059	-5.0434469	-3.5381837
H	-3.6220738	-2.4402934	-2.752364	H	-6.810287	-6.2986364	-4.7563136
H	-2.681314	-0.2242559	-2.3965847	H	-5.3652004	-5.269742	-4.8719806

Cartesian coordinates of **DBP-Ph-TFP*** in the neutral state

C	4.2879702	-0.7506755	-0.6199564	C	-0.2688414	3.1251313	0.2088752
C	4.1266901	0.627454	-0.6913733	C	-0.7570601	4.0628282	1.1309742
C	2.8793853	1.2159683	-0.4517937	C	-0.5261339	5.4159477	0.9663707
C	1.8037332	0.4071578	-0.1236762	C	0.1899098	5.8874449	-0.1412327
C	1.9774853	-1.0054913	-0.019988	C	0.6735013	4.9653474	-1.0768545
C	3.2101134	-1.5787666	-0.2849995	C	0.4462759	3.6126412	-0.893371
C	0.3997136	0.6707169	0.1793575	C	1.3891586	-3.8532972	1.2442123
C	-0.2316379	-0.6133104	0.4626917	C	1.1627583	-5.2023812	1.444581
C	0.6809599	-1.628379	0.3527436	C	0.0087496	-5.8145277	0.9392794
C	-0.4881771	1.6936745	0.3812767	C	-0.9144924	-5.0371405	0.2301464
C	-1.7678235	1.0757958	0.8163147	C	-0.6819211	-3.6851454	0.0479384
C	-1.607725	-0.3412363	0.8684137	N	-0.1606944	-7.1784875	1.1687319
C	-2.9909595	1.6516011	1.1175661	N	0.3803227	7.2627977	-0.2585549
C	-4.0534658	0.8278743	1.507754	C	1.0590032	7.9162461	-1.2270652
C	-3.8892101	-0.5494091	1.5882328	C	1.2099521	9.2640782	-1.2760789
C	-2.6657125	-1.1450596	1.260041	C	1.9423242	9.9506772	-2.3158296
C	0.4666101	-3.0576143	0.548904	C	2.596387	9.1536182	-3.4288283

O	2.0434958	11.1810224	-2.3062318	H	0.8067553	2.9166969	-1.6387541
C	-1.1894742	-7.9554609	0.764391	H	2.2794332	-3.4044806	1.6625408
C	-1.2837339	-9.2864435	1.0124364	H	1.8788857	-5.7959338	1.9996917
C	-2.3912446	-10.1058958	0.5756392	H	-1.8093472	-5.4750132	-0.1877412
O	-2.4185319	-11.313671	0.8300045	H	-1.3952587	-3.1044943	-0.5212327
C	-3.531975	-9.4781233	-0.2032076	H	0.5808344	-7.6310114	1.6865968
H	5.2556409	-1.1910725	-0.8238919	H	-0.0292965	7.8258299	0.4749631
H	4.9714488	1.2554841	-0.9444452	H	1.4942355	7.2904245	-1.9923336
H	2.7681103	2.288657	-0.5264275	H	0.7698	9.8891767	-0.5064343
H	3.3491454	-2.6504541	-0.2400849	H	3.0830979	9.8426814	-4.1156152
H	-3.136134	2.7213826	1.0523718	H	1.8590994	8.563236	-3.9766538
H	-5.0110879	1.2712013	1.7492941	H	3.3409692	8.4621209	-3.0288031
H	-4.7181731	-1.1727808	1.8985673	H	-1.9662543	-7.4468506	0.2126722
H	-2.5595275	-2.2195391	1.3133193	H	-0.5006255	-9.7936343	1.5658965
H	-1.2965603	3.7269639	2.0057109	H	-3.1734332	-9.0366839	-1.1354701
H	-0.8971953	6.1225269	1.6986976	H	-4.0122785	-8.6851446	0.3736521
H	1.2182851	5.2904585	-1.9512467	H	-4.2662544	-10.2472546	-0.4326744

Cartesian coordinates of **DBP-Ph-TFP*** in the radical cation state

C	4.7110729	0.1021346	0.0015778	C	0.319951	-5.0940532	0.1602198
C	4.2888291	1.4061813	-0.2008362	C	0.3023851	-3.7300406	0.0040797
C	2.9248946	1.7336212	-0.1686278	N	1.3802586	-7.0452412	1.1834487
C	2.0004129	0.7399023	0.0929632	N	-0.5968186	7.1245809	-0.5934642
C	2.4340967	-0.6019955	0.2991718	C	-1.4346234	7.9219806	-1.3073382
C	3.7802062	-0.9191693	0.2356774	C	-1.3206452	9.2669883	-1.3571063
C	0.5363701	0.7070004	0.1282689	C	-2.2570723	10.066579	-2.1480351
C	0.1241998	-0.6516722	0.2787415	C	-2.0315421	11.5579594	-2.131715
C	1.2433377	-1.4790079	0.4156799	O	-3.1734173	9.5677461	-2.7959484
C	-0.582104	1.536762	-0.0002265	C	0.5521327	-8.0063292	0.6978134
C	-1.7731182	0.6600781	0.1098387	C	0.6960532	-9.3222377	0.9697374
C	-1.3402028	-0.6828256	0.313815	C	-0.2353871	-10.3034214	0.4146269
C	-3.1193943	0.9776681	0.1698582	O	-1.16795	-9.9914882	-0.3220544
C	-4.0513443	-0.0436758	0.3981655	C	-0.0097599	-11.7417525	0.8122634
C	-3.6300469	-1.3482642	0.599647	H	5.765487	-0.137603	-0.0374745
C	-2.2662376	-1.6763016	0.5708472	H	5.0147546	2.1837114	-0.4003342
C	1.2721091	-2.897373	0.607654	H	2.6191945	2.7493658	-0.3699971
C	-0.6045131	2.958158	-0.1669379	H	4.1245902	-1.9379192	0.3386336
C	0.3690977	3.773887	0.4541587	H	-3.46286	1.9968116	0.0677751
C	0.3477538	5.137325	0.3167517	H	-5.1058678	0.1961485	0.4334373
C	-0.6373299	5.7583843	-0.479661	H	-4.3568506	-2.1259707	0.7951756
C	-1.610664	4.961966	-1.1154249	H	-1.9622128	-2.6926541	0.7713656
C	-1.5979473	3.5988465	-0.9433352	H	1.1083947	3.3188855	1.0962542
C	2.2701232	-3.5177602	1.3944144	H	1.0809084	5.7512378	0.8240566
C	2.2796166	-4.8765729	1.5837121	H	-2.3666806	5.4075752	-1.7444026
C	1.3093865	-5.6921244	0.9660634	H	-2.3344482	3.0033272	-1.4615971

H	3.0061598	-2.9123436	1.9016327	H	-1.0682796	11.7869935	-2.5967359
H	3.0289918	-5.3355668	2.2157823	H	-2.8263638	12.0668868	-2.6727569
H	-0.421987	-5.6941932	-0.3446471	H	-1.9839056	11.9249712	-1.1036664
H	-0.4382346	-3.2880157	-0.6455225	H	-0.2572059	-7.6783222	0.0610088
H	2.1374049	-7.3558426	1.7801155	H	1.5007356	-9.6691587	1.606501
H	0.1486619	7.5862895	-0.0866292	H	-0.5104835	-12.4077048	0.1119093
H	-2.2259402	7.4316643	-1.8561041	H	-0.4344339	-11.899628	1.8089557
H	-0.5346754	9.7753422	-0.811549	H	1.0529769	-11.9807304	0.8703286

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